GEK-73906B

DRIVE SYSTEM INSTRUCTIONS

DIRECTO-MATIC® PLUS

DLC+ SEQUENCER OPERATOR MANUAL

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

DIRECTO-MATIC® PLUS
DLC + SEQUENCER OPERATOR MANUAL

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The information contained herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, U.S.A.

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HOW TO USE THIS MANUAL

This manual is a guide to operating and programming General Electric's Directo-Matic® Plus Distributed Logic Control (DLC+) Sequencer. The manual also contains information on how to operate the DLC+ Programming Module and its cassette tape drive and printer.

We have written this manual for two groups of people: the system programmer and the system troubleshooter. The system programmer is responsible for maintaining DLC + Sequencer application software (relay strings). The system troubleshooter is responsible for troubleshooting and maintaining the system's hardware. Both groups will use the Programming Module, but for different purposes. Since each of these groups has different needs, we have included a wide range of information in this manual. Some of the written material may not apply to your particular job.

We strongly suggest that you read the entire manual before attempting to operate or program the DLC+ Sequencer. However, if you need a quick guide to operating the DLC + Programming Module, read Chapter 3 and Appendix A. Chapter 3 provides a very basic, tutorial introduction to the programming module. Appendix A is a description of the programming module keyboard functions.

We have included several programming examples in this manual to help you better understand the equipment operation. As you read through these examples, try programming them on a sequencer. We have also included a glossary in Appendix F which includes unfamiliar or specialized terms.

There are no prerequisite manuals to this one, but the following General Electric publications are suggested as reference material:

- Directo-Matic @Plus DLC + Sequencer Maintenance Manual (GEK-65477)
- Control Signal Highway Maintenance Manual (GEK-65411)

WARNING

To avoid potential electrical shock and misoperation, the DLC+ Programming Module must be properly grounded through the three-prong power plug. Do not use two-prong "cheater plugs". Also ensure that the electrical outlet in use has been properly grounded.

LIST OF CHANGED PAGES

Listed below are pages of this instruction book that have been changed, the type of change (page revised, page added, or page deleted), and the revision during which the change took place. If a page is rearranged due to changes on a previous page or if simple typographical errors are corrected, the changed page will not be listed below.

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CHAPTER 1
INTRODUCTION

General Electric's DLC + Sequencer is a microprocessor-based programmable controller that performs decision-making and arithmetic functions under software control. It is part of the Directo-Matic® Plus family of integrated industrial control equipment. Since the sequencer is user-programmable, you can easily modify its software to reflect changes in process or equipment. The sequencer is supported by a programming module, a cassette tape drive, and a printer.

The programming module allows you to inspect and modify sequencer software in the form of relay "strings." Relay strings are 5 by 7 matrixes displayed on the programming module in relay ladder diagram form; each string is a portion of the application software within the DLC + Sequencer. In *a* sense, each string is a "rung" of the relay ladder,diagram that represents the sequencer application software. Strings may be observed and modified while the sequencer is on line. String modifications can be tested on line before making them a permanent part of the sequencer memory. You can also use the programming module to run sequencer diagnostics.

The cassette tape drive is used to load application programs into the sequencer memory. It can also be used to dump the contents of sequencer memory to cassette tape for storage.

The printer is used to produce copies of application software in relay ladder diagram form along with a list of contact and coil cross references. The printer can also be used to produce an I/O map of the control system.

INTRODUCTION TO PROGRAMMABLE CONTROLLERS

As mentioned, the DLC+ Sequencer is a programmable controller. A programmable controller is a specialized computer used for sequence control of an industrial process. Programmable controllers are rugged, reliable, cost effective, and easy to program. A small programmable controller can perform the control and sequencing jobs that once required hundreds of relays.

Programmable controllers have thousands of potential applications. The DLC+ Sequencer can be used to control rolling mills, blast furnaces, process lines, paper mills, and similar industrial equipment. A complete, customized control system can be built using standard hardware and software building blocks.

Figure l-l is a simplified block diagram of a programmable controller and its supporting modules. The **memory** stores customized application software for the particular process being controlled. The **input interface** accepts inputs from process hardware such as limit switches, operator controls, and sensors and converts these inputs to logic signals for use by the processor. The **processor** scans these input signals and sends commands to the output boards under application software control. The **output interface** converts these commands to signals understood by devices such as motor starters, contactors, solenoids, and operator indicators. The **programming module** is used by an operator or programmer to monitor and modify the application program in memory or to troubleshoot a system problem. The **tape drive** is used to load application programs into memory or dump the contents of memory to cassette tape.

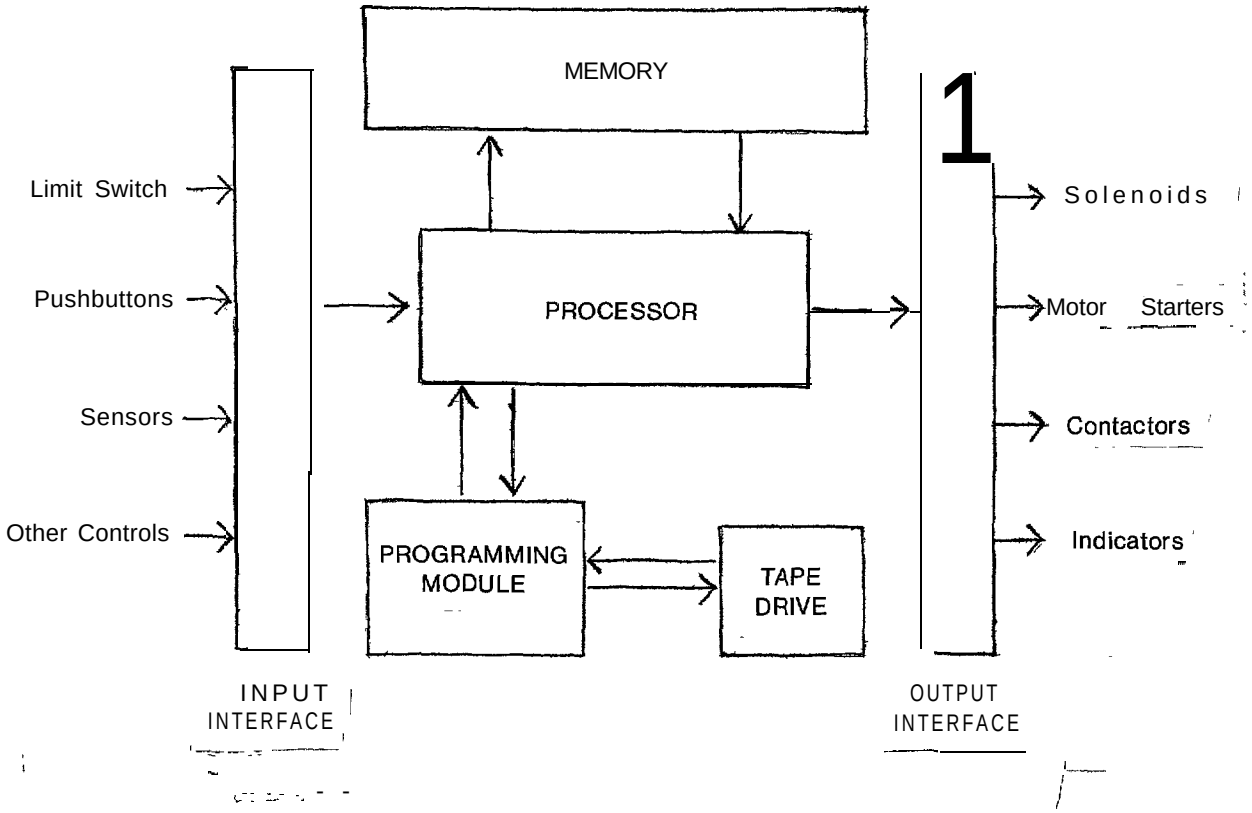


Figure l-l
Programmable Controller Simplified Block Diagram

The programmable controller constantly scans its inputs and updates its outputs. When the processor detects a change in an input, it uses application software instructions to determine what action to take. For instance, if a motor start button on an operator console (an input device) is pressed, the processor (under application software control) may check to see if all appropriate interlocks are closed. If they are all closed, the processor sends a signal to a motor control center (an output device) telling it to start the desired motor.

Since the process-controlling logic in a programmable controller is in application software, equipment additions or process changes are accomodated by simply changing the software, thus avoiding expensive rewiring. In the DLC + Sequencer, application software is easily modified by means of industry-standard relay ladder diagrams.

INTRODUCTION TO RELAY LADDER DIAGRAMS

A relay ladder diagram is a way to represent application software in visual form on a terminal screen or printer, using contacts, coils, and horizontal and vertical connections. A typical DLC + Sequencer relay ladder diagram as it appears on the DLC + Programming Module is shown in Figure l-2.

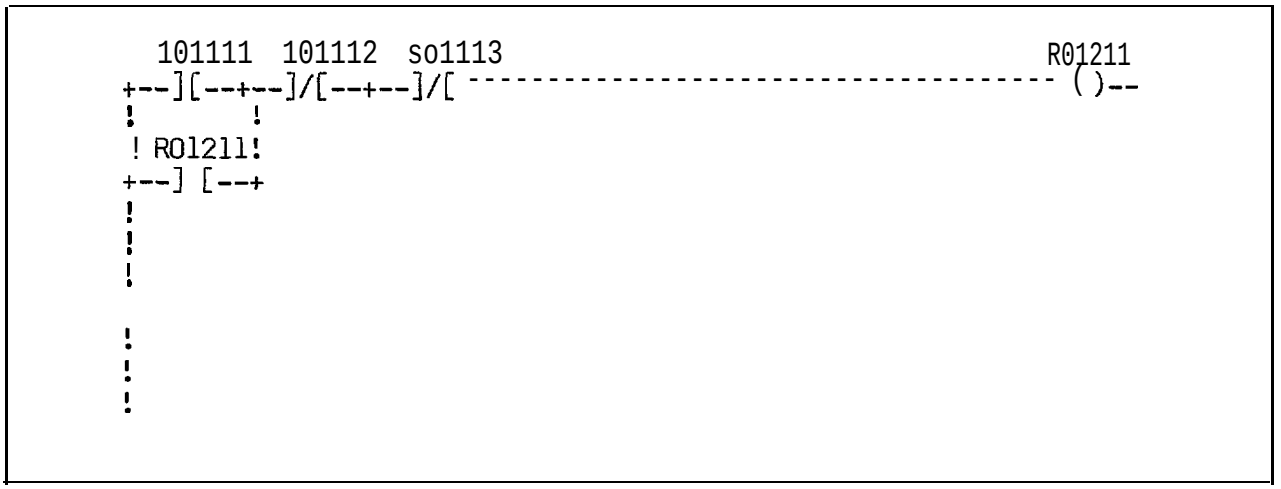


Figure 1-2.
Typical Relay Ladder Diagram

When a current path (that is, a complete connection) exists from the left bus (the leftmost part of the diagram) through any contacts to the coil, the coil is energized. The normally open and normally closed contacts are usually controlled by inputs to the programmable controller. For instance, contact 101111 (normally open) might be closed when a "start" button is pressed by an operator. Contact 101112 (normally closed) might be opened when a "stop" button is pressed. Contact R01211 (normally open) might be a seal contact. The coil is usually considered an output from the programmable controller. In this case, energizing coil R01211 might activate a motor starter and at the same time seal contact R01211. Of course, this is only a very simple example of the type of control routine that can be represented in relay ladder diagram format.

The contact or coil identification located just above each element on the relay ladder diagram is an input and output device address. This address ties the contact or coil (which exists in software) to a physical input or output device through an input or output interface.

Powerful DLC+ Sequencer software functions such as timers, counters, arithmetic operations, and time delays can be represented simply in relay ladder diagram format. Since such diagrams can be printed as well as viewed, they are essentially self-documenting. A hard-copy "picture" of the application software for an entire process can be produced and saved to aid in later troubleshooting.

DLC+ SEQUENCER AND PROGRAMMING MODULE OVERVIEW

General Electric's DLC+ Sequencer and DLC + Programming Module are part of a growing family of Directo-Matic ®Plus control equipment. Some of the characteristics of these DLC+ products are listed below.

- Both of these products are microprocessor-based, which means that product operation is controlled primarily by software or firmware rather than hardware.
- Automatic board-level diagnostics identify hardware faults on all microprocessor-based boards through a yellow IMOK light located on the front edge of the board. If the light goes out, a hardware fault has been detected.
- A wide range of I/O boards allows a variety of input and output devices to be connected to the sequencer, including digital and analog devices.
- The sequencer is capable of remote communications over time-shared wiring with I/O devices, other sequencers, and process control computers.
- The programming module is powerful, portable, and easy to use. Application programs are displayed in easy-to-understand relay ladder diagram form.

The remainder of this manual covers the operation and programming of the DLC+ Sequencer using the DLC + Programming Module.

CHAPTER 2
EQUIPMENT DESCRIPTION

This chapter provides a basic description of the DLC + Sequencer and DLC+ Programming Module hardware. More complete information can be found in the ***Directo-Matic® Plus DLC+ Sequencer Maintenance Manual (GEK-65477)***. **As** you read this chapter it may be necessary to refer to Appendix D, which contains a list and description of printed wiring boards used in the DLC + Sequencer.

SEQUENCER EQUIPMENT

The DLC + Sequencer is a highly structured programmable controller consisting of a number of hardware and software "building blocks." As additional functions or capabilities are needed, the sequencer is expanded by adding extra boards, modules, firmware (in PROM), and software (in RAM). Because of its modularity, the sequencer can handle a wide range of process control and sequencing needs.

The DLC + Sequencer hardware can theoretically handle from 32 to 2000 I/O points, depending on the size of memory and the number of I/O boards installed. An I/O point is equivalent to a single I/O signal such as a limit switch input or an output to a solenoid. The sequencer can also handle groups of I/O points configured as BCD or integer values.

Internally, the sequencer can store and manipulate from 600 to 3600 elements within a number of relay strings. An element consists of a contact or coil plus its associated signal address. The maximum number of elements available in a particular installation depends on the size of sequencer memory.

The sequencer processes I/O signals through a wide variety of I/O boards. These I/O boards can handle logic level (+ 5VDC) signals, high level AC and DC signals, and analog to digital and digital to analog conversion. I/O devices can be located up to 4000 feet from the sequencer module.

Typical Sequencer Configurations

Although the DLC + Sequencer can be used in a variety of process control applications, this manual will cover only two basic configurations: the DLC+ Sequencer as a stand-alone programmable controller and as part of a hierarchical process control system.

In its simplest configuration, the DLC+ Sequencer operates as a stand-alone programmable controller to coordinate local and remote I/O devices, drives, and operator desks. As shown in Figure 2-1, the sequencer can communicate with local I/O devices directly, and with remote I/O devices over a Control Party Line. In addition, the sequencer can communicate directly with Siltron™ power conversion modules.

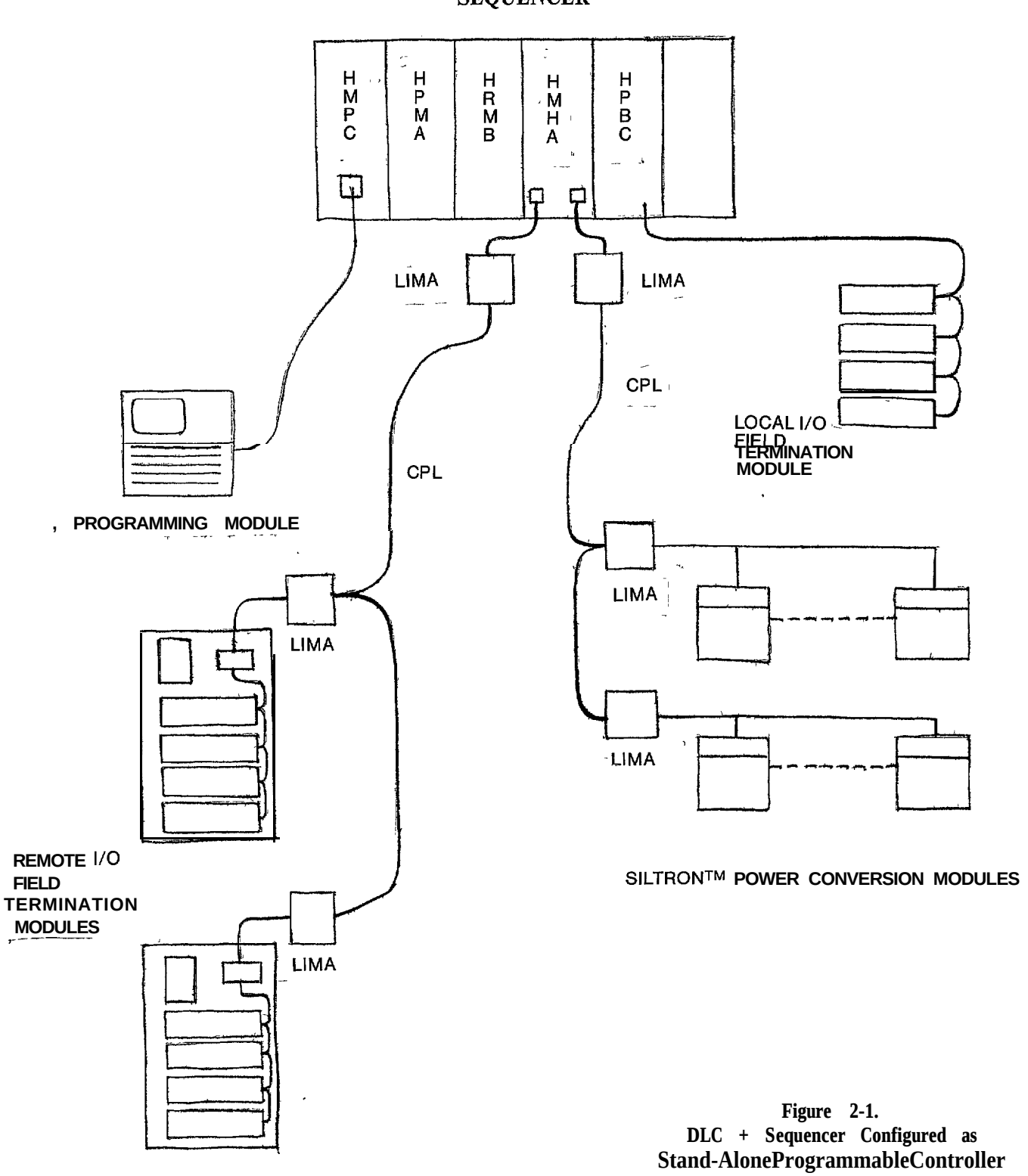


Figure 2-1.
DLC + Sequencer Configured as Stand-Alone Programmable Controller

The sequencer can also be tied into a hierarchical process control system as shown in Figure 2-2. In this configuration, the sequencer becomes a remote slave station under the master control of a process computer. Communications take place over a Control Signal Highway. The sequencer still coordinates and controls all I/O devices attached to it, but does so under the overall control of a process computer.

Sequencer Boards

A variety of printed wiring boards (each briefly described in Appendix D) make up the hardware building blocks used in the DLC+ Sequencer. The boards are plugged into an electronics module (see "Sequencer Packaging" below) and require a power supply for operation. The physical location of each type of board in the electronics module is requisition-dependent and is covered in the system elementary sent with the equipment.

A minimum sequencer requires the following boards:

- HMPC or HMPF Microprocessor Board - the "heart" of the DLC + Sequencer
- HPMA PROM Memory Board(s) - stores product software
- HRMB RAM Memory Board(s) - stores application software
- One or more I/O boards to interface I/O devices with the sequencer

The HMPC board contains an Intel 8086 microprocessor, which processes both application and product software during operation. The HMPC board contains an Intel 8086 microprocessor, which processes both application and product software during operation. The HMPF board can be used instead of the HMPC, the board is functionally equivalent to the HMPC. In most cases, the application program resides in battery-backed RAM memory for ease in programming changes during field installation and equipment modification. The product software (DLC + Sequencer operating system) is supplied in PROM memory.

The I/O boards for a very small system would probably be located in the same electronics module where the microprocessor and memory boards are. Additional hardware allows the I/O boards to be located outside the module and even thousands of feet away. See the section below titled "Sequencer I/O Structure" for more information. Appendix D contains brief descriptions of the digital and analog I/O boards most commonly used in DLC + Sequencer systems.

When required, the DLC+ Sequencer can communicate with a high-level process control computer over a Control Signal Highway. Additional boards are required to convert the sequencer into a CSH slave station.

Sequencer Bus Structure

The DLC+ Sequencer hardware is designed around four buses:

- **L-bus (Local bus)** - The main sequencer bus through which the microprocessor (HMPC) board communicates with boards located within a single electronics module. These boards might include memory (HPMA and HRMB) boards, Control Party Line master (HMHA) boards, and module I/O boards. The L-bus is located within the backplane of the electronics module.
- **E-bus (Express bus)** - A data bus used for communications between two or more DLC + Sequencer HMPC boards, or between an HMPC board and a Control Signal Highway (CSH) slave interface.
- **F-bus (Full bus)** - A sequencer bus containing all address, data, and control lines. Not commonly used in current sequencer implementations.

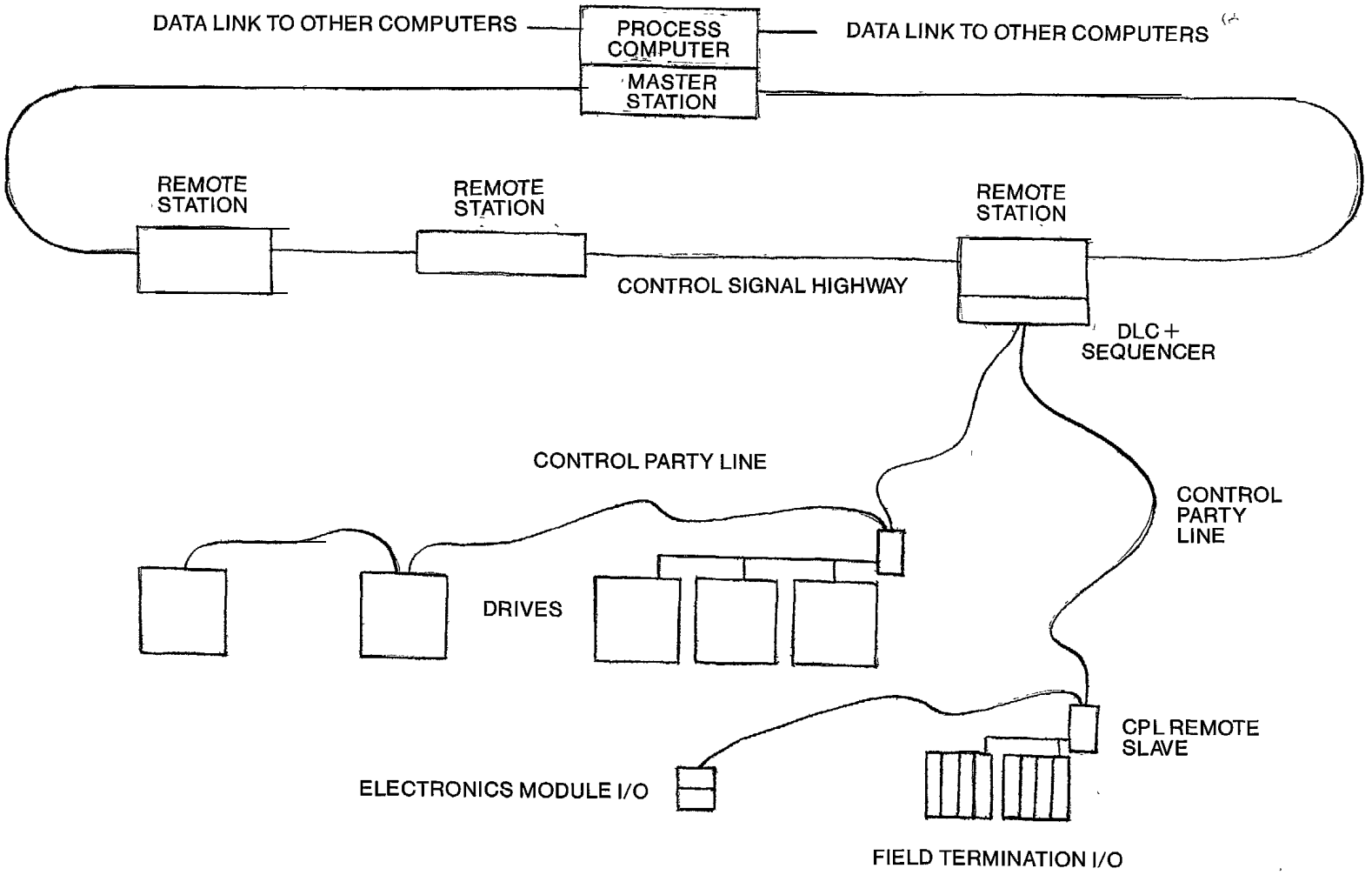


Figure 2-2.
DLC + Sequencer in Hierarchical Process Control System

Sequencer I/O Structure

The DLC + Sequencer handles three different types of I/O, classified according to the distance between the I/O board and the sequencer module:

1. Module I/O - I/O boards located in the same module as the sequencer
2. Local I/O - I/O boards located in a separate module within 50 feet of the sequencer module
3. Remote I/O - I/O boards located in a separate module more than 50 feet from the sequencer module

When module I/O is used, the I/O boards are inserted in the same card rack as the sequencer boards, as shown in Figure 2-4. The I/O boards are interrogated directly by the HMPC and no buffering is required. Addressing is handled automatically by backplane wiring.

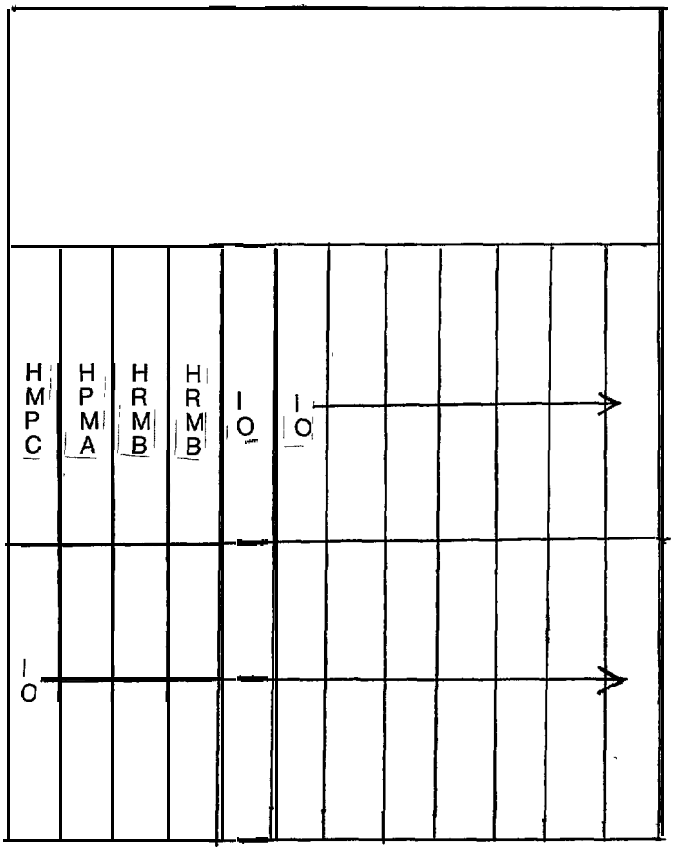


Figure 2-4.
Module I/O Configuration

When local I/O is used, the I/O boards are inserted into a separate card rack from the sequencer boards, as shown in Figure 2-5. This card rack can be an electronics module or a field termination module and can be located up to 50 feet from the sequencer module.

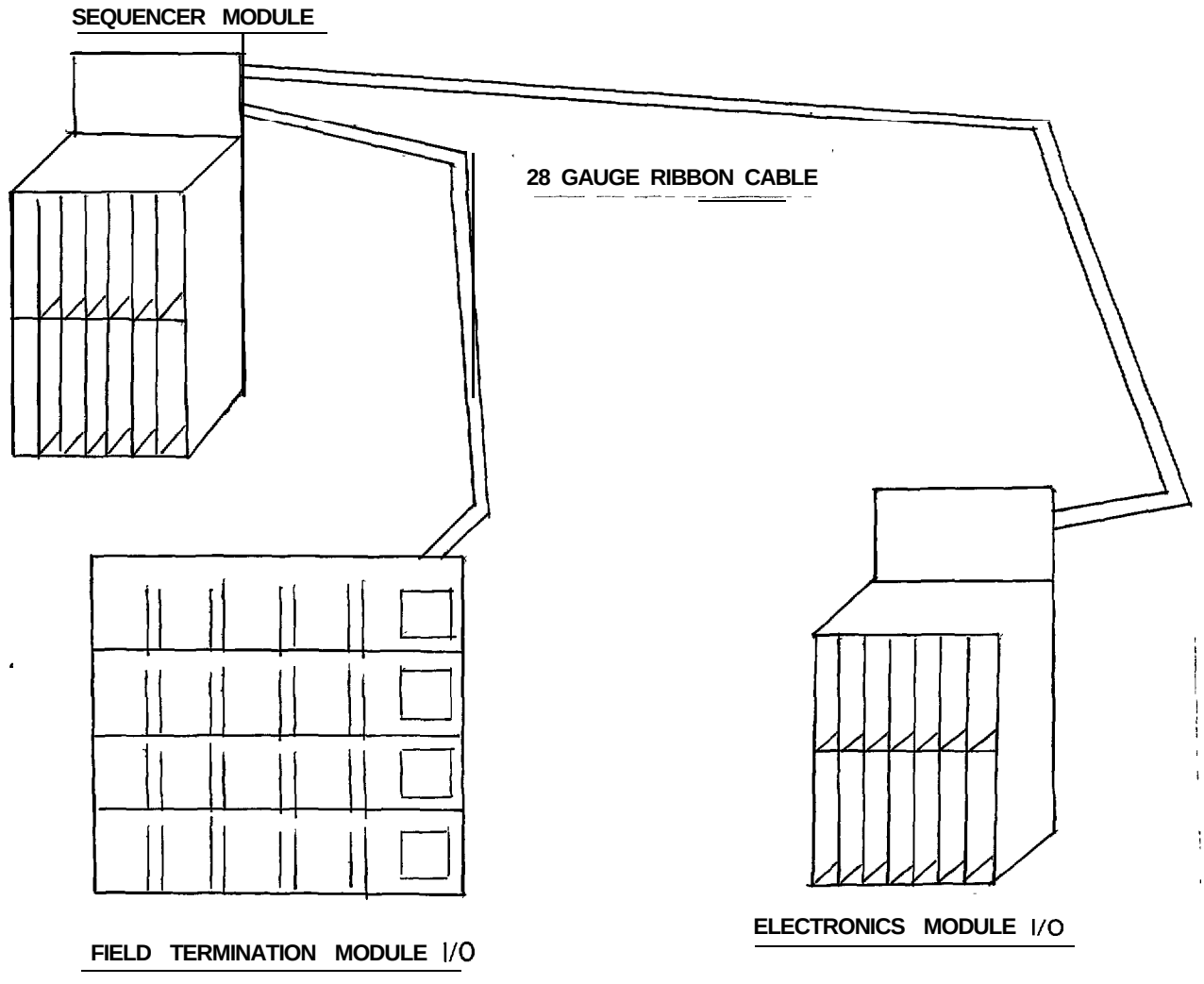


Figure 2-5.
Local I/O Configuration

Remote I/O communication is accomplished by incorporating one or more Control Party Line (CPL) communication links into the sequencer system as shown in Figure 2-6. A CPL is implemented by an HMHA master board in the sequencer module connected to one or more more I/O modules or Siltron™ power conversion modules by two twisted-pair shielded cables. Theoretically, up to 64 modules can be handled by one CPL link, although the practical limit is much less. The modules can be located up to 4000 feet from the sequencer module. If remote I/O stations are over 300 feet from the sequencer module, the CPL channels are optically coupled via LIMA modules to improve signal isolation and noise immunity. Up to four HMHA boards (and thus, four Control Party Lines) can be handled by one sequencer. Each HMHA allows full duplex communication over two CPL channels at a 48 Khz. rate on each channel. Data transmitted over the Control Party Line is automatically checked for validity and is rejected if an error occurs.

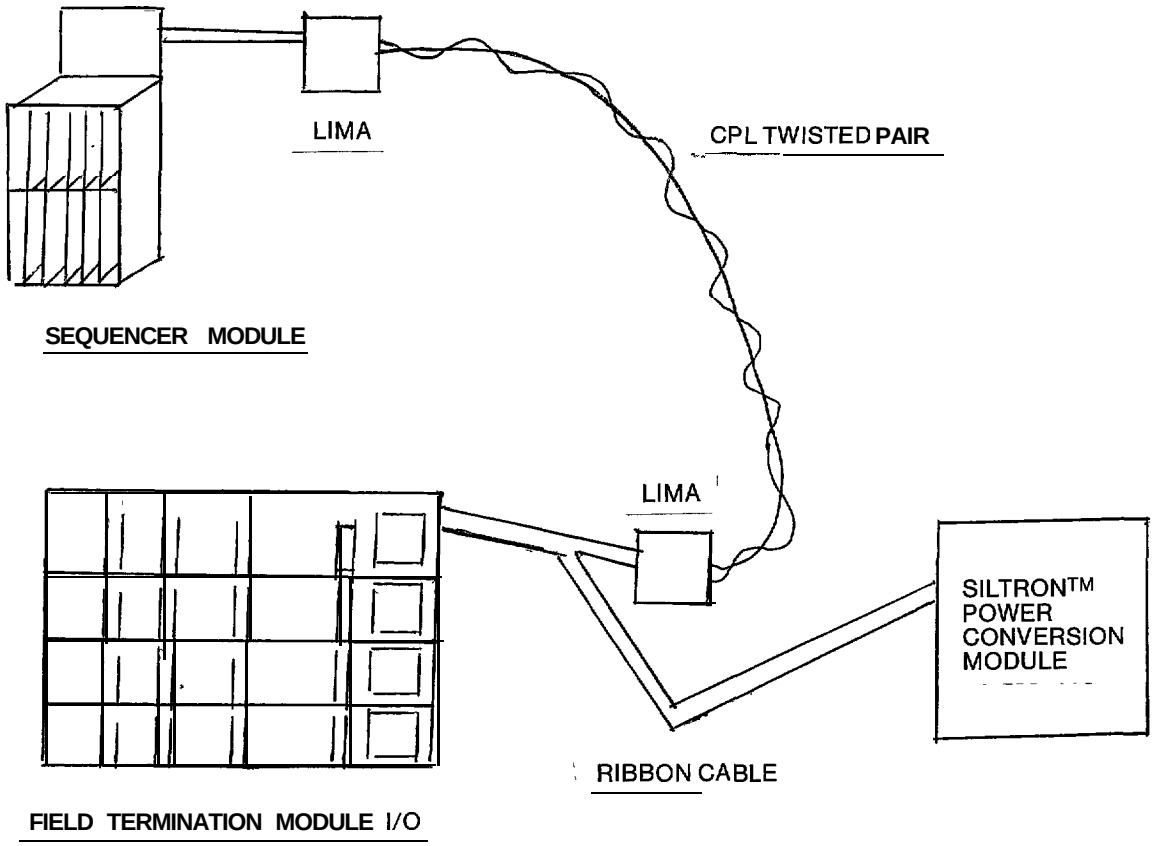


Figure 2-6.
Remote I/O Configuration

Sequencer Packaging

Two types of card racks are available for mounting DLC + Sequencer and I/O boards: the electronics module and the field termination module. The electronics module can contain either sequencer or I/O boards, while the field termination module contains only I/O boards.

A typical electronics module is shown in figure 2-7. There are three different sizes of electronics modules available:

- TEM (Tiny Electronics Module) - holds up to 6 boards
- SEM (Small Electronics Module) - holds up to 12 boards
- MEM (Medium Electronics Module) holds up to 24 boards

When an I/O board is installed in an electronics module, the wiring to the board from external devices is terminated with a 17 pin connector which plugs onto the front of the board. I/O boards in operator stations are typically installed in electronics modules rather than field termination modules due to the higher density of I/O signals in the operator station.

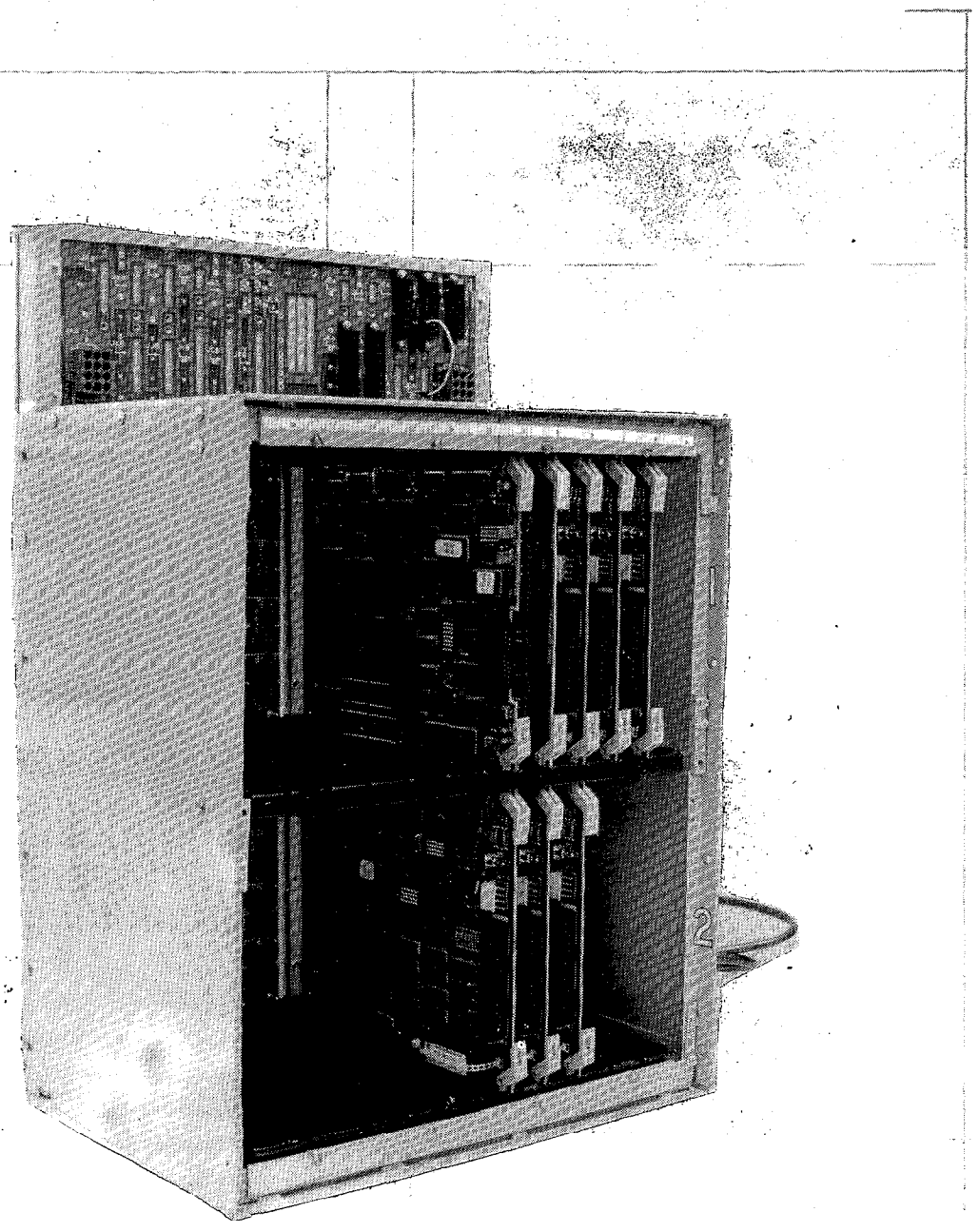


Figure 2-7
Typical Electronics Module

Figure 2-8 shows a typical configuration of field termination modules, which can be used for either local or remote I/O.. Each module holds 4 or 8 I/O boards and up to 4 modules can be ganged together to hold up to 30 I/O boards (2 slots are reserved for interface boards). This arrangement is ideal for field termination of signal wires, since wire troughs and terminal strips for terminating 12 gauge wire are also supplied. I/O boards may be inserted and removed without disturbing the field wiring and terminations.

The field termination module accepts digital I/O boards; analog I/O boards can be used with some restrictions. Digital or analog I/O boards can be used without restrictions in an electronics module.

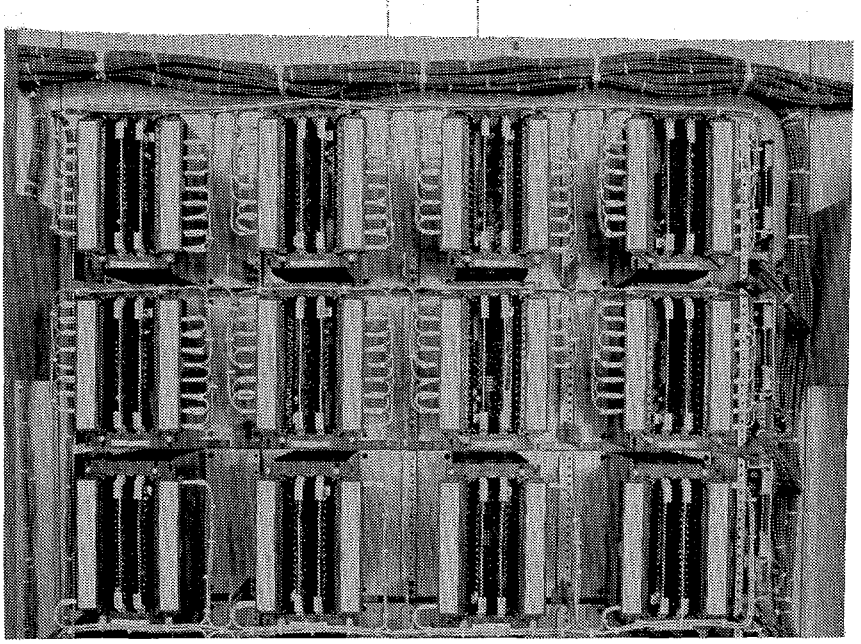


Figure 2-8.
Typical Field Termination Module

Power Supplies

General Electric provides most power supplies required by sequencer components. DC digital input power is provided from -105VDC unregulated power supplies in current ratings of 2 and 4 Amperes. AC digital output power (for use with solenoids and other such devices) is available from an AC source in the motor control centers. DC power required by board electronics is provided by a power supply that provides +5VDC at 40 Amperes, ±15VDC at 2 Amperes, and +28VDC at 5 Amperes. Other power supply ratings are available for special applications. All power supplies require a single-phase, 115 VAC, 50/60 Hz. source.

Sequencer Diagnostic Concepts

The DLC+ Sequencer was designed with system uptime as a prime consideration. The components used in the sequencer were selected with reliability in mind and the total number of system components was minimized to reduce the probability of failure. However, since any electronic system will occasionally fail, powerful diagnostic and maintenance aids were designed into the DLC + Sequencer to enable quick location and replacement of faulty hardware. Some of these diagnostic aids are listed below.

- The sequencer and programming module perform self-tests on power up and can report faults via the programming module screen.

- Each board containing a microprocessor (including the HMPC, HMHA, and HSHA boards) is equipped with a yellow IMOK light. As long as the board operates properly, the IMOK light remains on, but if a failure occurs, the IMOK light goes off. This makes troubleshooting many problems simple.

- All I/O boards contain a status indicating light for each input or output signal. An operator or maintenance man can tell at a glance whether a specific signal line is energized or deenergized.

- A DSHA daughter board containing maintenance aids is available for the HSHA CPL slave board. Using this daughter board, an operator or maintenance man can monitor or exercise I/O boards that are controlled by the HSHA slave board.

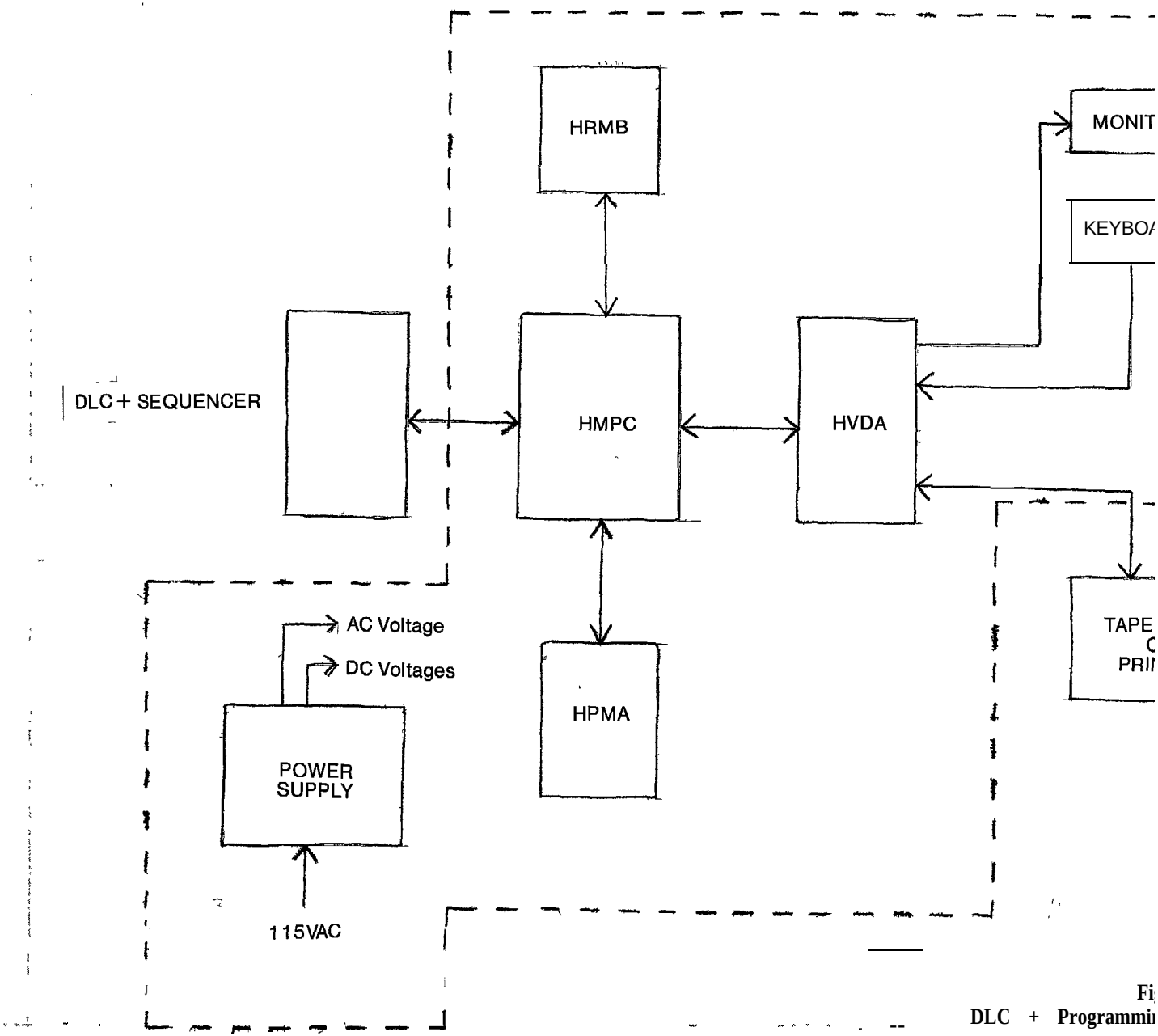
- The DLC+ Sequencer Programming Module allows on-line monitoring of inputs and outputs, the ability to force outputs, and the ability to run sequencer tests.

- A Highway Test Module is available for testing and troubleshooting the Control Signal Highway.

The ***Directo-Matic® Plus DLC+ Sequencer Maintenance Manual (GEK-65477)*** contains more information on these diagnostic and maintenance aids.

PROGRAMMING MODULE EQUIPMENT

As shown in Figure 2-9, the DLC+ Programming Module contains standard DLC+ Sequencer boards along with a power supply, video monitor, and keyboard. The HMPC microprocessor board coordinates the operation of the programming module under control of product software stored in PROMS on the HPMA PROM memory board. The HMPC also provides an interface for communication with the DLC + Sequencer. The HRMB RAM memory board stores strings being created and also acts as a communications buffer for the programming module. The HVDA video display board acts as an interface for the video monitor and keyboard. In addition, the HVDA interfaces a tape drive or printer and provides a video outputjack for an external video monitor. The power supply provides necessary AC and DC voltages for proper operation of the programming module.



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DLC + Programmi

CHAPTER 3
BASIC PROGRAM-MING MODULE OPERATION

This chapter covers the basic operation of the DLC+ Programming Module and the optional cassette tape drive and printer. We will cover initial set-up, keyboard layout and functions, screen formats, and simplified operation. Chapter 4 contains more detailed information on programming module operation, including information on the operating modes.

PROGRAMMING MODULE OVERVIEW

The programming module is a portable device used for programming and troubleshooting the DLC+ Sequencer. A cassette tape drive is used with the programming module for storing the contents of sequencer memory on tape and for loading strings or diagnostic programs into sequencer memory. A printer is also available for printing the contents of sequencer memory in relay ladder diagram form.

You can use the programming module to perform three specific tasks:

1. Create or modify strings in an application program. This is usually done when the process changes or equipment is added or deleted.

2. Monitor the process or troubleshoot a suspected process problem. This usually involves monitoring the operation of on-line strings in real time plus forcing contacts and observing the results.

3. Perform supervisory functions such as dumping memory to tape, loading memory from tape, printing strings in relay ladder form, clearing memory, and running system tests.

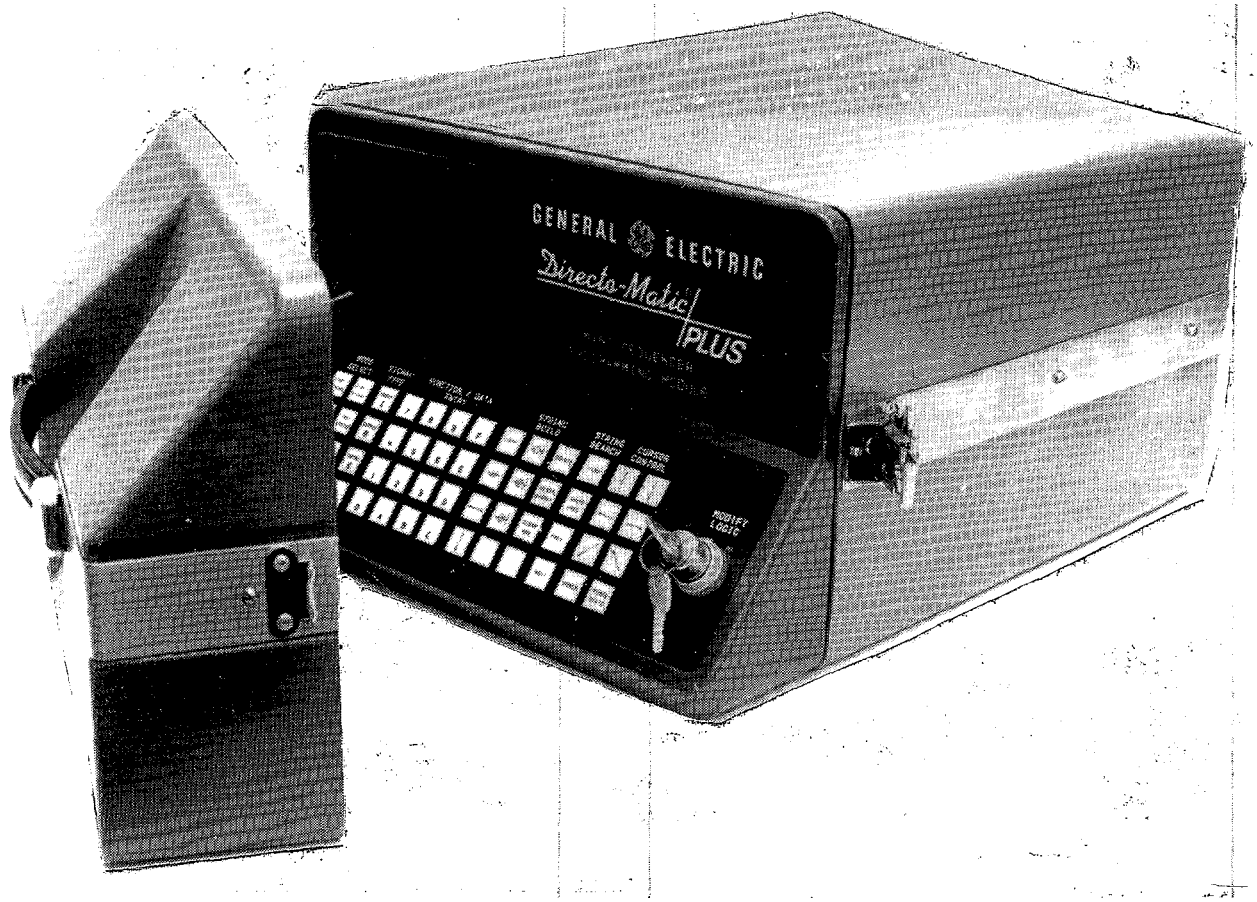
Creating a string involves several steps. First a skeleton string or control block must be created in an off-line memory area and displayed on the programming module screen. Then the string or block is modified to suit the process by changing or deleting contacts, assigning signal addresses, and sometimes inserting numerical values into control blocks. Next the string or block is tested on-line to ensure its correct operation. Finally, the string or block is permanently entered into sequencer memory. Optionally, a printed copy of the new string can be produced on a printer for future reference.

Since the programming module lets you look at and modify the states of signals, contacts, and coils, troubleshooting is easy. Both application program and hardware faults can be tracked down using the various display modes and signal forcing capabilities of the programming module.

The many supervisory functions available on the programming module support the main functions of string creation, modification, and troubleshooting. Supervisory functions include tape and printer operations, sequencer controls such as clear memory and resequence strings, and diagnostic operation.

As shown in Figure 3-1, the programming module consists of a high resolution CRT screen and a multi-function keyboard built into a rugged carrying case. The keyboard is used to select the operating mode of the programming module, to build various types of relay strings, to enter numeric data, to force the state of existing relay coils or contacts, and to control the display cursor. The CRT screen displays data according to the operating mode of the terminal: relay strings are displayed as 5 by 7 matrixes, signal information is shown in tabular form, and operating procedures are displayed in a menu format.

The programming module is designed for reliable, simple operation. The module operates on single phase, 115 VAC power. It can be connected or disconnected from the DLC+ Sequencer module while the sequencer is operating. An automatic self-test verifies proper operation of the module each time it is powered up and detects and reports any hardware errors. The cassette tape drive and printer plug into the back of the sequencer.



(DSF-53968)

Figure 3-1
DLC + Programming Module

INTRODUCTION TO THE OPERATING MODES

The DLC+ Programming Module operates in five different, key-selected modes which are listed below. The operating modes are covered in detail in Chapter 4. Each mode is used for a specific purpose: building strings, testing strings, observing string operation, performing auxiliary functions, and so on. The operating modes are:

Build String mode — Used for creating new relay strings and modifying old ones. Strings are built or modified in an off-line memory area and must be tested before being permanently entered into sequencer memory.

Test String mode — Used for testing newly-created or modified strings before entering them into sequencer memory. This mode is actually a part of Build String mode.

Display String mode — Used for displaying single strings as they are executing on-line. Strings cannot be modified in this mode, but signals can be forced or unforced. Forcing is explained later in this chapter.

Display Signal mode — Used for displaying up to eight input, output, or internal signal states in tabular form. Signals may be forced and unforced in this mode.

Supervisory mode — Used for auxiliary functions unrelated to building and displaying strings. Some examples of these functions are running diagnostics, clearing memory, and controlling printer and tape operations.

INITIAL SET-UP

Before using the programming module, it must be powered up and properly connected to the DLC+ Sequencer module. In addition, the tape drive or printer (if used) should be cabled to the programming module and prepared for operation. As mentioned, the programming module can be connected to the sequencer while the sequencer is on-line; it is not necessary to power down the sequencer. In addition, the programming module itself can be running while being connected.

Figure 3-2 is a block diagram of the proper way to connect the programming module, tape drive, printer, and sequencer module. The programming module, tape drive, and printer have their own power cords and require 115VAC from standard convenience outlets. The sequencer module obtains its power from a power supply through jack JAA.

The signal cables connecting the units are all RS-232-compatible with 25-pin connectors at each end. The cables must not be longer than 50 feet. As shown in Figure 3-3, all programming module signal connections are made to boards behind a swing-out panel at the back of the module. The cable running between the sequencer module and the programming module should be connected to the jack on the HMPC board in both modules. The cable running from the tape drive or printer to the programming module should be connected to the jack on the programming module HVDA board. **Note that only one of these devices (tape drive or printer) can be connected to the programming module at a time. However, the signal cables may be plugged and unplugged during operation to change from one device to another.**

After the programming module and peripheral devices have been properly cabled, turn on power to the programming module using the power switch on the back of the module. You should hear a blower inside the module come on and the module should perform a self-test (described in the next section of this manual). If the module will not power up, check to make sure the power cord is plugged into a good outlet and that the line fuse (located just beneath the power switch) is good. If the module still won't operate, refer to the DLC+ Sequencer Maintenace Manual for troubleshooting procedures. Once the programming module is operating, prepare the tape drive or printer for operation according to its instruction manual.

Note that throughout this manual, any specific instructions concerning set-up or operation of the tape drive will refer to the Electronic Processors Inc. STR ® LINK IIA or STR ® LINK III tape drive. Other tape drives that are compatible with the DLC+ Sequencer may have different operating procedures. The main differences between the STR @LINK IIA and the STR @LINK III tape drives are:

- Speed - the STR ® LINK IIA operates at speeds up to 1200 baud, while the STR ® LINK III operates up to 2400 baud.
- Capacity - the STR ® LINK IIA stores approximately 800 strings on a cassette tape, while the STR ® LINK III stores up to 1400 strings on a tape cartridge.

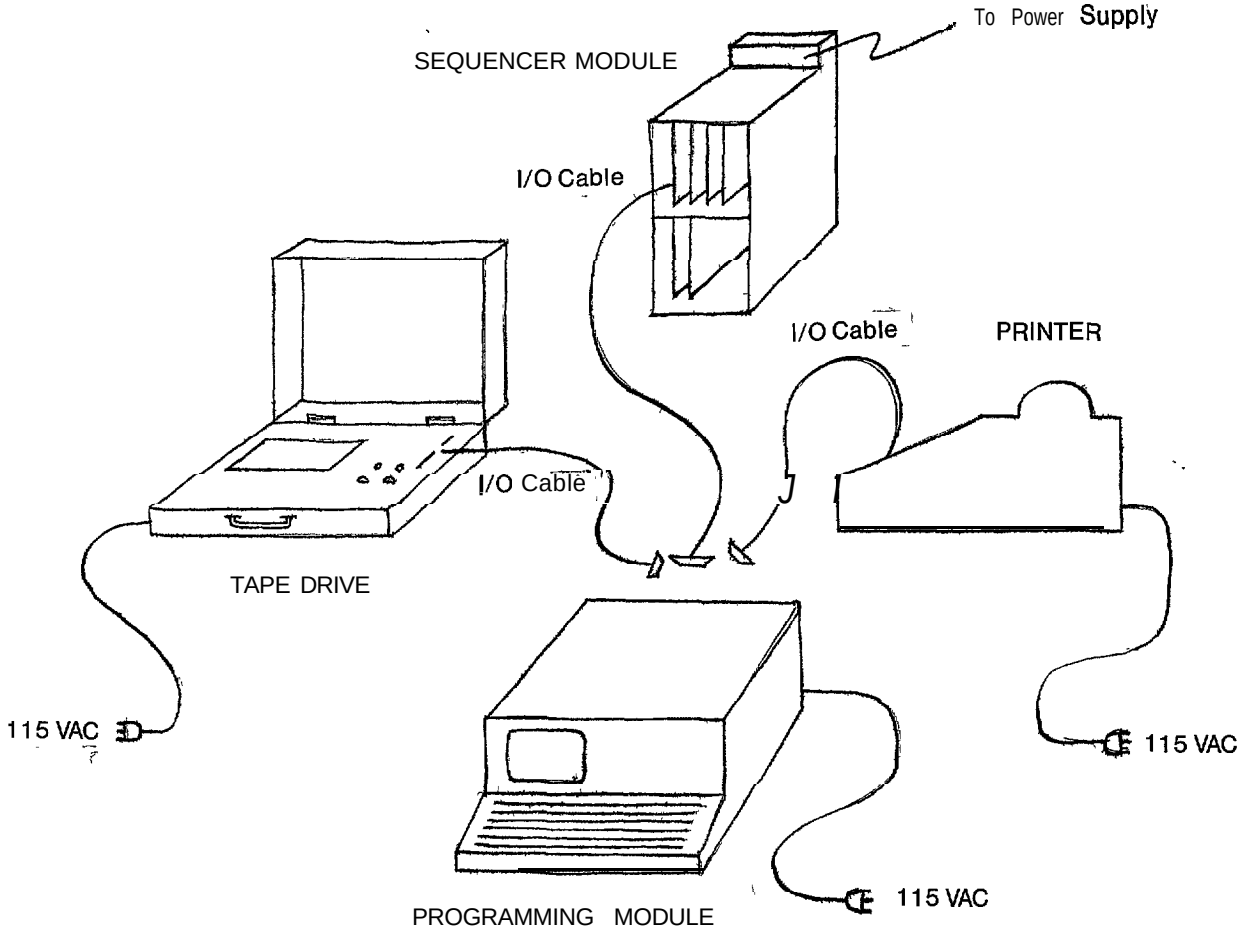


Figure 3-2
DLC+ Programming Module Cabling

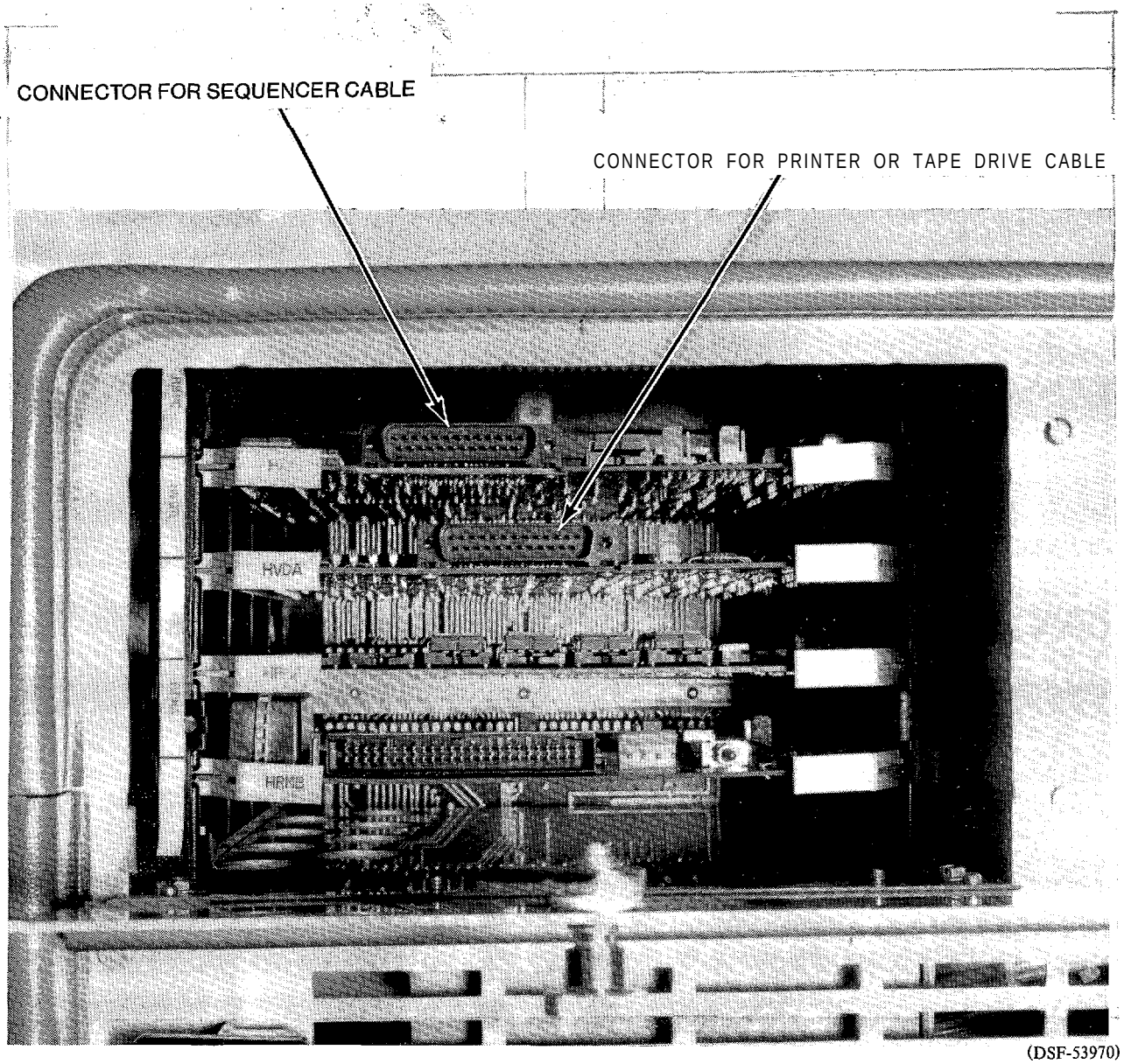


Figure 3-3.
DLC + Programming Module Signal Connections

PROGRAMMING MODULE SELF-TEST

The programming module executes a self-test when it is first powered up. During the self-test, the READY, COIL, and ELEMENT LEDs on the keyboard will flash and an array of numbers and letters will appear on the CRT screen, indicating that the self-test is executing. If no errors are detected during the self-test, the programming module displays the message "PLEASE PRESS ANY KEY ON THE KEYBOARD". Once a key is pressed, the programming module displays an initial screen image (described below) and waits for the operator to press a mode key. If the self-test does not execute correctly, refer to the programming module self-test information in the ***Directo-Matic® Plus DLC+ Sequencer Maintenance Manual (GEK-65477)*** for troubleshooting information.

The programming module self-test can detect errors during operation as well as during the power-up sequence. In this case, the error message **SELF TEST ERROR** NN will appear at the center of the screen, where NN is an error code indicating' the nature of the error. The self-test error codes are listed and explained in Appendix C.

The programming module self-test checks most of the functions of the programming module by performing internal memory (RAM) verification, internal firmware (PROM) verification, internal microprocessor verification, 'and internal I/O (card-to-card) verification

Note that this self-test does not check out the sequencer, only the programming module. The sequencer performs its own self-test as described in Appendix B. If a sequencer fault occurs while the programming module is connected to it, the programming module will display an error code at the bottom center of its screen **only if the operator presses the BUILD STRING key on the keyboard.**

SCREEN FORMAT

After the programming module self-test has successfully executed, the module automatically initializes and waits for the operator to press a mode select key. The module screen will appear as shown in Figure 3-4.

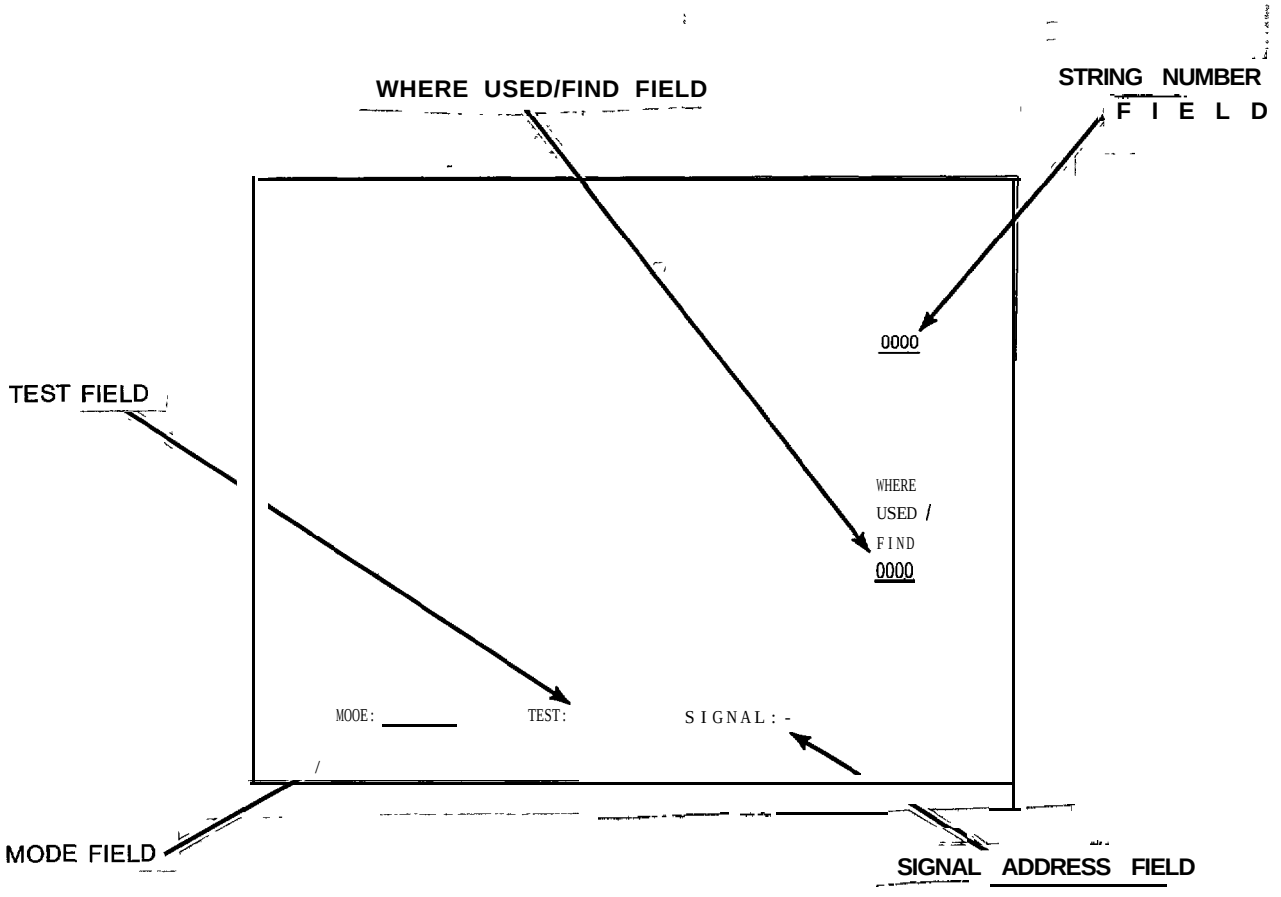


Figure 3-4.
Programming Module Initial Screen Format

This initial screen contains live inverse video fields that are used while programming or viewing strings in all modes except Supervisory mode. ***Note that in this manual, we indicate an inverse video field on a figure by underlining that field.*** Some of the fields allow the operator to enter data into them, while others are "read-only" fields used for displaying information.

The **Mode field** is a read-only field that displays the mode in which the programming module is currently operating. This field is visible only in Build String, Display String, and Test modes. This field is blank when the module is first initialized.

The **Test field** is a read-only field that displays the string number of a string currently under test. This field is visible only in Build String, Display String, and Test modes. When a string is placed into the test state, its string number is automatically displayed in this field. As long as the string is under test, its string number will appear in the Test field, even if the module is in another mode. Pressing the Build String key causes the string whose string number is in the Test field to be displayed. This field is blank when the module is first initialized or if there is no string under test.

The **Signal Address field** is a programmable field used only in Display Signal mode. The operator enters a signal address into this field when he wishes to display that signal in tabular form. This field is blank when the module is first initialized.

The **Where Used / Find field** is a programmable field used in conjunction with the Find and Where Used keys, which are explained later in this chapter. This field contains zeros when the module is first initialized.

The String Number field is a programmable field that displays the string number of the string currently displayed or allows the operator to assign a string number to a string he is creating. This field is visible only in the Build String, Display String, and Test modes. This field contains zeros when the module is first initialized.

The blank area of the screen normally contains the image of a string being built or displayed. This area is blank when the module is first initialized or if no strings are currently present in sequencer memory.

Figure 3-5 shows the programming module screen as it appears when displaying a string. Besides the five fields discussed above, there are other elements visible when a string is being displayed. Remember that these elements are actually software elements; the coils, contacts, and connections exist only in sequencer memory, not in actual hardware.

The **normally open contacts** interrupt the "current flow" from left to right until they are closed by the signal whose address appears directly above the contact.

The **normally closed contacts** allow current flow until opened by the signal whose address appears directly above the contact.

The **coil** provides an output through the signal whose address appears directly above the coil. This output depends on the state of the coil's associated string. If a current path exists from the left bus through contacts and connections to the coil, the coil is energized and the signal becomes active. Otherwise, the coil is deenergized and the signal is inactive.

The **signal address** located above each contact and coil ties that element to a physical input or output device or to another coil or contact. See the next section of this manual for more information on signal addresses.

The **vertical** and **horizontal shorts** act just like electrical shorts. They can be used to shunt current horizontally or vertically.

The **cursor** (not shown) is a small, flashing indicator on the screen that shows where data entered from the keyboard will appear.

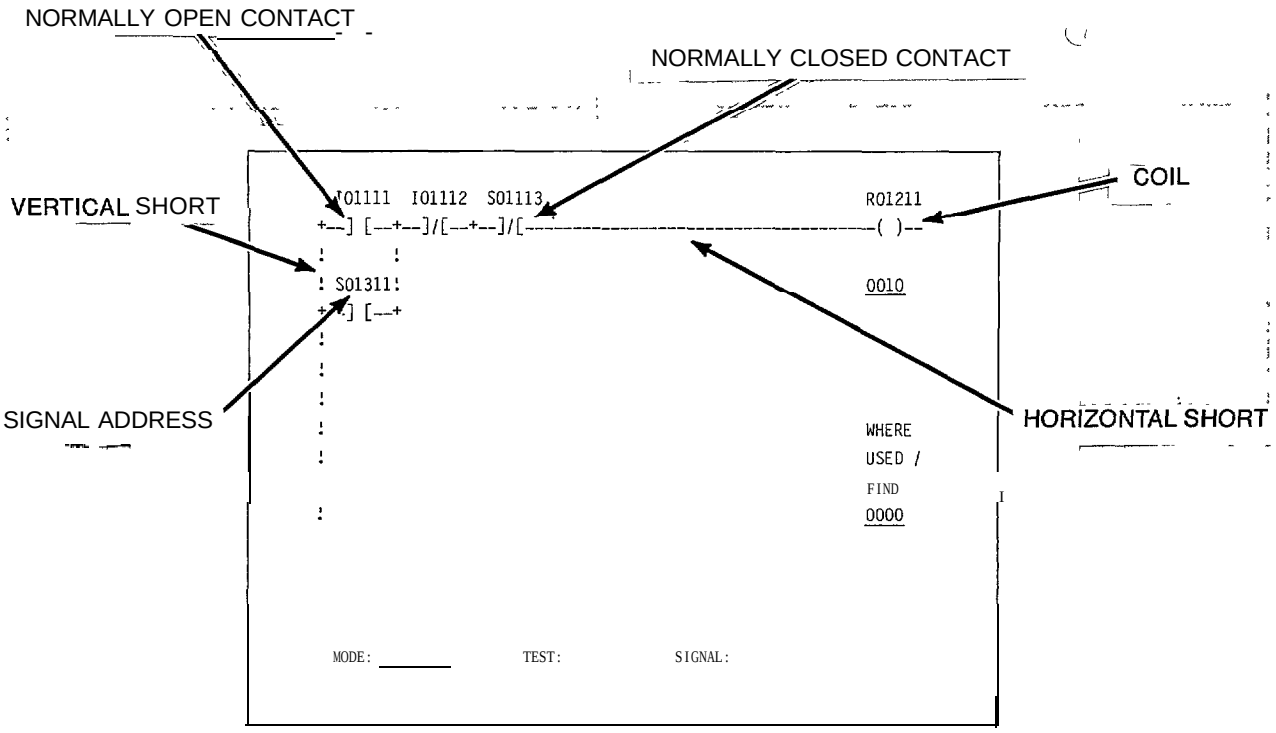


Figure 3-5.
Screen Format with String Displayed

SIGNAL IDENTIFICATION

The method of signal identification in the DLC + Sequencer provides a direct relationship between the signal address (shown above each coil and contact on the programming module screen or hard copy printout) and the physical input or output device. By looking at the signal address of a coil or contact, an operator or programmer can tell:

- **What kind** of signal the signal address represents: an input signal, an output signal, or an internal status signal;
- What **type** of signal the signal address represents: a point-type signal (used for input and output) or a value-type signal (used to represent BCD or integer (count) values).
- The physical **origin** or **destination** of the signal. The DLC + Sequencer signal identification scheme is patterned after the physical structure of I/O hardware found in typical industrial control systems:
    - Several I/O **signals** (from devices) are wired into a board or **group;**
    - **Several groups** are contained in a card **rack;**
    - Several **racks** are contained in a **station;**
    - Several **stations** may be coordinated by one **sequencer.**

Figure 3-6 shows the relationship between the sequencer signal addressing structure and the physical I/O structure. Refer to Chapter 2 for more information on the physical I/O structure.

All DLC + Sequencer signal addresses can be represented in general by the character string **KSSRGP,** where:

K = Kind of signal (I = Input, R = Output, S = Internal Status)
Input = signal originating in external hardware
Output = signal destined for external hardware
Internal Status = signal used internally by the sequencer

SS = Station address (hexadecimal 01 — FE)
A station is a logical major grouping of signals. Stations may be either physical stations or virtual stations. Physical stations consist of hardware (typically up to 16 card racks) mounted in one panel or distributed among several panels. Virtual stations exist only within sequencer memory and are used for transferring data between sequencers or from a sequencer to a process computer.

R = Rack address (hexadecimal 0 — F)
A rack is typically a card rack (electronics module or field termination module) containing several (up to 32) I/O boards. A rack can contain up to 16 groups of inputs and 16 groups of outputs, plus 16 groups of internal status signals.

G = Group address (hexadecimal 0 — F)
A group is typically an I/O board containing 8 input or output points. Some I/O boards can handle up to 16 inputs or outputs and must be addressed as two groups. A group may contain only internal status signals,

P = Point address (0 — 7) or type of signal (hexadecimal A — F). **P** values of 8 and 9 are not allowed.
The **P** specifier is used in one of two ways:
(1) If the value of **P** is from 0 to 7, it means that the signal address specifies a point-type signal. A point is usually a single input or output signal wired into an I/O board through a terminal board or connector. A point may also be a single bit internal status signal.
(2) If the value of **P** is from A to F, it means that the signal address specifies a value-type signal containing 16 points that should be interpreted as follows:
A=Reserved for future use
B=Interpret signal as 4 BCD (Binary Coded Decimal) digits. Takes 2 groups (16 points).
C=Interpret signal as an integer (count) value from -32,768 to +32,767. Takes 2 groups (16 points).
D =Reserved for future use.
E =Reserved for future use.
F=Reserved for future use.
The group address given in a value-type signal address is used as the lower order group address. For example, the value-type signal I0132B is a BCD signal that uses both group address 2, which is given, and 3, which is not given. In a value-type signal address, the given group address must always be even.

Below are some examples of valid signal address&:

- Signal address **IO4126** refers to an input **(I)** signal coming from station 04, rack **1,** group 2, and point 6.
- Signal address **R12431** refers to an output (**R**) signal going to station 12, rack 4, group 3, point **1.**
- Signal address **S2136B** refers to an internal status (**S**) signal with a station address of 21, a rack address of 3, and a group address of 6; the **B** indicates that this signal is a 16 point BCD value and that the group address includes both groups 6 and 7 (16 points = 2 groups).

Figure 3-6 summarizes the DLC + Sequencer signal addressing structure. The DLC + Sequencer can address up to 256 stations (this represents a range of addresses from hexadecimal 00 to FF) , but station addresses 00 and FF are reserved for special purposes. The remaining 254 stations (hexadecimal 01 to FE) can be either physical or virtual stations. Physical stations consist of hardware while virtual stations exist only within sequencer memory. A station can address up to 16 racks of I/O equipment, each rack can address up to 16 I/O boards, and each I/O board can address up to 8 points.

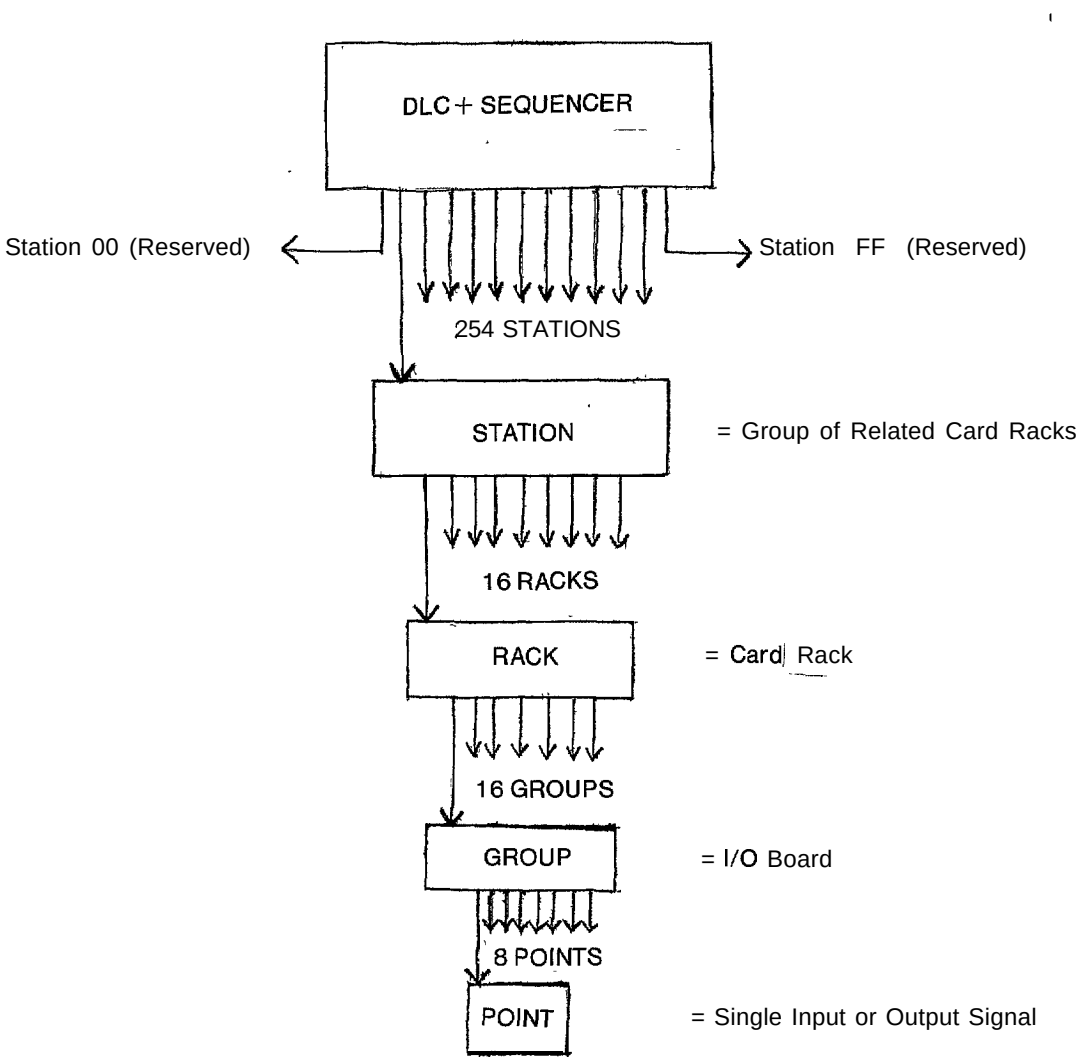


Figure 3-6.
DLC + Sequencer Addressing Structure

In this addressing structure, each input, output, or status signal is uniquely identified. For physical input and output signals, addressing of a specific input or output point is accomplished by hardware strapping on the I/O boards. The DLC+ Sequencer contains a configuration table, called the Key Information Block (KIB) that defines the specific stations to be addressed by a sequencer and the route or channel by which each station is accessed (through module I/O, local I/O, or over a Control Party Line). Virtual stations also have station addresses assigned within the KIB.

A channel (mentioned above) is a port through which a specified number of I/O signals is transferred to and from the DLC+ Sequencer. In effect, signal addresses are grouped into channels depending on where the signals originate or terminate. Specific channels are reserved for module I/O (boards located within the DLC+ Sequencer electronics module), local I/O (boards located within 50 feet of the DLC+ Sequencer electronics module), and remote I/O (boards located more than 50 feet from the DLC+ Sequencer electronics module and connected to the sequencer by a Control Party Line).

The current DLC + Sequencer implementation recognizes 16 channels:

Channel	Reserved For
0	Signals going to or coming from module I/O boards
1	Signals going to or coming from local I/O boards
2 - 9	Signals going to or coming from a master HMHA board (Control Party Line)
10 - 11	Signals transmitted between sequencers over the E-bus or signals sent to and from the Control Signal Highway network
12 - 15	Signals sent or received by a slave HMHA board

Channels are allocated through the KIB function. Assigning a particular signal address to a channel means that all I/O to or from that signal address will be expected to come through that channel.

Station, rack, group, and point addresses are set up to simpify the addressing, locating, and troubleshooting of physical devices. Using hard copy relay ladder diagrams and other forms of documentation, a programmer or maintenance technician can trace a signal from the application software to the actual physical device.

Although the theoretical address capacity of the DLC + Sequencer is enormous, the actual capacity of a single DLC+ Sequencer is limited by hardware constraints and depends on the needs of the control system. A single station may include sections of input, output, and internal status signals, each with the full address range provided by the signal address. For example, a physical input and output station dealing with process signals can theoretically incorporate 16 racks, each with 16 groups of 8 input points and 16 groups of 8 output points. This is a total station capacity of 2048 inputs and 2048 outputs which is substantially greater than most applications require. The large DLC + Sequencer address space was created for two reasons:

- To provide easy-to-understand system partitioning
- To provide the capability to uniquely identify all I/O devices in large, multi-sequencer control systems.

For example, the following might be a partial listing of the station assignments and used capacities within a rolling mill system which includes a number of DLC+ Sequencers and a process computer. Note that each station address relates to a particular location or function within the system.

Station Address	Location/Function/Capacity
1	Remote I/O line-up for fluid systems, containing two cabinets of field termination modules, 1 rack per cabinet, 16 groups of inputs and outputs in each rack, and 8 points per group.
4	Virtual station used for interchange of signals between the process computer and one of the DLC + Sequencers over a Control Signal Highway.
5	Remote I/O line-up for coil handling, containing a field termination module to handle limit switches and solenoids.
A	Local I/O for main operator station, mounted in the same cabinet as one of the DLC+ Sequencers in the system.
B	Remote I/O for floor-mounted operator station to control roll change.
C	Remote I/O for roll change limit switches (inputs) and solenoids (outputs).
10	Single auxiliary drive (Siltron™ power conversion module) assigned as 1 rack containing 1 group of input points and 2 groups of output points. Each single drive using an HIOB board must be assigned a unique station address in the range 01 to 7F.
11-16	Single auxiliary drives (Siltron™ power conversion modules) with HIOB boards, each addressed as a separate station.
1E	Local I/O mounted in the same cabinet as one of the DLC + Sequencers in the system.
2B	Main drive lineup, consisting of 4 Silcomatic® drives, 2 racks per drive containing various numbers of input and output signals, including digital and analog signals.

Purpose of Station FF

Station FF has special uses within the DLC + Sequencer and cannot be used for other purposes. The input (I), output (R), and internal status (S) signals within station FF have special meanings as outlined below. The signals can be set and interrogated by application software.

Input (I) Signals. The first two racks (256 points) of input to station FF record the station numbers of stations that have exceeded the consecutive I/O error limit since their channel was reset. The consecutive I/O error limit is set as a system parameter within the Supervisory mode KIB function and represents the maximum number of consecutive Control Party Line transmission errors detected by the HMHA board that can occur before the system is notified of an error. A station could exceed the consecutive I/O error limit due to a hardware failure or loss of power. (Note that a station will continue to execute I/O regardless of its error status, but inputs to the sequencer will not be used if an error occurred on the last sweep of the I/O points.)

Each point in each group represents one station. If the point is active (that is, if the bit is set), then the corresponding station has exceeded the consecutive I/O error limit. The two input racks in station FF contain signals IFFOOO through IFF1F6 and represent stations 0 through FE as shown below:

These Signals	Represent These Stations
Rack 0	
IFFOOO-IFF007	o-7
IFF010-IFF017	10 - 17
IFF020-IFF027	20-27
↓ ↓	↓ ↓
IFFOFO-IFFOF7	FO-F7
Rack 1	
IFF100-IFF107	S-F
IFF110-IFF117	18 - 1F
IFF120-IFF127	28-2F
↓ ↓	↓ ↓
IFF1F0-IFF1F7	FX-FE

Output (R) Signals. Signals RFFOOO and RFFOlO can be used to enable and disable sequencer output or' cause the sequencer outputs to send all zeros. Setting signal RFFOOO to a 1 (active) disables sequencer outputs, while setting RFFOOO to a 0 (inactive) enables sequencer outputs. This has the same effect as the Output Status function in Supervisory mode. Setting signal RFFOlO to 1 (active) will cause sequencer outputs to send all zeros, while setting RFFOlO to 0 (inactive) allows normal output.

Signals RFFlOO, RFF110 and RFF120 through RFF127 are used in conjunction with the special block available within the DLC + Sequencer (see Chapter 4 for more information on special blocks). Setting signal RFFlOO to an active state enables the use of special blocks (the sequencer comes up in this condition). Setting signal RFFlOO to an inactive state disables special blocks from executing. Signal RFF110 will become active when an error occurs in a special block. Signals RFF120 through RFF127 contain the function number (in hexadecimal) of the last special block that the sequencer tried to execute but which was undefined. The sequencer sets these signals automatically.

Internal Status (S) Signals. Eight groups of internal status signals are used as eight bit counters to specify how many times the sequencer software had to wait on an I/O channel before continuing operation (that is, the HMHA channel I/O was not complete when it was needed). The information in these counters is a breakdown of the I/O overflow count given in the KIB system performance display. The counter values are reset to zero when the sequencer is powered up.

The signal groups and channels they represent are listed below. Note that since channels 0 and 1 do not contain HMHA boards, the sequencer software will never have to wait for channel I/O on those channels; hence, channels 0 and 1 are not included in this list.

These Signals	Represent This Station
SFF020-SFF027	2
SFF030-SFF037	3
SFF040-SFF047	4
SFF050-SFF057	5
SFF060-SFF067	6
SFF070-SFF077	7
SFF080-SFF087	8
SFF090-SFF097	9

KEYBOARD LAYOUT AND FUNCTIONS

The programming module keyboard, shown in Figure 3-7, allows the operator to control the programming module through keys, indicators, and a keyswitch. The keys are arranged by function and fall into six categories:

- Mode Select Keys
- Signal Type Keys
- Function/Data Entry Keys
- String Build Keys
- String Search Keys
- Cursor Control Keys

This section briefly covers the use of the keys in each category, as well as the keyswitch and status lights. For more information on a specific key, refer to Chapter 4 and Appendix A. Chapter 4 contains information on key usage in each of the five operating modes, while Appendix A contains a summary of each key's function.

Note that each of the keys described below will automatically repeat their functions if held depressed. This feature is useful only with certain keys such as some of the String Build and Cursor Control keys.

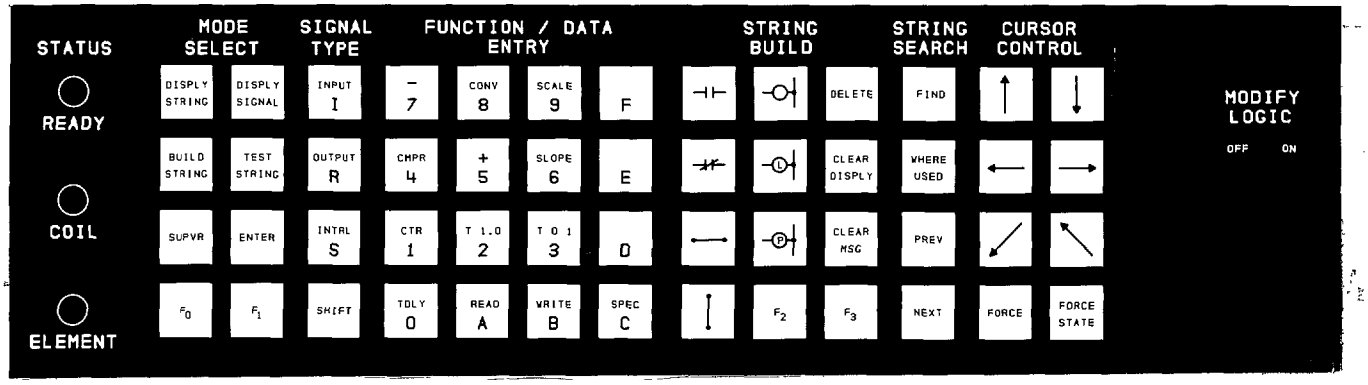


Figure 3-7
DLC + Programming Module Keyboard

Mode Select Keys

The Mode Select keys are used to select the operating mode of the programming module. Operating modes are discussed later in this manual. (The ENTER key is not associated with mode selection, but is used in some modes for data entry or menu selection.) When the programming module has initialized after power up, you must press one of the Mode Select keys before the module will respond. The Mode Select keys can be used to change modes at any time ***except when executing*** a ***Supervisory function.*** If the module is executing a Supervisory function (that is, if the main Supervisory menu is not visible) and you wish to change modes, you must first cancel the Supervisory function by pressing the DELETE key. It is sometimes necessary to press DELETE several times to exit some Supervisory functions.

DISPLY STRING — Puts programming module into Display String mode, which allows string logic to be displayed in relay ladder diagram form.

DISPLY SIGNAL — Puts programming module into Display Signal mode, which allows the states of up to eight signals to be displayed in tabular form.

BUILD STRING	Puts programming module into Build String mode, which allows creation of a new string or modification of an existing string.
TEST STRING	Puts programming module into Test String mode, which allows currently displayed string to be tested on-line without being entered into the sequencer memory. A string **must** be tested before it can be entered into memory. Test mode can only be entered from Build String mode.
SUPVR	Puts programming module into Supervisory mode and displays a menu of Supervisory functions. These functions are not directly involved with building or viewing strings.
ENTER	Not involved in mode selection, this key is used primarily to execute Supervisory menu selections, confirm data entry, and enter strings into sequencer memory.
F0	This key will force the VDM into a reset condition. The current operation will be aborted and the VDM self tests will be initiated. This key must be pressed twice in succession before the reset will be activated.
F1	Currently unused

Signal Type Keys

The Signal Type keys are used to specify the type of coil or contact signal being programmed (in Build String mode) or displayed (in Display Signal mode). These keys are used only when the cursor appears within an inverse video field. The SHIFT key is not involved in specifying signal types but is used in conjunction with the Function/Data Entry keys to create programmable control blocks.

INPUT (I)	Specifies an input signal originating in external hardware such as limit switches, thumbwheels, and tachometers. When this key is pressed, an I will appear at the cursor location.
O U T P U T (R)	Specifies a signal originating in the sequencer and destined for external hardware such as solenoids, indicators, or speed reference circuits. When this key is pressed, an R will appear at the cursor location.
INTRL (S)	Specifies a signal used internally by the sequencer (in other words, a signal both generated and used within the sequencer). When this key is pressed, an S will appear at the cursor location.
SHIFT	Not involved in specifying signal type. Used primarily in building control blocks and verifying critical operations such as clearing memory or entering strings into memory.

Function/Data Entry Keys

The Function/Data Entry keys are used to enter alphanumeric data, create programmable control blocks, or make menu selections. These keys are used in all operating modes. Most of these keys have two functions: when pressed by itself, each key generates an alphanumeric character (0 through F); when pressed after pressing the SHIFT key, most keys generate a programmable control block. Chapter 4 contains more information on control blocks. Programmable control blocks can be created only in Build String mode, only if the message "No Logic Assigned" is visible on the screen, and only if no other keys have been pressed after entering Build String mode.

O/TDLY	Inserts a numeric 0 at the cursor location when entered by itself. Builds a skeleton time delay string when entered after the SHIFT key.
1/CTR	Inserts a numeric 1 at the cursor location when entered by itself. Builds a skeleton counter block when entered after the SHIFT key.
2/T1.O	Inserts a numeric 2 at the cursor location when entered by itself. Builds a skeleton one second timer block when entered after the SHIFT key.
3/T0.1	Inserts a numeric 3 at the cursor location when entered by itself. Builds a skeleton one-tenth second timer block when entered after the SHIFT key.
4/CMPR	Inserts a numeric 4 at the cursor location when entered by itself. Builds a skeleton compare block when entered after the SHIFT key.
5/+	Inserts a numeric 5 at the cursor location when entered by itself. Builds a skeleton add block when entered after the SHIFT key.
6/SLOPE	Inserts a numeric 6 at the cursor location when entered by itself. Builds a skeleton slope/offset block when entered after the SHIFT key.
7/-	Inserts a numeric 7 at the cursor location when entered by itself. Builds a skeleton subtract block when entered after the SHIFT key.
8/CONV	Inserts a numeric 8 at the cursor location when entered by itself. Builds a skeleton convert block when entered after the SHIFT key.
9/SCALE	Inserts a numeric 9 at the cursor location when entered by itself. Builds a skeleton scale block when entered after the SHIFT key.
A/READ	Inserts an alphanumeric A at the cursor location when entered by itself. Builds a skeleton read block when entered after the SHIFT key, but this block is currently not usable.
B/WRITE	Inserts an alphanumeric B at the cursor location when entered by itself. Builds a skeleton write block when entered after the SHIFT key, but this block is currently not usable.
C/SPEC	Inserts an alphanumeric C at the cursor location when entered by itself. Builds a skeleton special block when entered after the SHIFT key.
D	Inserts an alphanumeric D at the cursor location when entered by itself. Causes an error when entered after the SHIFT key.

E	Inserts an alphanumeric E at the-cursor location when entered by itself. Causes an error when entered after the SHIFT key.
F	Inserts an alphanumeric 0 at the cursor location when entered by itself. Causes an error when entered after the SHIFT key.

String Build Keys

The String Build keys are used to create, modify, and delete skeleton strings and contacts and to clear error messages from the screen. The DELETE key is also used to display the previous menu in Supervisory mode. All keys except DELETE and CLEAR MSG are used only in Build String mode. The F2 and F3 keys are currently undefined.

N. 0. CONTACT	Inserts a normally open contact at the cursor location if the cursor is within a string and the module is operating in Build String mode. If the cursor is on a vertical rather than a horizontal connection, the cursor moves to the next horizontal position without inserting a contact. After inserting the contact, the signal address can be entered by pressing a Signal Type key followed by the rest of the address.
N. C. CONTACT	Inserts a normally closed contact at the cursor location if the cursor is within a string and the module is operating in Build String mode. If the cursor is on a vertical rather than a horizontal connection, the cursor moves to the next horizontal position without inserting a contact. After inserting the contact, the signal address can be entered by pressing a Signal Type key followed by the rest of the address.
HORIZONTAL SHORT	Inserts a horizontal short at the cursor location if the cursor is within a string and the module is operating in Build String mode. If the cursor is on a vertical rather than a horizontal connection, the cursor moves to the next horizontal position without inserting a horizontal short.
VERTICAL SHORT	Inserts a vertical short at the cursor location if the cursor is within a string and the module is operating in Build String mode. If the cursor is on a horizontal rather than a vertical connection, the cursor moves to the next vertical position without inserting a vertical short.
SKELETON STRING	Creates a skeleton string (a 5 by 7 matrix of normally open contacts, vertical and horizontal shorts, and an output coil). This key works only in Build String mode and only if the message "No Logic Assigned" is visible.
LATCH RELAY	Builds a skeleton string with a latch relay output rather than a standard coil output. This key works only in Build String mode and only if the message "No Logic Assigned" is visible.
PULSE RELAY	Builds a skeleton string with a pulse relay output rather than a standard coil output. This key works only in Build String mode and only if the message "No Logic Assigned" is visible.
DELETE	Deletes displayed data, depending on mode. In Build String mode, this key deletes contacts and shorts and can be used with the SHIFT key to delete strings from sequencer memory. Also used in Display Signal and Supervisory modes.

CLEAR DISPLY	In Build String mode, erases all elements from the screen beginning at the cursor to the bottom of the screen.
CLEAR MSG	Clears messages from the screen in all modes.
F2, F3	Currently unused.

String Search Keys

The String Search keys are used to find and display strings located in sequencer memory. These keys are used primarily in Display String and Display Signal modes.

FIND	In Display String mode, finds and displays the string whose string number has been entered in the Where Used/Find field. Entries in the Where Used/Find field can be made manually or by using the WHERE USED key.
WHERE USED	If the cursor is located on a **contact,** pressing this key causes the Where Used/Find field to display the string number of the string containing the related coil. If the cursor is located on a **coil,** the Where Used/Find field displays the string number of the string(s) containing related contacts.
PREV	Finds and displays the string whose string number occurs prior to the currently displayed string.
NEXT	Finds and displays the string whose string number occurs just after the currently displayed string.

Cursor Control Keys

The Cursor Control keys control the movement of the programming module's cursor. The FORCE and FORCE STATE keys do not affect the cursor, but are used to force and unforce contacts, coils, and signals.

UP	Moves the cursor one position directly **above** the current cursor location.
DOWN	Moves the cursor one position directly **below** the current cursor location.
LEFT	Moves the cursor one position to the **left** of the current cursor location.
RIGHT	Moves the cursor one position to the **right** of the current cursor location.
NEXT LINE	Moves the cursor to the leftmost position on the next line.
HOME	Moves the cursor alternately to the upper left contact or the Where Used/Find field.
FORCE	Places the contact, coil, or signal at the cursor location under force control. The FORCE STATE key is then used to control the contact, coil, or signal.
FORCE STATE	Alternately energizes and deenergizes the contact, coil, or signal at the cursor location if that contact, coil, or signal is under force control.

Keyswitch and Status Lights

MODIFY LOGIC	When this keyswitch is ON, strings in sequencer memory can be deleted or modified or new strings can be entered in sequencer memory. When the switch is OFF, sequencer memory cannot be modified.
READY	When this light is on, the programming terminal is powered on and ready for operation.
COIL, ELEMENT	These lights monitor the state (energized or de-energized) of coils and elements while in the Display String mode. They are also used on powerup/reset to indicate that a diagnostic test completed or failed.

LEARNING TO USE THE PROGRAMMING MODULE

This section of the manual provides a simplified, step-by-step procedure for learning to operate the DLC+ Programming Module. The procedure is especially useful if you have never operated the programming module before. The procedure guides you through the most common programming module functions: building a sample string and entering the string into memory, displaying strings and signals, using the String Search keys, forcing and unforcing signals, and performing printer and tape operations. Try performing each step on a programming module to help you learn its proper operation.

We have included a section below titled "Initial KIB Set-up." Since you probably won't have to configure a Key Information Block when you begin using the DLC+ Programming Module, you may skip this section on first reading.

Before starting these procedures, you should have connected the programming module to the sequencer and turned on the module's power switch. The module self-test should have successfully executed and the screen should appear as shown in Figure 3-4. We assume that the sequencer you are programming does not have application software loaded into memory.

In the procedures below, all programming module key names are in upper case letters. In order to avoid needless repetition, if a procedure tells you to press ENTER, it means to press the ENTER key.

Initial KIB Set-up

Before you can use the programming module to build and enter strings, you must set up the Key Information Block (KIB) within the sequencer as described below. The KIB determines the I/O configuration of the sequencer you are programming, including the station, rack, and group addresses recognized by the sequencer. Improper KIB set-up can result in errors during string building and testing.

1. Once the self-test has executed, press SUPVR to display the Supervisory menu.

2. The sequencer must be off-line before modifying the KIB. Press 9 and then ENTER to select the On/Off Line function. If "On Line" appears on the screen, press SHIFT and then ENTER so that "Off Line" now appears, otherwise press DELETE to return to the Supervisory menu.

3. We want to perform the KIB function on the Supervisory menu. Press 7 and then ENTER to display the KIB menu.

4. We now want to define the stations, racks, and groups recognized by the sequencer. Press 1 and then ENTER to display the Rack Definition menu. The message "No Entries in Rack Table" should appear at the bottom of the screen.

5. Since we want to add to the rack table, press 4 and then ENTER. The cursor positions itself at an inverse-video field next to the word "Channel". You can now begin entering KIB information as shown below using the Function/Data Entry keys. After you have made an entry in an inverse-video field, the cursor will automatically jump to the next field. If you need to go back to a prior field to make a correction, simply use the Right and Left Cursor Control keys to position the cursor to the desired field.

   A. Enter the two digit **channel** address (00 through 16).
   B. Enter the two digit **station** address (01 through FE).
   C. Enter the single digit **rack** address (0 through F).
   D. Enter the number of **groups** used (01 through 16).
   E. Enter the maximum number of groups **(max groups)** used (00 through 16). This number **must** be greater than or equal to the number of **groups used.**
   F. Enter the single digit **type.** Enter a 2 if this rack is a Control Signal Highway interface that is both read and written. Enter a 1 if this rack is a Siltron™ HIOB board. Enter a 0 if this rack is any other type of I/O hardware.
   G. Press SHIFT and then ENTER to enter the values into the KIB. You have now completed entries for one rack. If more than one rack must be defined, press 4 and then ENTER and repeat this procedure from step A for each rack. The number next to the word "Rack" on the screen will increment each time you complete a rack entry.

The values you enter into the rack table depend on the I/O configuration of the control system you are working on. For purposes of this tutorial, enter the following values into the rack table. Note that assigning a rack to channel 01 assumes the presence of **an** HMHA board in the sequencer.

Entry 1 - Channel **01** Station **01** Rack 1 Groups Used 4 Max Groups 6 Type 0
Entry 2 - Channel **01** Station 01 Rack 2 Groups Used 4 Max Groups 6 Type 0
Entry 3 - Channel **01** Station 01 Rack 3 Groups Used 4 Max Groups 6 Type 0

In the following examples, you will use the rack defined in entry 1 for input signals and the rack defined in entry 2 for output signals. In an actual installation, the rack defined in entry 1 and entry 2 would probably contain both input and output I/O boards, but we are simplifying things for this tutorial. You will use the rack defined in entry 3 for internal status signals, although such signals could "share" the racks defined in entries 1 and 2.

Note: Be very careful when entering the Max Groups value. Once entered, the value cannot be changed without deleting the rack entry, which involves deleting all strings that reference the rack entry.

6. When all rack entries are complete, press DELETE twice to display the Supervisory menu.

7. If you wish to put the sequencer back on-line before starting to build strings, press 9 and then ENTER to select the On/Off Line function. "Off Line" should appear on the screen. Press SHIFT and ENTER so that "On Line" now appears. You are now ready to build strings.

Building a String

This procedure tells you how to create a simple string and insert appropriate signal addresses. The signal addresses will be based on the KIB entries you made. All input signals will be from station 01 rack 1, all output signals will be to station 01 rack 2; and all internal status signals will use station 01 rack 3. There are four available groups (32 points) in each rack.

1. Press BUILD STRING to put the programming module in Build String mode. Assuming that the string number in the String Number field is other than that of an existing string, the screen will appear as shown in Figure 3-4 with the message "No Logic Assigned" at the bottom of the screen.

2. Press SKELETON STRING to create a 5 by 7 matrix of contacts and shorts. You could have created a latch or pulse relay with the appropriate String Build keys or a control block using SHIFT plus a Function/Data Entry key, but we assume you have created simply a skeleton string. See Chapter 4 for more information about these other kinds of strings.

3. Use the Cursor Control keys to position the cursor to the String Number field in the upper right portion of the screen. Enter the desired string number for the string you are about to build (for example, 00 10).

4. Assume you want to program a normally open contact at the first contact location and that you want that contact to be controlled by an operator switch tied into the sequencer at point 1, group 1, rack 1, station 01. Position the cursor to that contact location and press INPUT to specify an input signal address. The cursor will jump to a point directly above the contact and an I will appear. Enter 01111 to complete the desired signal address (station 01, rack 1, group 1, point 1), then press the NORMALLY OPEN CONTACT key, which moves the cursor to the next contact location.

5. Assume you want to program a normally closed contact at the next contact location and that you want that contact to be controlled by an interlock switch tied into the sequencer at point 2, group 1, rack 1, station 01. Press INPUT to specify an input signal address. The cursor will jump to a point directly above the contact and an I will appear. Enter 01112 to complete the desired signal address (station 01, rack 1, group 1, point 2), then press the NORMALLY CLOSED CONTACT key, which moves the cursor to the next contact location.

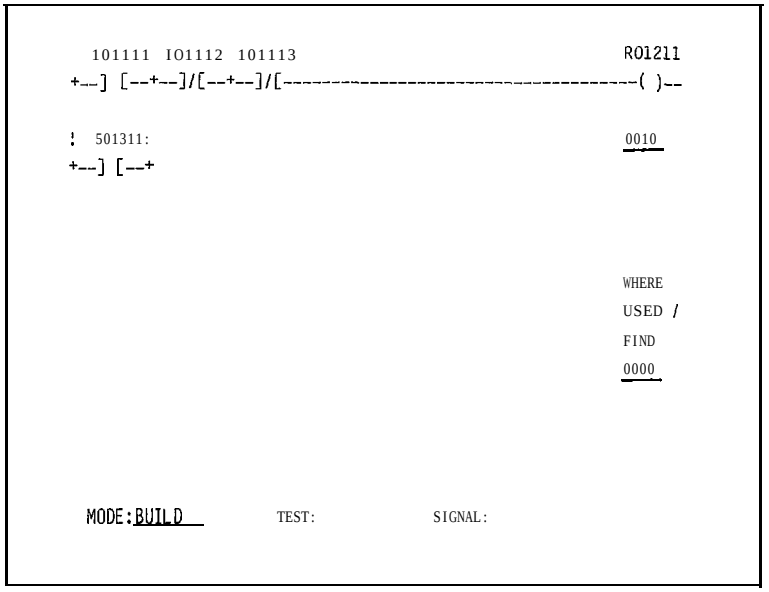
6. Repeat step 5 to create a normally closed contact in the next position with a signal address of 101113.

7. Press the HORIZONTAL SHORT key several times to create a horizontal short between contact IO1113 and the coil.

8. Assume you want the output coil to control a motor starter through point 1, group 1, rack 2, station 01. Position the cursor to the coil and press OUTPUT to specify an output signal address; at this point an R will appear. Enter 01211 to complete the desired signal address (station 01, rack 2, group 1, point 1).

9. Assume you want to create a normally open contact in parallel with contact 101111, perhaps to act as a seal for that contact. Also assume that the new contact should be controlled by the output coil of another string through internal status signal SO131 1 (that is, when the other coil is energized, the new contact will close). Position the cursor to first contact location on the next line and press INTRL to specify an internal status signal. The cursor will jump to a point directly above the contact and an S will appear. Enter 01311 to complete the desired signal address (station 01, rack 1, group 3, point 1).

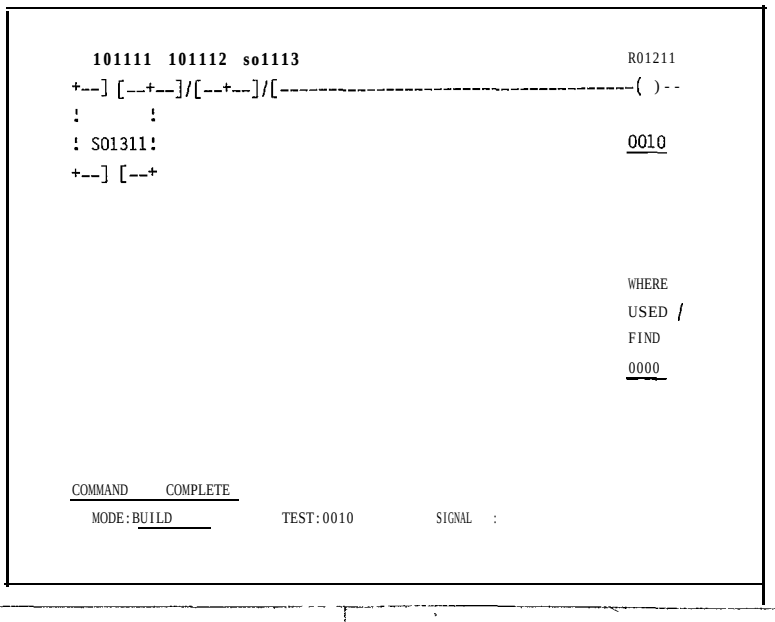
10. Since you are finished programming this string, you may want to clear the unused contacts from the screen. Position the cursor to the next contact location and press CLEAR DISPLY to clear the rest of the screen. The programming module screen should now appear as shown below:



Testing a String

Once a string has been built, it must be tested before being entered into sequencer memory. When a string is tested, the sequencer first checks to make sure all signal addresses are valid (that is, that the station, rack, and group addresses used in the signal address are present in the KIB rack entry tables). Then the string is temporarily placed on-line and allowed to execute along with the application software.

1. At this point, the programming module should be in Build String mode and a completed string visible on the screen. Make sure that the MODIFY LOGIC keyswitch is in the ON position so that the string can be placed on-line. Press TEST STRING to cause the string to go into the test state. The string number of the string under test now appears in the Test field at the bottom center of the screen and the message "Command Complete" appears as shown below:



2. While in Test mode, the string cannot be modified. You can, however, remove the string from Test mode by pressing TEST STRING again, make modifications, then put the string back in Test mode. Only one string at a time can be tested using Test mode.

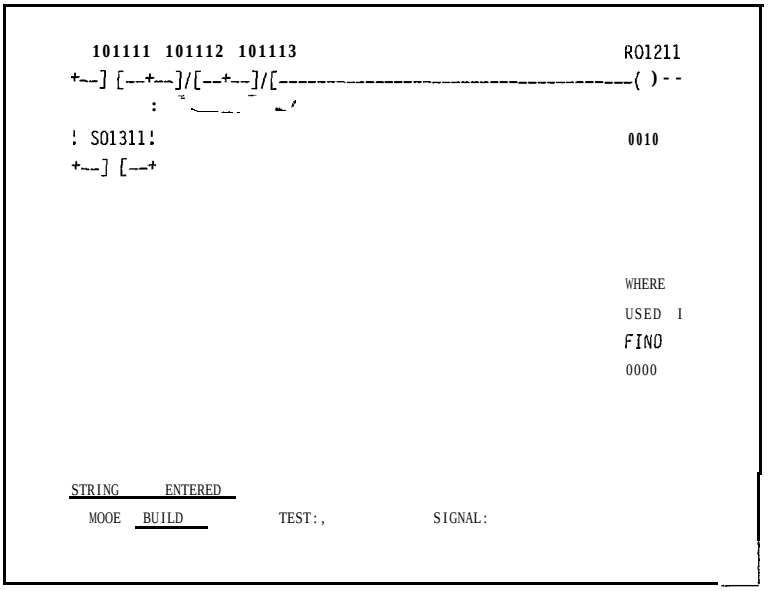
3. You can operate the programming module in other modes while the string is being tested. For instance, you could select Display String mode to display the string being tested or select Supervisory mode to perform other functions. The Test field will continue to display the string number of the string under test while the programming module is operating in Build String and Display String modes.

Entering a String Into Sequencer Memory

Strings can be permanently entered into sequencer memory after they have been tested. If the string number of a new string matches the string number of a string already in memory, the new string takes the place of the old string.

1. Make sure that the MODIFY LOGIC keyswitch is in the ON position so that the string can be entered into memory. The programming module must be *in* Build String mode with a string under test.

2. Press SHIFT and then ENTER to enter the string into sequencer memory. The message "String Entered" appears at the bottom of the screen and the string number in the Test field disappears as shown below:



Deleting a String from Sequencer Memory

Strings can be deleted from sequencer memory at any time, although this is not commonly done with the sequencer on-line. Before deleting a string be sure that you have a copy of the string on tape and that the process will not be adversely affected.

1. Make sure that the MODIFY LOGIC keyswitch is in the ON position so that the string can be deleted from memory. The programming module must be in Build String mode.

2. Enter the string number of the string to be deleted in the String Number field and press BUILD STRING. The string will appear on the programming module screen.

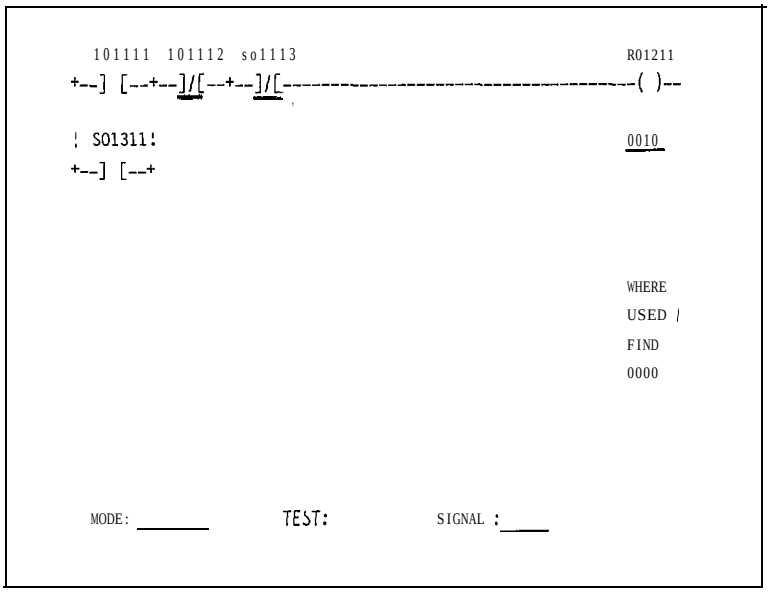
3. Press SHIFT and then DELETE to delete the string from sequencer memory. The message "Command Complete" appears at the bottom of the screen to indicate that the string has been deleted.

Displaying Strings

Display String mode allows you to view (but not modify) any on-line string in real time. This means that the status (energized or deenergized) of coils and contacts on the screen is continually updated. Both strings in memory and strings under test can be displayed in Display String mode. In addition, the WHERE USED and FIND keys enable you to find and display strings where known coils or contacts are used.

1. Press DISPLY STRING. The screen will display either a string or the message "NO LOGIC ASSIGNED", meaning that no string exists with the string number appearing in the String Number field. At this point you can press NEXT to display the next string in memory or PREV to display the previous string in memory. You can also position the cursor to the Where Used/Find field in the lower right of the screen and enter the string number of the string you wish to display. For purposes of this example, enter 0010.

2. Press FIND and the desired string will be displayed on the programming module screen as shown below:



Note the following screen characteristics of the displayed string:

A. All closed contacts and energized coils are displayed in inverse video (shown underlined in the preceding figure). Thus, normally closed contacts are displayed in inverse video until they are energized.
B. All open contacts and deenergized coils are displayed in normal video. Thus, normally open contacts are displayed in normal video until they are energized, at which time they appear in inverse video.
C. When the cursor is positioned on a coil or contact, an inverse video field surrounds that coil or contact.
D. If a contact or coil is under force control, its signal address appears in inverse video. See the section on forcing below for more information.

E, The screen image of coils and contacts may not show every state change. The COIL and ELEMENT LEDs on the keyboard, however, display coil and contact status in near-real time (after a two-second delay) and will show every state change. The COIL LED shows the state of the displayed string's coil (LED on = coil energized, LED off = coil de-energized). The ELEMENT LED displays the state of the contact selected by the cursor (LED on = contact closed, LED off = contact open). If the cursor is not on a contact, the ELEMENT LED will remain off.

3. The PREV and NEXT keys can be used to display other strings in sequencer memory. Pressing PREV displays the string just in front of the currently displayed string in string number sequence. Pressing NEXT displays the next string in string number sequence following the currently displayed string.

4. The FIND and WHERE USED keys can be used to display strings whose string numbers are unknown if a related coil or contact is known. These keys can be used three ways:

   A. Position the cursor on a contact and press WHERE USED. The string number of the string containing the related coil (if any) will appear in the Where Used/Find field. Press FIND to display that string.

   B. Position the cursor on the coil and press WHERE USED. The lowest string number of a string using a contact of that coil will appear in the Where Used/Find field. Each depression of WHERE USED causes the next higher related string number to appear in the Where Used/Find field. If a contact occurs twice in a string, the string number will appear twice in the Where Used/Find field. Pressing FIND at any time displays the string whose string number appears in the field.

   C. Enter a string number into the Where Used/Find field and then press FIND to display the desired string.

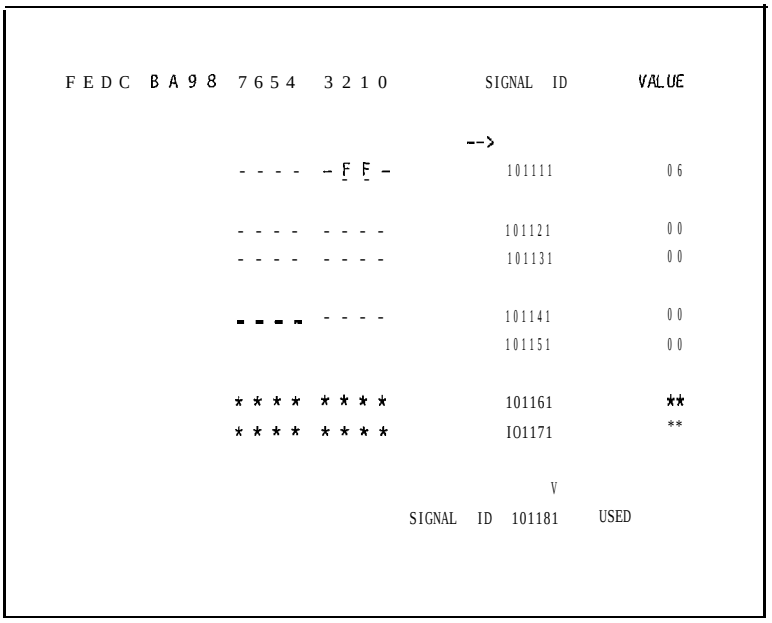
Displaying Signals

In Display String mode, you can view only one string at a time. Display Signal mode allows you to view a large number of signals at once in an easy-to-read tabular form. You can view up to eight groups of signals, with each group containing up to 16 points. Like Display String mode, Display Signal mode provides continual updating of the status of displayed signals. Contacts are shown closed or open and coils are shown energized or deenergized.

1. Press DISPLY SIGNAL to enter Display Signal mode. The screen displays three empty columns: one labelled with the hexadecimal digits F through 0, the next labelled "Signal ID", and the third labelled "Value". A pointer appears in the Signal ID column and at the bottom of the screen are two inverse video fields labelled "Signal ID" and "Used".

2. Press INPUT and then enter a signal address (such as 01111). This tells the programming module that you wish to display signal 101111. You could have entered OUTPUT or INTRL to display output (R01 111) or internal status (S01 111) signals. The signal address appears in the Signal ID inverse video field. Note that a pointer just above the Signal ID field moves from left to right as you enter the address.

3. Press ENTER to display the signal. The signal address appears at the pointer location under the Signal ID header. The status of all points in that group appear under the hexadecimal digits. The hexadecimal equivalent value of the points in the group appears under the Value header. Continue to press ENTER until the screen appears as shown below. Note that as you press ENTER, the signal address in the Signal ID field increments by group number and the signal address, status, and value appear in the appropriate columns.



4. The Display Signal screen uses the following formats:

   A. The signal address appearing in the Signal ID column represents a group of points. The status of each point is shown under the hexadecimal digits. For instance, the status of point 101110 would be found under the 0 digit, while the status of point IO1116 would be found under the 6 digit. Groups containing point-type signals require 8 status bits while groups containing value-type signals require 16 status bits.

   B. If the status of a point is shown as a dash, it means that the contact is open or coil is deenergized. If the status is an inverse video dash, it means that the contact is closed or the coil energized. If the status is an asterisk, it means that the point has not been defined in the KIB rack entry table. If the status is an F, it means that the contact or coil is under force control (see below for more information on forcing). Notice that a contact or coil defined in the KIB but not used in a string appears inactive on the screen (open or deenergized).

   C. The two or four digit number appearing in the Value column is the decimal equivalent of any status bits that are active (displayed in inverse video). The status bits are weighted in a binary fashion according to their position: bit 0 is equivalent to a 1, bit 1 is equivalent to a 2, bit 3 is equivalent to 4, and so on. For instance, if point 0 is active, a 01 appears in the Value column. If both point 0 and point 1 is active, a 03 appear in the value column.

   D. The pointer in the Signal ID column shows where the next signal address will appear. The pointer in the Signal ID field shows where a number entered from the keyboard will appear. The RIGHT and LEFT Cursor Control keys control the pointer in the Signal ID field, while the UP and DOWN keys control the pdinter in the Signal ID column.

5. Pressing DELETE deletes from the screen the signal address located at the Signal ID column pointer.

6. The WHERE USED key can be used to find the string number of a string where a coil or contact signal address is used. To use this function, enter the signal address of a coil or contact and press WHERE USED. The string number of the string where that coil or contact is used will appear in the Used field. If the coil or contact is not used in any string, the word “None” will appear in the Used field.

Forcing

Forcing is the ability to change the state of an on-line contact or coil from the programming module keyboard. A contact can be forced open or closed and a coil can be forced energized or deenergized regardless of the state of the associated string logic. Forcing is useful during troubleshooting of suspected system problems such as defective solenoids or bad wiring. Forcing can also be used to test newly programmed strings to ensure their proper operation under a variety of circumstances. Only input (I) and output (R) signals can be forced; internal status (S) signals cannot be forced.

The sequencer application program responds only to the forcing of input signals, not output signals. Forcing output signals (coils) affects only output devices. In other words, if you force a coil with an output (R) address, the external device will be affected, but any internal (software) contacts tied to that coil will not be affected.

All signals are automatically unforced when the programming terminal is either unplugged from the sequencer or powered off.

Contacts and coils can be forced in either Display String or Display Signal mode. The following procedure tells you how to force in both modes.

To Force in Display String Mode:

1. Display the string containing the coil or contact you wish to force. See the previous section titled "Displaying a String" for information on how to display strings.

2. Position the cursor to the coil or contact you wish to force and press FORCE. The signal address of the coil or contact will change to inverse video to indicate that the element is now under force control.

3. **Press FORCE STATE to** change the state of the coil or contact. Each time you press FORCE STATE, the state of the coil or contact changes. For instance, if you had placed a normally open contact under force control and pressed FORCE STATE, the contact would be closed. Pressing **FORCE STATE** again causes the contact to open.

4. After 'you have forced the desired contacts and coils in one string, you-can display other strings to force other coils and contacts or observe the results of the forcing.

5. To unforce a coil or contact, move the cursor to the coil or contact and press FORCE again. If you want to unforce all forced coils and contacts in the system, press SHIFT and then FORCE.

To Force in Display Signal Mode:

1. Display the signal you wish to force (the screen will display a group of 8 or 16 points). See the previous section titled "Displaying a Signal" for information on how to display signals.

2. Enter the signal address of the specific point you wish to force into the Signal ID field at the bottom of the screen. The group containing this point must be displayed before you can force that point.

3. Press FORCE to put that point under force control. An F will appear at the display signal table location representing that point. When the F is displayed in inverse video, it indicates the contact is closed or the relay is energized.

4. Press FORCE STATE to change the state of the point. Each time you press FORCE STATE, the state of the coil or contact changes. For instance, if you had placed a normally open contact under force control (the F would be displayed in normal video) and pressed FORCE STATE, the contact would be closed (the F would be displayed in inverse video). Pressing FORCE STATE again causes the contact to open.

5. After you have forced the desired points in one group, you can display other groups to force other coils and contacts or observe the results of the forcing.

6. To display all forced signals in the system, insert FFFFF into the Signal ID field and press ENTER. The screen will display up to eight groups containing forced signals. If there are more than eight groups containing forced signals, press NEXT to display the other groups.

7. To unforce a coil or contact, enter the proper signal address into the Signal ID field and press FORCE again. If you want to unforce all forced coils and contacts in the system, press SHIFT and then FORCE.

CHAPTER 4
PROGRAMMING MODULE OPERATING MODES

This chapter provides detailed operating information on the five DLC + Programming Module modes. Before reading this chapter, you should be familiar with Chapter 3, which contains basic operating information for these modes.

BUILD STRING MODE

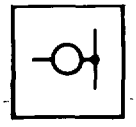
The main purpose of Build String mode is to enable you to create and modify relay strings. This section of the manual provides general programming instructions for creating and modifying strings in Build String mode. This section also contains specific programming information for each of the different types of control blocks available via the Function/Data Entry keys. More information on this mode is found in Chapter 3.

General Programming Instructions

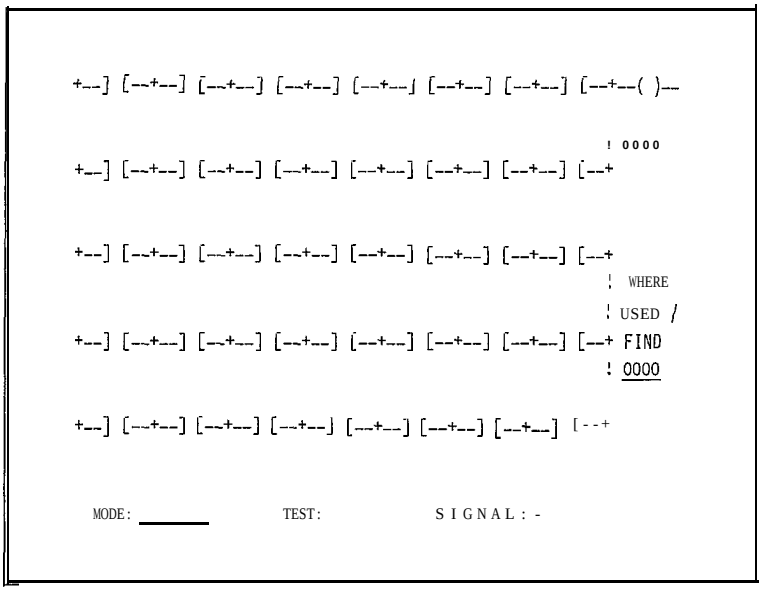
- Once a skeleton string or control block has been created, it must be edited to form the desired logic network.
- All changes made to a string in Build String mode occur in an off-line buffer area. Modifications to sequencer memory are not made until after the string has been tested and entered into memory.
- "Current flow" in a string must always be from left to right. You should not insert a connection that allows current flow from right to left.
- Programmable fields are modified by positioning the cursor to the field and inserting the desired values or signal addresses. The L***** field always contains a signal address, while the V***** field always contains a constant. Unless otherwise stated, a programmable field can only accept constants or signal addresses with a decimal value of -32,768 to + 32,767.
- A new skeleton string or control block cannot be created unless the message "No Logic Assigned" is visible on the screen.
- The string number assigned to the string determines its execution sequence; that is, string 0010 will execute before string 0015.
- A contact can be modified (that is, changed from normally open to normally closed or vice versa) without changing its signal address. Simply position the cursor to the contact and press the appropriate key.
- New strings can be inserted between existing strings while in Build String mode. The Resequence function of Supervisory mode spaces the strings in regular intervals of 10 to allow easy insertion of new strings.
- A string can be deleted from sequencer memory by displaying the string in Build String mode and then pressing SHIFT and DELETE. The Modify Logic keyswitch must be ON to delete a string.
- All control block parameters (PV, CV, FV, A, B, C, D, etc.) are assumed to be count values in all blocks except the convert and special block.

Skeleton String

PRESS THIS KEY:



TO BUILD THIS STRING:



Programming Notes!

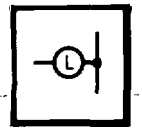
1. Contacts can be programmed as input, output, or internal status points (I, R, or S prefix).
2. The coil can be programmed as an output or internal status point (R or S prefix).

Operation:

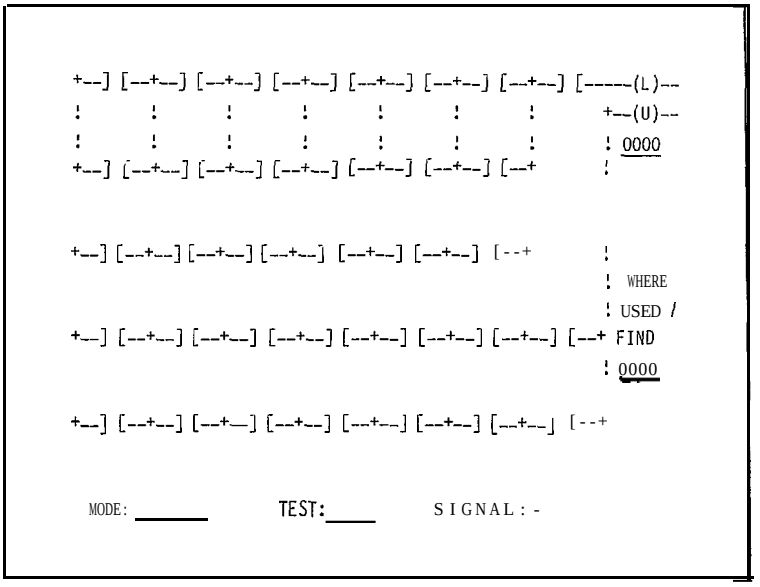
The coil operates as any relay would. When a current path exists from the left bus to the coil, the coil is energized. When the current path is interrupted, the coil is deenergized.

Latch Relay String

PRESS THIS KEY:



TO BUILD THIS STRING:



Programming Notes:

1. Contacts can be programmed as input, output, or internal status points (I, R, or S prefix).
2. The coils can be programmed as output or internal status points (R or S prefix).

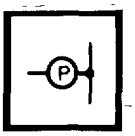
Operation:

The latch (L) coil is energized when a current path exists from the left bus to the L coil and remains energized even after its current path is interrupted (even through a power interruption). The L coil is deenergized when its current path is interrupted and the unlatch (U) coil is energized. The U coil is energized when a current path exists from the left bus to the U coil and is deenergized when that current path is broken.

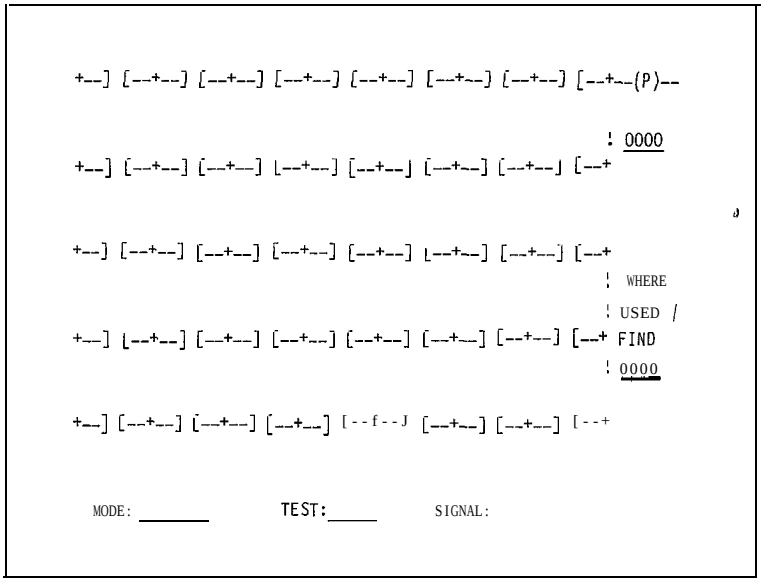
Note that there are two separate string networks in this block – the upper network controls the latch (L) coil while the lower network controls the unlatch (U) coil.

Pulse Relay String

PRESS THIS KEY:



TO BUILD THIS STRING:



Programming Notes:

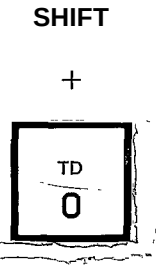
1. Contacts can be programmed as input, output, or internal status points (I, R, or S prefix).
2. The coil can be programmed as an output or internal status point (R or S prefix).

Operation:

The coil is energized for one sequencer sweep time each time current begins flowing to the coil (that is, whenever the logic solution of the string network results in a change of state from false to true). After energizing for one sweep the coil is deenergized until the next change of state, hence the term “pulse relay.” The length of the sequencer sweep is established by the KIB function. In effect, the pulse relay operates like a single shot.

Time Delay String

PRESS THIS KEY:



TO BUILD THIS STRING:

```
+--] [--+--] [--+--] [--+--] [--+--] [--+--] [--+--] [--+--(D)--
                                                    ! V*.*
                                                    ! 0000
+--] [--+--] [--+--] [--+--] [--+--] [--+--] [--+--] [--+--] [--+

+--] [--+--] [--+--] [--+--] [--+--] [--+--] [--+--] [--+--] [--+
                                                    : WHERE
                                                    ! USED /
+--] [--+--] [--+--] [--+--] [--+--] [--+--] [--+--] [--+--] [--+ FIND
                                                    : 0000

+--] [--+--] [--+--] [--+--] [--+--] [--+--] [--+--] [--+--] [--+

MODE:_____        TEST:        SIGNAL:-
```

Programming Notes:

1. Contacts can be programmed as input, output, or internal status points (I, R, or S prefix).
2. The coil can be programmed as an output or internal status point (R or S prefix).
3. The time delay value (V) must be a decimal value no greater than 99.9. This value can be modified by placing the cursor in the V*.* field and inserting a new value.

Operation:

The coil is energized V seconds after a current path exists from the left bus to the coil. V is the time delay value programmed into the block. If the current path is interrupted prior to the completion of the time delay, the time count is reset to zero. If the current path is interrupted or a power failure occurs while the coil is energized, the coil is deenergized and the time count reset to zero. The current time delay value is not displayed in the V*.* field during a time delay countdown; only the original value is displayed there.

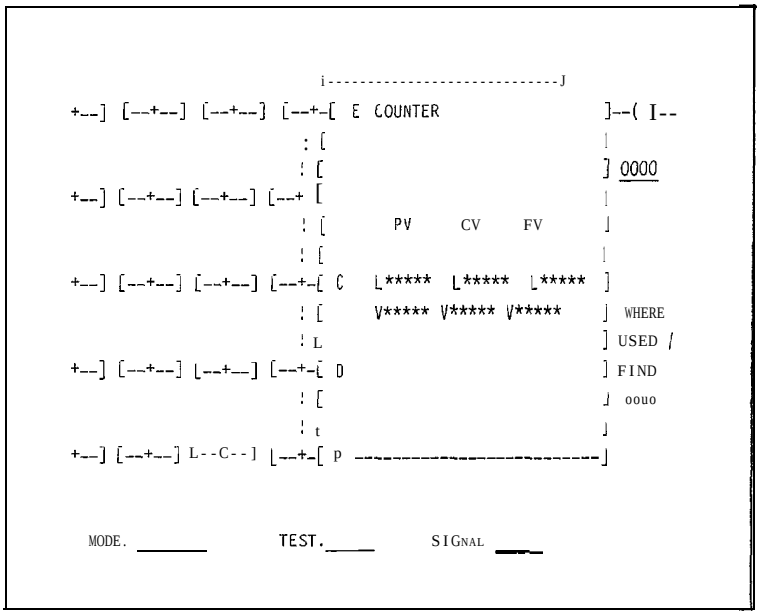
Counter Block

PRESS THIS KEY: TO BUILD THIS STRING:

SHIFT

+

CTR
1



Programming Notes:,'

1. Contacts can be programmed as input, output, or internal status points (I, R, or S prefix).
2. The coil can be programmed as an output or internal status point (R or S prefix).
3. The preset value (PV) can be programmed as a constant by inserting the desired value into the V***** field. If PV is programmed as a constant, the signal address (L*****) must not be defined. PV can also be programmed as an input (I) or internal status (S) count signal by inserting the proper signal address into the L***** field. BCD inputs to PV must be converted to count signals. PV can be any decimal value in the range -32,768 to + 32,767.
4. The current value (CV) can be programmed as an internal status (S) count signal or an output (R) count signal by inserting the proper signal address into the L***** field. The V***** field cannot be modified. CV can be any decimal value in the range -32,768 to + 32,767.
5. The final value (FV) is programmed the same as the preset value (PV) .

Operation:

When current is provided to the enable (E) input, the counter is enabled and begins counting each closure of the count (C) input string. Each time the C input string closes (goes from a no-current condition to a current condition), the counter increments or decrements the count value (CV). CV is decremented if the down (D) input string is closed and incremented if the D string is open. If the preset (P) string is closed, the current value (CV) is set equal to the preset value (PV); the preset function overrides all other functions. As long as the P string is closed, the value of CV is held at the preset (PV) value, even if the counter is enabled and counting. When the current value (CV) becomes greater than or equal to the final value (FV) and the E string is closed, the coil is energized. The coil can only be deenergized when CV becomes less than FV while the E string is closed. If current to the E string is interrupted, the coil remains in its previous state.

Timer Block

PRESS THIS KEY:

SHIFT

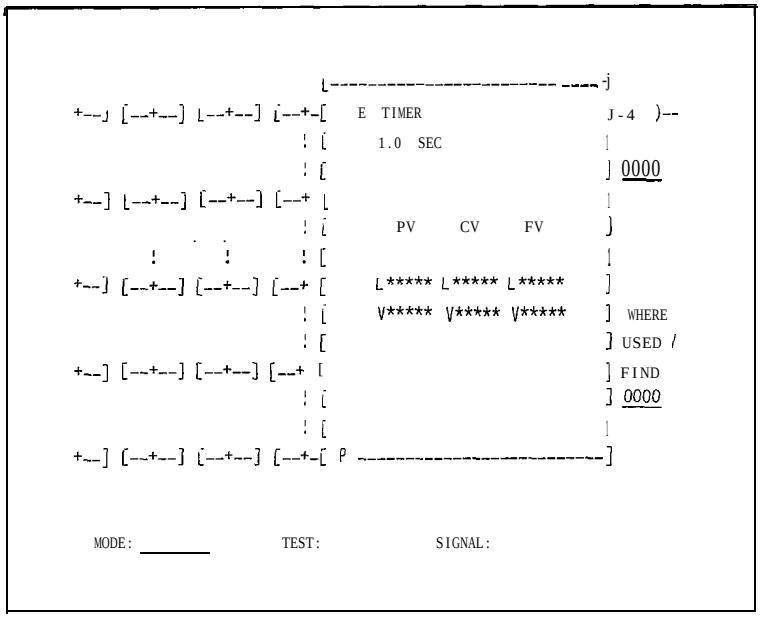
+

T 1.0
2

OR

T 0.1
3

TO BUILD THIS STRING:



Programming Notes:'

1. A 1 .O second timer is created by pressing SHIFT and 2, while a 0.1 second timer is created by pressing! SHIFT and 3.
2. Contacts can be programmed as input, output, or internal status points (I, R, or S prefix).
3. The coil can be programmed as an output or internal status point (R or S prefix).
4. The preset value (PV) can be programmed as a constant by inserting the desired value into the V***** field. If PV is programmed as a constant, the signal address (L*****) must not be defined. PV can also be programmed as an input (I), output (R) , or internal status (S) count signal by inserting the proper signal address into the L***** field. PV can be any decimal value in the range 0 to +32,767.
5. The current value (CV) can be programmed as an internal status (S) count signal or an output (R) count signal by inserting the proper signal address into the L***** field. The V***** field cannot be modified. CV can be any decimal value in the range 0 to + 32,767.
6. The final value (FV) can be programmed as a constant by inserting the desired value into the V***** field. If FV is programmed as a constant, the signal address (L*****) must not be defined. FV can also be programmed as an output (R) or internal status (S) count signal by inserting the proper signal address into the L***** field. FV can be any decimal value in the range 0 to + 32,767.
7. If PV, CV, or FV is to be used as a BCD readout, that value must be converted from a count signal to a BCD signal (see section titled "Convert Block" for more information).

Operation:

When current is provided to the enable (E) input, the timer begins incrementing the current value (CV) every second (for 1 .O second timer) or tenth of a second (for 0.1 second timer). When current is provided to the preset (P) input, the current value (CV) is set equal to the preset value (PV); the preset function overrides all other functions. As long as the P string is closed, the value of CV is held at the preset (PV) value, even if the timer is enabled. When the current value (CV) becomes greater than or equal to the final value (FV) , the coil is energized. The coil can only be deenergized when CV becomes less than FV while the E string is closed. If the E string is open, the coil remains in its previous state.

Compare Block

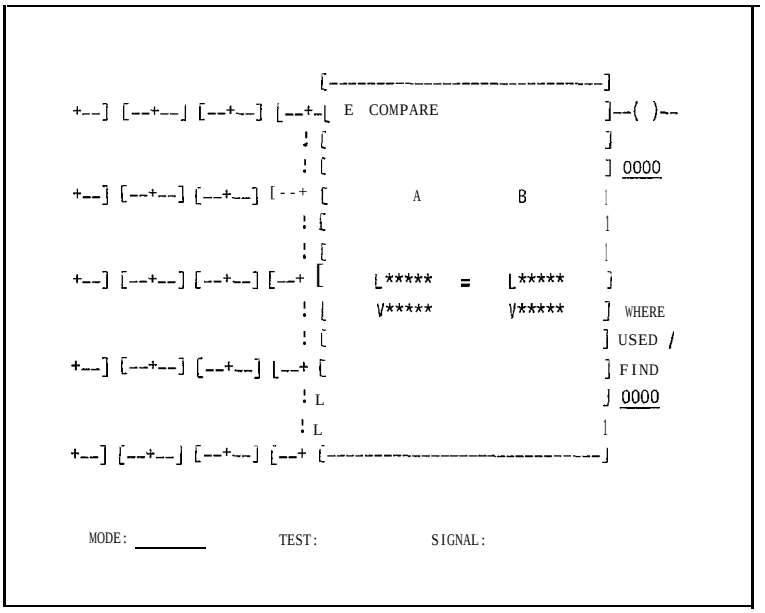
PRESS THIS KEY:

SHIFT

+

CMPR
4

TO BUILD THIS STRING:



Programming Notes!

1. Contacts can be programmed as input, output, or internal status points (I, R, or S prefix).
2. The coil can be programmed as an output or internal status point (R or S prefix).
3. A and B can be programmed as constants by inserting the desired value into the respective V***** field. If A and B are programmed as constants, their signal addresses (L*****) must not be defined. A and B can also be programmed as input (I), output (R), or internal status (S) count signals by inserting the proper signal address into the respective L***** field. A and B can be any decimal value in the range -32,768 to +32,767.
4. The relational operator is initially an equal sign (=). This operator can be changed by positioning the cursor to the equal sign and pressing SHIFT and 4. Each depression of SHIFT and 4 causes a different relational operator to appear: less than, greater than, less than or equal, greater than or equal, and not equal.

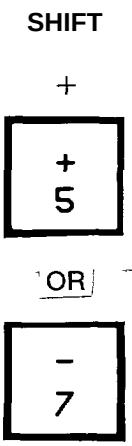
Operation:

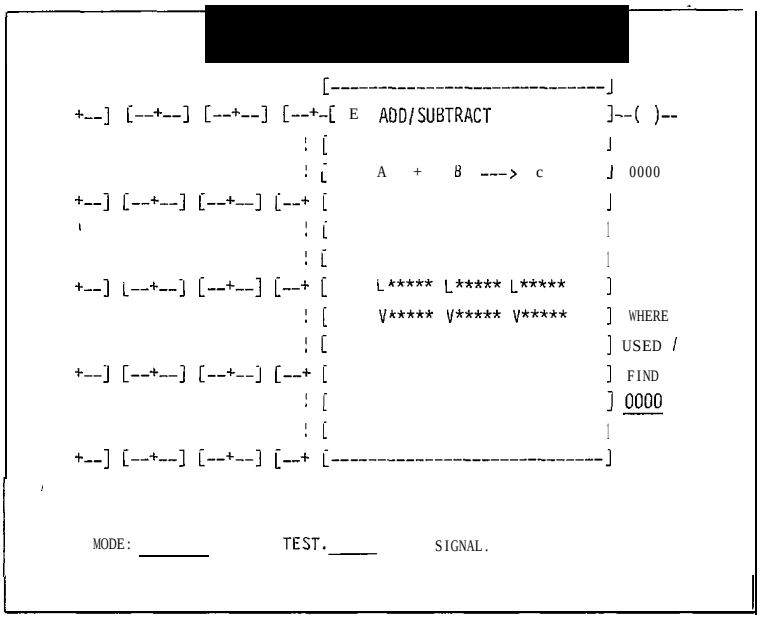
When current is supplied to the E input, the values of A and B are compared according to the relational operator. If the two values compare the coil is energized. If the two values do not compare, the coil is deenergized. If current to the E string is interrupted, the coil remains in its previous state.

Add/Subtract Block

PRESS THIS KEY:

TO BUILD THIS STRING:





Programming Notes:

1. An add block (A + B) is created by pressing SHIFT and 5, while a subtract block (A − B) is created by pressing SHIFT and 7. If you want to change an already existing add or subtract block to its complement (i.e, change an add block to a subtract block or vice versa), position the cursor to the plus or minus sign and press the appropriate keys (SHIFT and 5 to produce an add block, SHIFT and 7 to produce a subtract block).
2. Contacts can be programmed as input, output, or internal status points (I, R, or S prefix).
3. The coil can be programmed as an output or internal status point (R or S prefix).
4. A and B can be programmed as constants by inserting the desired value into the respective V***** field. If A and B are programmed as constants, their signal addresses (L*****) must not be defined. A and B can also be programmed as input (I), output (R), or internal status (S) count signals by inserting the proper signal address into the respective L***** field. A and B can be any decimal value in the range -32,768 to +32,767.
5. C can be programmed as an output (R) or internal status (S) count signal by inserting the proper signal address into the L***** field. The V***** field cannot be modified. C can be any decimal value in the range -32,768 to +32,767.
6. The add block can be used in place of a convert block for moving one 16 bit signal (internal, output, or internal status) to another 16 bit signal (output or internal status). Using an add block for such a move requires less internal memory space than using a convert block. To do this, program A with the signal address of the signal to be moved, set B to zero, and program C with the signal address of the target signal.

7. The add block can also be used to send the status of a cluster of I/O points to another cluster of I/O points, assuming all of the points are within the same 16 bit group. This would ordinarily be done using a separate string for each point. For instance, assume you wish to send the status of points RlllOO – R11110 to points R11180 – R11190. You could set up sixteen strings, each with a single contact (using signal addresses RlllOO – R11110) feeding a single coil (using signal addresses R11180 – R11190). The status of points **RlllOO** – R11110 would now be transmitted to points R11180 – R11190. A simpler and less memory-intensive method of transferring the same information would be to set up an add block with R1110C as the A value, zero as the B value, and R1118C as the C value. The transfer of status occurs when the block is enabled. When using this method the range of points receiving the status (the C value - R11 180 – R11190) must not be used in any other strings.

Operation:

When current is supplied to the enable (E) input, the values of A and B are added (add block) or subtracted (subtract block) and the result is placed in C. If the result is outside the range of C, an overflow occurs and the coil is energized. The coil remains energized until the result of the addition or subtraction does not cause an overflow. The coil will indicate the overflow condition of the C value even when current is interrupted to the E input. When power is first applied to the sequencer, the coil is deenergized and the value of C set to zero.

Slope/Offset Block

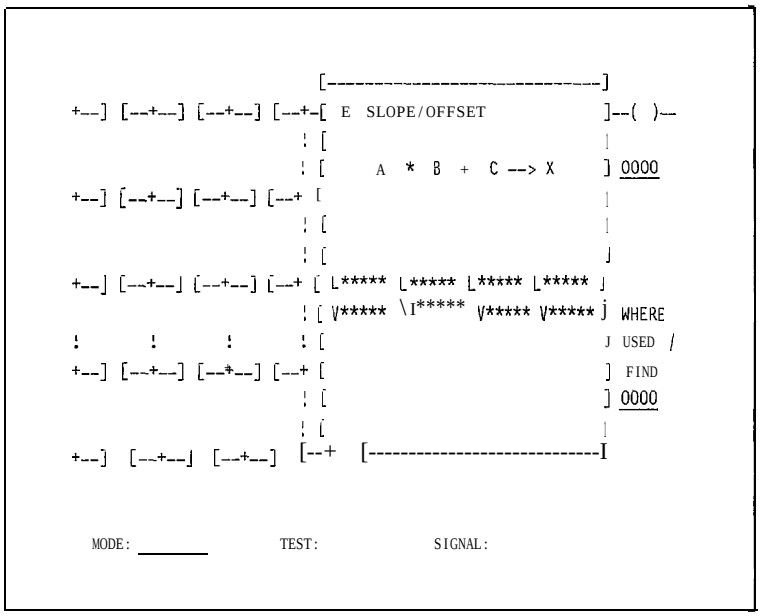
PRESS THIS KEY:

SHIFT

+

SLOPE
6

TO BUILD THIS STRING:



Programming Notes:

1. Contacts can be programmed as input, output. or internal status points (I, R, or S prefix).
2. The coil can be programmed as an output or internal status point (R or S prefix).
3. A, B, and C can be programmed as constants by inserting the desired value into the respective V***** field. If A, B, or C is programmed as a constant, its signal address (L*****) must not be defined. A, B, and C can also be programmed as input (I), output (R), or internal status (S) count signals by inserting the proper signal address into the respective L***** field. A, B, and C can be any decimal value in the range -32,768 to $32,767.
5. X can be programmed as an output (R) or internal status (S) count signal by inserting the proper signal address into the L***** field. The V***** field cannot be modified. X can be any decimal value in the range -32,768 to +32,767.

Operation:

When current is supplied to the enable (E) input, the values of A and B are multiplied, the value of C is added to the product, and the result is placed in X. If the result is outside the range of X, an overflow occurs and the coil is energized. The coil remains energized until the result of the slope/offset calculation does not cause an overflow. The coil will indicate the overflow condition of the X value even when current is interrupted to the E input. When power is first applied to the sequencer, the coil is deenergized and the value of X set to zero.

Convert Block

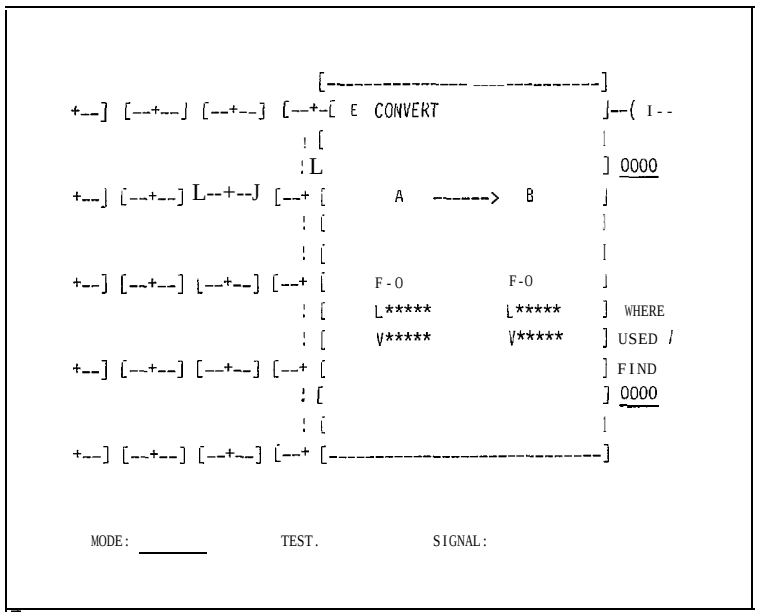
PRESS THIS KEY:

SHIFT

+



TO BUILD THIS STRING:



Programming Notes:

1. Contacts can be programmed as input, output, or internal status points (I, R, or S prefix).
2. The coil can be programmed as an output or internal status point (R or S prefix).
3. The two programmable fields (A and B) are programmed to correspond to the desired conversion operation, A is typically an input or internal status signal, while B is typically an output or internal status signal or a constant. If B is programmed as a constant, its signal address (L*****) must not be defined. A and B may represent either BCD or count values.
4. The F-O field represents the range of signal points assigned to the convert operation. The two F-O fields must represent an equal number of points. The field may be modified to reflect the requirements of the particular conversion. For instance, the range F-O represents all 16 points in a BCD or count signal (hexadecimal F through 0), while the range 7-O represents the least significant 8 points in a BCD or count signal. Proper use of this field enables masking and shifting of data during a convert operation. As an example, you could cause the middle 8 bits in a BCD signal to be moved to the lower 8 bits of another BCD signal by using values of B-4 and 7-O respectively in this field. Any unused bits will be masked; this means that the upper 8 bits of the destination BCD signal will remain unchanged.

Operation:

When current is supplied to the enable (E) input, the convert function specified in the block is performed. The following conversions are possible:

- Direct move of BCD input (I or S) signal to BCD output (R or S) signal. A = BCD input, B = BCD output.
- Direct move of count input (I or S) signal to count output (R or S) signal. A = count input, B = count output.
- Convert count input (I or S) signal to BCD output (R or S) signal. A = count input, B = BCD output.
- Convert BCD input (I or S) signal to count output (R or S) signal. A = BCD input. B = count output.
- Set a BCD or count signal equal to a decimal value. A = BCD or count signal, B = constant (-32,768 to +32,767).

Up to four BCD digits can be handled by the convert block.

The coil is energized if any of the following invalid conversions are attempted:

- Signal types are not either BCD (signal address ends in B) or count (signal address ends in C)
- The count value is greater than hexadecimal 270F (decimal 9999) or is negative during a coversion from count to BCD.
- The BCD value is illegal (contains digits other than 0 - 9) during a conversion from BCD to count.
- The F-O fields do not represent the same number of points (for example, you attempt to move 4 bits into 8 bits).
- You attempt a direct move of less than 16 bits between different signal types (for example, you attempt to move 8 bits in a BCD signal to 8 bits in a count signal).

Scale Block

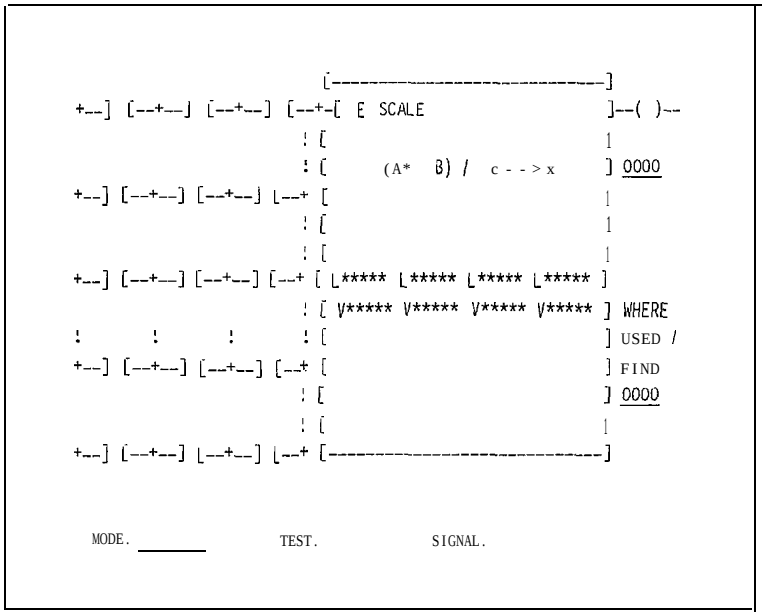
PRESS THIS KEY:

SHIFT

+

SCALE
9

TO BUILD THIS STRING:



Programming Notes:

1. Contacts can be programmed as input, output, or internal status points (I, R, or S prefix).
2. The coil can be programmed as an output or internal status point (R or S prefix).
3. A, B, and C can be programmed as constants by inserting the desired value into the respective V***** field. If A, B, or C is programmed as a constant, its signal address (L*****) must not be defined. A, B, and C can also be programmed as input (I), output (R), or internal status (S) count signals by inserting the proper signal address into the respective L***** field. A, B, and C can be any decimal value in the range -32,768 to +32,767.
5. X can be programmed *as* an output (R) or internal status (S) count signal by inserting the proper signal address into the L***** field. The V***** field cannot be modified. X can be any decimal value in the range -32,768 to +32,767.

Operation:

When current is supplied to the enable (E) input, the values of A and B are multiplied, the product is divided by the value of C, and the result is placed in X. If the result is outside the range of X, an overflow occurs and the coil is energized. The coil remains energized until the result of the scaling calculation does not cause an overflow. The coil will indicate the overflow condition of the X value even when current is interrupted to the E input. When power is first applied to the sequencer, the coil is deenergized and the value of X set to zero.

Special Block

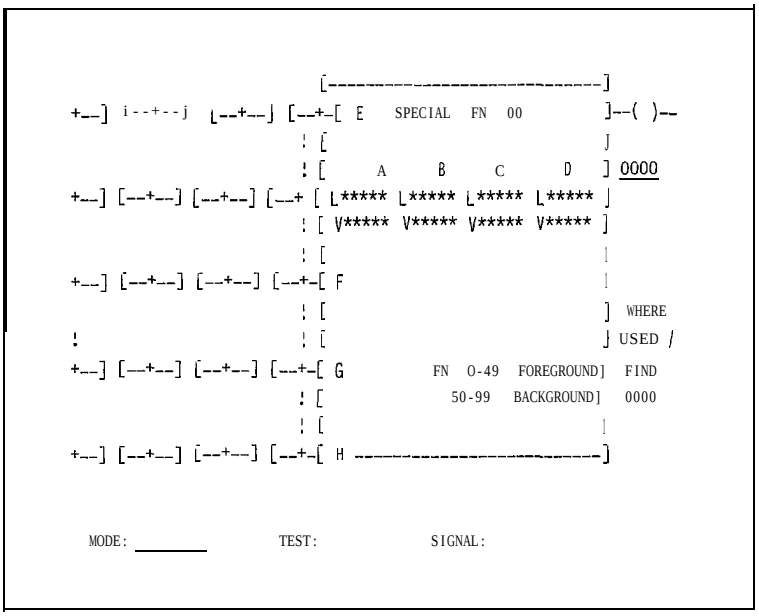
PRESS THIS KEY:

SHIFT

+

SPEC
C

TO BUILD THIS STRING:



Programming Notes:

1. Contacts can be programmed as input, output, or internal status points (I, R, or S prefix)
2. The coil can be programmed as an output or internal status point (R or S prefix).
3. A, B, C, and D can be programmed as constants by inserting the desired value into the respective V***** field. If A, B, C, or D is programmed as a constant, its signal address (L*****) must not be defined. A, B, C, and D can also be programmed as input (I), output (R), or internal status (S) count signals by inserting the proper signal address into the respective L***** field. A, B, C, and D can be any decimal value in the range -32,768 to + 32,767.

Operation:

When current is supplied to the enable (E) input, the special block causes a software routine within the sequencer called a "function code module" to be executed. Function code modules are written in assembly language or a high-level language rather than in relay ladder form and are typically supplied in product software PROMs. As such, function code modules are not user-programmable. Any function code modules in your system are described in the system documentation. The function code module can accept logic signals from the special block F, G, and H inputs, as well as values or signals programmed into the A, B, C, and D fields. In addition, the function code module can return values to the A, B, C, and D fields The use of these fields is also explained in the system documentation.. The two digit FN number in the upper right of the screen identifies the individual function code module requested by the special block. In addition, the number determines whether the function code will execute in **foreground** or **background** mode. If the number is from 00 and 49, the function code module will automatically execute in foreground mode, while a number from 50 to 99 means the function code module will automatically execute in background mode. Foreground execution means that the function code module executes immediately and within one sequencer sweep when it is called by the special block. Background execution means that the function code module executes concurrently with relay ladder execution and may take several sequencer sweeps to complete execution. The status of the coil is controlled by the function code module in foreground mode, while in background mode the coil is energized as long as the function code module is executing.

TEST STRING MODE

The main purpose of Test String mode is to enable you to test new or modified strings before placing them permanently into sequencer memory. This section of the manual provides general instructions for using Test String mode. More information on this mode is found in Chapter 3.

- A string must be displayed in Build String mode before it can be put under test. Only one string at a time can be under test.
- If a modified string (that is, a string whose string number already exists in sequencer memory) is placed under test, the modified string replaces the original string in the execution sequence.
- A string under test can be displayed and signals can be forced in Display String and Display Signal modes.
- The test is automatically aborted if a power interruption occurs or if the programming module is unplugged from the sequencer.
- The Modify Logic keyswitch must be ON before a string under test can be entered into sequencer memory.

DISPLAY STRING MODE

The main purpose of Display String mode is to enable you to view new or modified strings as they are executing on-line. This section of the manual provides general instructions for using Display String mode. More information on this mode is found in Chapter 3.

- While in Display String mode, pressing FIND causes the programming module to find and display the string whose string number is in the Where Used/Find field. If no string exists with that number, the message "No Logic Assigned" is displayed. PREVIOUS and NEXT can be used to display strings whose string numbers occur just prior to or after the currently displayed string's string number.
- An open contact is displayed in normal video, while a closed contact is displayed in inverse video. Likewise a deenergized coil is displayed in normal video, while an energized coil is displayed in inverse video.
- The COIL and ELEMENT LEDs on the keyboard are active in this mode and are used to display the status (energized or de-energized) of coils and contacts. The COIL LED shows the state of the displayed string's coil (LED on = coil energized, LED off = coil de-energized). The ELEMENT LED displays the state of the contact selected by the cursor (LED on = contact closed, LED off = contact open). If the cursor is not on a contact, the ELEMENT LED will remain off. The COIL and ELEMENT LEDs display coil and contact status in near-real time (after a two-second delay) and will show every state change, while the screen image of coils and contacts may not show every state change.
- The current value of constants within programmable fields is visible in real time. For instance, the current value (CV) in an on-line timer block can be seen to change on the screen.
- You can force up to 16 contacts and coils at one time in this mode. When a contact or coil is under force control, its signal address will be displayed in inverse video. Pressing SHIFT and then FORCE clears all forced signals in the system.

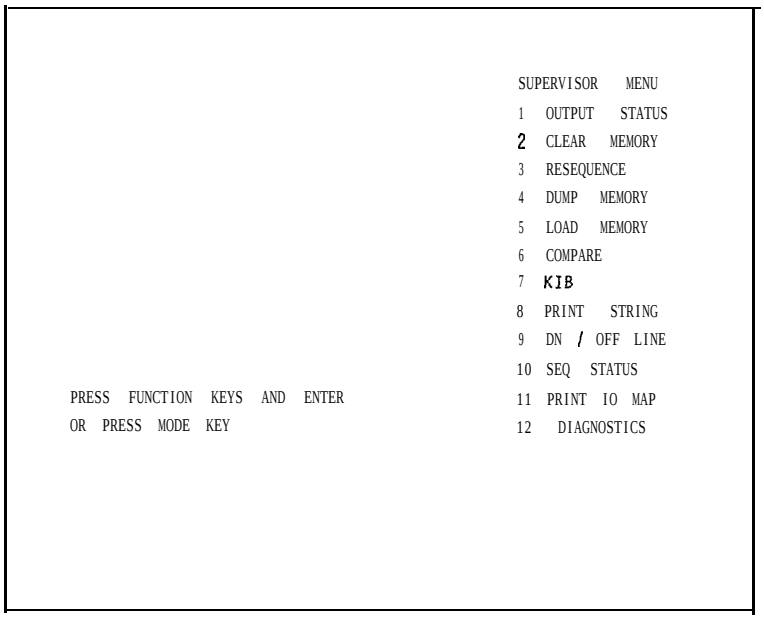
DISPLAY SIGNAL MODE

The main purpose of Display Signal mode is to enable you to view the status of several signals at once in a tabular format. This section of the manual provides general instructions for using Display Signal mode. More information on this mode is found in Chapter 3.

- Entering a signal address and pressing ENTER causes the signal status (that is, status of all points in that group) to be displayed in the signal table. Repeated depressions of ENTER increments the signal address and displays the next group of points.
- The Up and Down Cursor Control keys move the vertical pointer up and down on the screen, while the Right and Left Cursor Control keys move the Signal Address field pointer right and left on the screen.
- If you place the programming module in another mode and then go back to Display Signal mode, the original display will still be on the screen.
- You can force up to 16 signals at one time in this mode. When a signal is under force control, an F appears at the corresponding location on the signal table. Pressing SHIFT and then FORCE clears all forced signals in the system.
- All signals currently under force control can be displayed by entering FFFFF in the Signal ID field and pressing ENTER.

SUPERVISORY MODE

A wide variety of functions are available in Supervisory mode. This section of the manual provides background and operating information for each function. We cover each function in the order it is listed on the Supervisory mode menu listing, which is reproduced below:



Output Status-

Function:

Allows you to view the current output status and change it if necessary. Output status refers to the ability of the sequencer to update its output devices. With outputs enabled, the sequencer reads inputs, executes application program instructions, and updates outputs normally. With outputs disabled, the sequencer continues to read inputs and execute application program instructions, but no longer updates output devices. In this case, output devices remain in their original state (unless forced to zeros by station FF) and forcing is no longer possible.

Operation:

1. Press the SUPVR key to display the Supervisory mode menu, then press the 1 key and the ENTER key. The current output status (Outputs Enabled or Outputs Disabled) will appear on the screen.

2. If you wish to change the output status, press SHIFT and ENTER. The new output status will appear.

3. If you do not wish to change the output status, press DELETE.

Clear Memory

Function:

Clears application program memory (RAM and BRAM). This function must be done while the sequencer is off-line. Since all data in memory (including the KIB) is lost, it is usually a good idea to make a tape copy of memory (if desired) before using this function.

Operation:

1. Press the SUPVR key to display the Supervisory mode menu, then press the 2 key and the ENTER key.

2. Press SHIFT and ENTER to clear memory or press DELETE if you do not want to clear memory.

Resequence

Function:

Changes the string numbers of the strings in sequencer memory so that they are multiples of 10. For instance, the first string number would be 0000, the second 0010, the third 0020, and so on. This is commonly done after new strings are assigned string numbers between already existing string numbers.

Operation:

1. Press the SUPVR key to display the Supervisory mode menu, then press the 3 key and the ENTER key.

2. Press SHIFT and ENTER to resequence memory. When the resequence function is complete, the final string number is displayed on the screen.

3. If you do not wish to resequence memory, press DELETE.

Dump Memory

Function:

Writes data from the sequencer's memory to a cassette tape. The operator selects the data to be written, which can be a range of strings and the KIB, a range of strings only, or the KIB only

Operation:

1. Prepare the tape drive for use as follows. First make sure the tape drive is connected to the terminal and powered on, and that a tape has been placed in the tape drive. Ensure that the write protect tab on the tape is set to allow recording. On the tape drive (which is assumed to be either an EPI STR® LINK IIA or STR® LINK III tape drive), set the BAUD RATE switch to 1200 (2400 for the STR® LINK III), the STOP BITS switch to 1, the I/O SELECT switch to RS 232, and the MODE SELECT switch to HALF DUPLEX. Press the REWIND button once to let the tape rewind, then press the button once again. Finally, press the REMOTE CONTROL switch so that the REMOTE LED is on.

2. The sequencer may be on-line or off-line for this function, although the dump occurs faster if the sequencer is off-line. Press the SUPVR key to display the Supervisory mode menu, then press the 4 key and the ENTER key. The screen will prompt you to identify the type of tape drive being used; enter a 2 if you are using a STR® LINK IIA or a 3 if you are using a STR @LINK III. After this, the following information will appear:

```
STR LINK III TAPE DUMP
                                DUMPING ID:

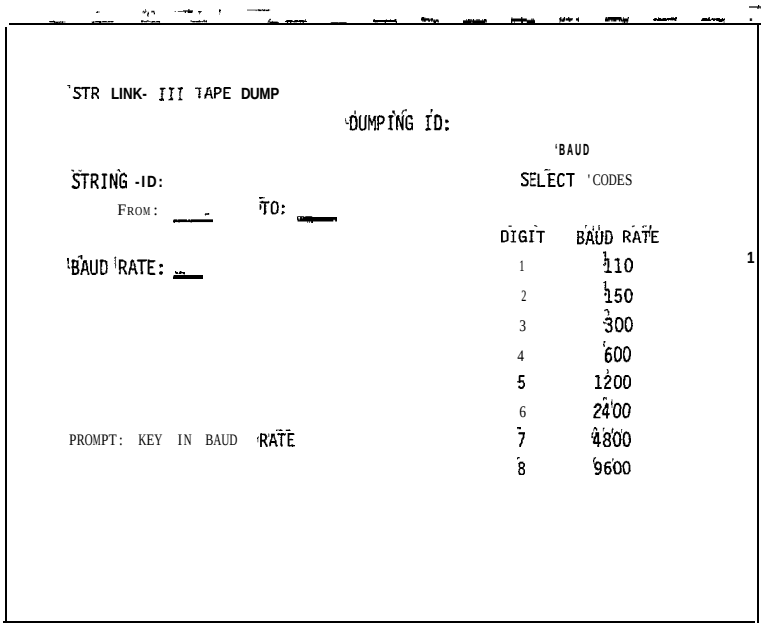
STRING ID:
     FROM:  ____     TO:  ____




 PROMPT: KEY IN STRING ID'S
ENTER SHIFT-7 TO DUMP KIB
```

3. If you wish to dump the KIB, press the SHIFT key and then the 7 key. Enter the FROM and TO string IDs (four decimal digits each) corresponding to the range of strings you wish to dump. The cursor "rolls" in the field after you have entered a full field of data. Press the ENTER key after each entry. If you wish to dump all strings in memory and are not sure of the string ID of the last string, simply *enter FROM 0000 TO 9999. Do not fry to dump more* ***than 400*** *strings on one side* ***of the*** *tape* ***(800 strings for the STR® LINK I...***

4. At this point, the screen displays the following information:



Enter the number corresponding to the desired baud rate (1200 baud for the STR® LINK IIA or 2400 for the STR @LINK III> and then press the ENTER key.

5. The screen now prompts you to key in a record ID. This ID consists of alphanumeric characters that will be recorded on the tape and displayed in the Record ID field when you perform a Load Memory operation using that tape. You could, for example, use the string number of the last string on the tape as a record ID. You can use any Function/Data Entry key (0 through F) plus the INPUT (I), OUTPUT (R), and INTRL (S) keys to create the record ID. The cursor "rolls" in the field after you have entered a full field of data; if you make a mistake entering the record ID, you must press DELETE and start the procedure over to correct the error. Press the ENTER key to confirm the record ID and start the Dump Memory operation.

6. As the dump proceeds, the Dumping ID field displays the string number currently being written to tape. Pressing DELETE at any time will cancel the operation. When all strings in the range you specified in step 3 have been dumped, the screen displays "Operation Complete" and returns to the Supervisory menu.

7. Perform the Compare function (described in this chapter) to ensure that all strings were correctly written to tape.

Load Memory

Function:

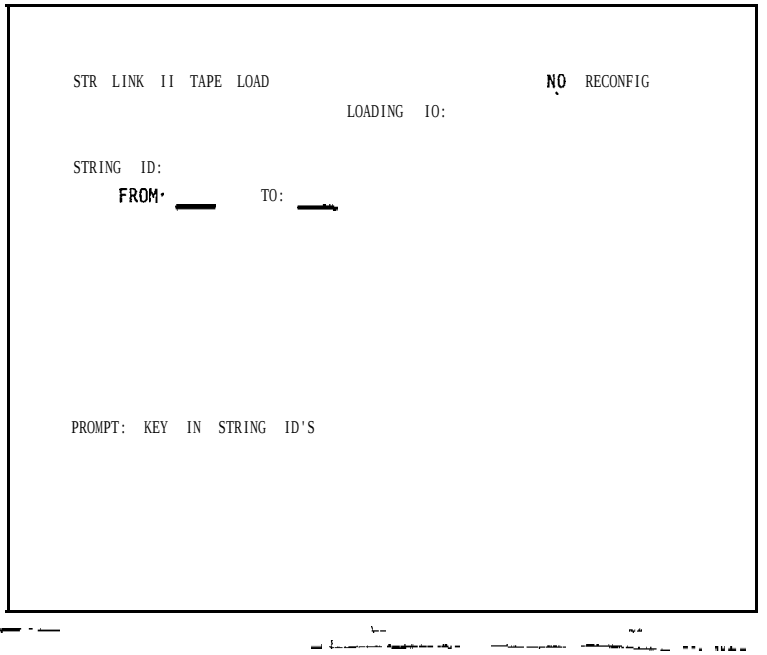
Transfers strings from a cassette tape to sequencer memory. The operator selects the range of strings to be transferred.

Operation:

1. Perform the Clear Memory function. Ensure that the sequencer is off-line.

2. Prepare the tape drive for use as follows. First make sure the tape drive is connected to the terminal and powered on, and that a tape containing the strings you wish to load into memory has been placed in the tape drive. Ensure that the write protect tab on the tape is set to prevent inadvertent recording. On the tape drive (which is assumed to be either an EPI STR ® LINK IIA or STR ® LINK III tape drive), set the BAUD RATE switch to 1200 (2400 for the STR @LINK III), the STOP BITS switch to 1, the I/O SELECT switch to RS 232, and the MODE SELECT switch to HALF DUPLEX. Press the REWIND button once to let the tape rewind, then press the button once again. Finally, press the REMOTE CONTROL switch so that the REMOTE LED is on.

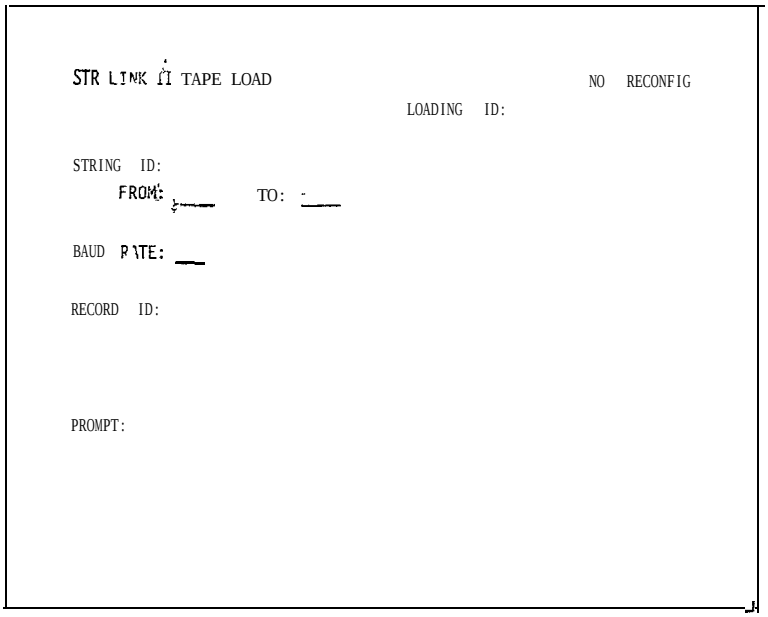
3. Press the SUPVR key to display the Supervisory mode menu, then press the 5 key and the ENTER key. The screen will first ask whether reconfiguration parameters should be loaded from tape (see the section titled "KIB" in this chapter for information about system configuration). Enter a 0 if these parameters should not be loaded or a 1 if they should be loaded. ***Note that loading reconfiguration parameters will clear system memory.*** The screen will next prompt you to identify the type of tape drive being used; enter a 2 if you are using a STR ® LINK IIA or a 3 if you are using a STR *LINK III. After this, the following information will appear:



4. Ensure that the appropriate KIB entries are on tape or have been keyed into the system through the programming module. Enter the FROM and TO string IDs (four decimal digits each) corresponding to the range of strings you wish to load. The cursor "rolls" in the field after you have entered a full field of data. Press the ENTER key after each entry. If you wish to load all strings on the tape and are not sure of the string ID of the last string, simply enter FROM 0000 TO 9999. ***Note that this will cause all strings numbered from 0000 to 9999 on the tape to be loaded to memory. If data on the tape has been overwritten at any time, there may be extra stringspresent that should not be written to memory.***

5. At this point, the screen displays a list of baud rates and asks for a selection as shown above under step 4 of "Dump Memory". Enter the number corresponding to the desired baud rate (1200 baud for the STR ® LINK IIA or 2400 for the STR ® LINK III) and then press the ENTER key to start the memory load operation.

6. The programming module screen displays the following information:



As the load proceeds, the Record ID field displays the record ID (if any) assigned when the data was dumped from memory. The Loading ID field displays the string number currently being written to memory. Pressing DELETE at any time will cancel the operation. If the programming module detects any errors while loading, an error count appears below the Record ID field. If any errors occur, you should restart the load from the beginning of the tape. When all strings in the range you specified in step 3 have been loaded, the screen displays "Operation Complete" and returns to the Supervisory menu.

7. Perform the Compare function (described below) to ensure that all strings were correctly written to memory.

Compare

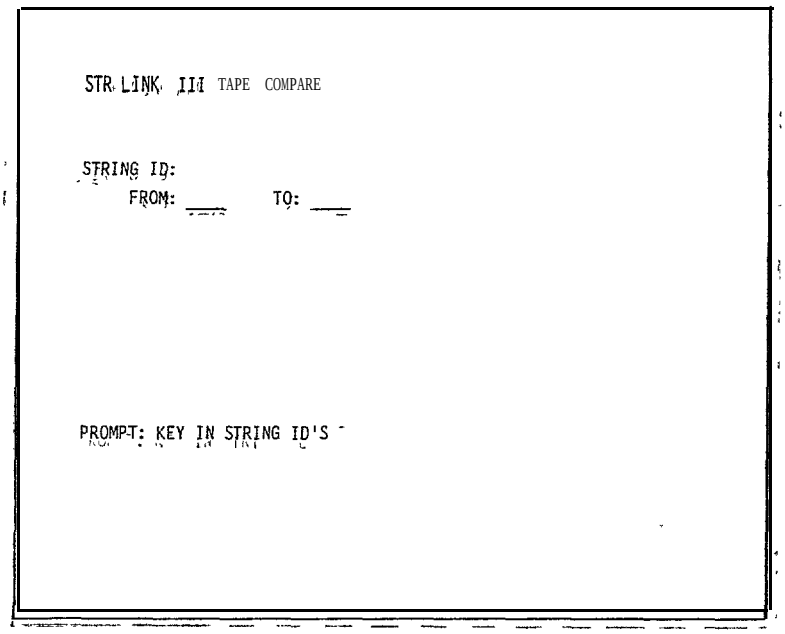
Function:

Compares data on a cassette tape with the data in sequencer memory. The compare function should be used after performing a Dump Memory or Load Memory function to ensure that the data was transferred accurately. If a KIB exists on the tape and in the sequencer, the two are automatically compared.

Operation:

1. Prepare the tape drive for use as follows. First make sure the tape drive is connected to the terminal and powered on, and that a tape containing the strings you wish to compare with memory has been placed in the tape drive. Ensure that the write protect tab on the tape is set to prevent inadvertent recording. On the tape drive (which is assumed to be either an EPI STR ® LINK IIA or STR ® LINK III tape drive), set the BAUD RATE switch to 1200 (2400 for the STR ® LINK III), the STOP BITS switch to 1, the I/O SELECT switch to RS 232, and the MODE SELECT switch to HALF DUPLEX. Press the REWIND button once to let the tape rewind, then press the button once again. Finally, press the REMOTE CONTROL switch so that the REMOTE LED is on.

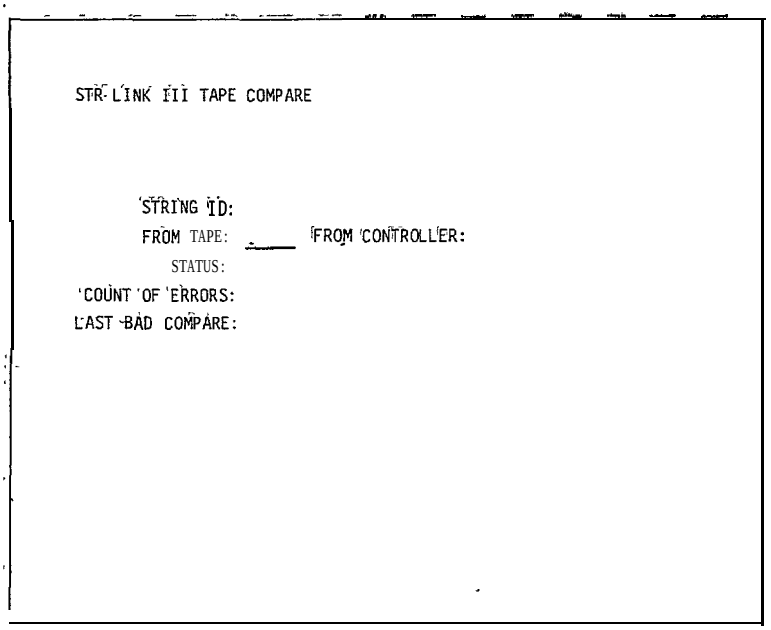
2. The sequencer may be on-line or off-line for this function, although the compare occurs faster if the sequencer is off-line. Press the SUPVR key to display the Supervisory mode menu, then press the 6 key and the ENTER key. The following information will appear:



3. Enter the FROM and TO string IDs (four decimal digits each) corresponding to the range of strings you wish to compare. The cursor "rolls" in the field after you have entered a full field of data. Press the ENTER key after each entry. If you wish to compare all strings on the tape and are not sure of the string ID of the last string, simply enter FROM 0000 TO 9999.

4. At this point, the screen displays a list of baud rates and asks for a selection as shown above under step 4 of "Dump Memory". Enter the number corresponding to the desired baud rate (1200 baud for the STR® LINK IIA or 2400 for the STR® LINK III) and then press the ENTER key to start the compare operation.

5. The programming module screen displays the following information:



As the compare proceeds, the String ID fields (FROM TAPE and FROM CONTROLLER) display' the tape and sequencer string numbers being compared. As long as the strings compare, the Status field displays GOOD. If the programming module detects any errors while comparing, the Status field displays BAD and the Count of Errors field increments to show the total number of errors detected. The Last Bad Compare field displays the string numbers of the last pair of strings that did not compare. Pressing DELETE at any time will cancel the compare operation. When all strings in the range you specified in step 3 have been compared, the screen displays "Operation Complete" and returns to the Supervisory menu.

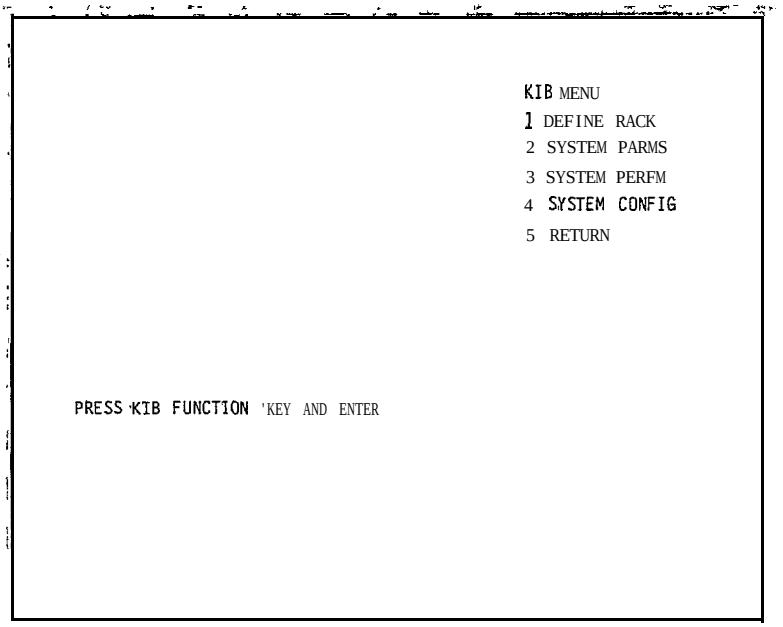
KIB

Function:

Used to configure sequencer I/O by identifying the channels, stations, racks, and groups recognized by the sequencer. Also used to set system parameters and system configuration as well as monitor system performance.

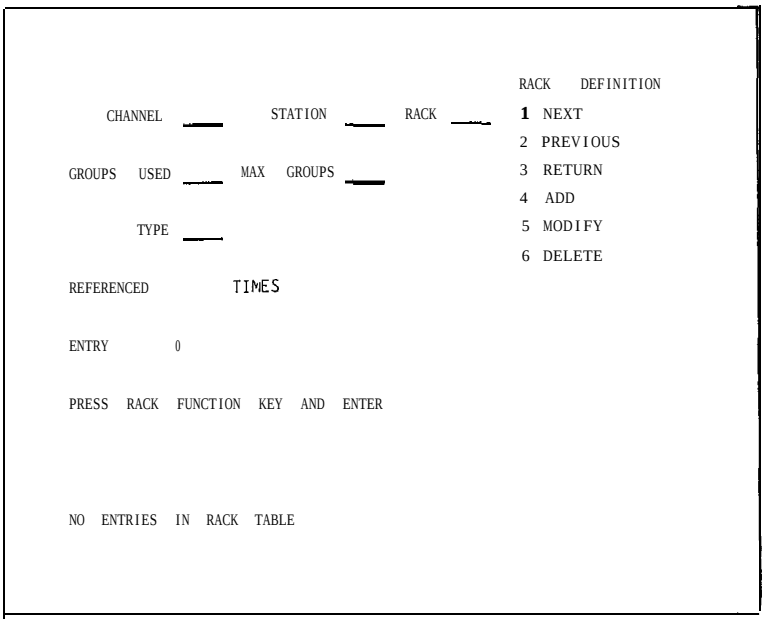
Operation:

1. Press the SUPVR key to display the Supervisory mode menu, then press the 7 key and the ENTER key. The KIB menu will be displayed as shown below:



2. If you want to display or modify the KIB rack table, press 1 and then ENTER. The KIB rack definition menu will appear as shown below. The sequencer must be off-line before attempting to create, modify, or delete a rack entry.

   The KIB rack table is used to configure sequencer I/O devices in terms of channels, stations, racks, and groups. See Chapter 3 for more information on signal identification and KIB configuration.



If any rack table entries exist, the screen displays the last rack entry that was viewed or modified using the KIB function. If no rack entries exist, the message "No Entries in Rack Table" appears at the bottom of the screen. The fields appearing on the screen are explained below. All fields can be modified except the REFERENCED TIMES and ENTRY fields.

CHANNEL - a two digit number (00 through 16) representing the channel through which I/O to and from the station is routed.

STATION - a two digit hexadecimal number (01 through FE) representing the station to which the rack is assigned.

RACK - a single digit hexadecimal number (0 through F) representing the rack to which all groups specified in this rack entry are assigned.

GROUPS USED - a two digit decimal number (01 through 16) specifying the number of groups that the sequencer scans during each sweep. This number can be modified at a later time without deleting and recreating the rack entry.

MAX GROUPS - a two digit decimal number (00 through 16) specifying the number of groups actually usable in the rack. This number determines the amount of sequencer memory set aside for handling data transfer between the rack and the sequencer. Once set, this number cannot be changed later. Typically, the maximum number of groups includes the number of groups used plus some number of "spare" groups for future expansion. The number in the Max Groups field must be greater than or equal to the value in the Groups Used field.

TYPE - a single digit (0, 1 , or 2) specifying the type of I/O handled by the rack. A 2 indicates a Control Signal Highway (E-bus) I/O point, a 1 indicates HIOB (Siltron™ power conversion module) I/O, while a 0 indicates any other type of I/O.

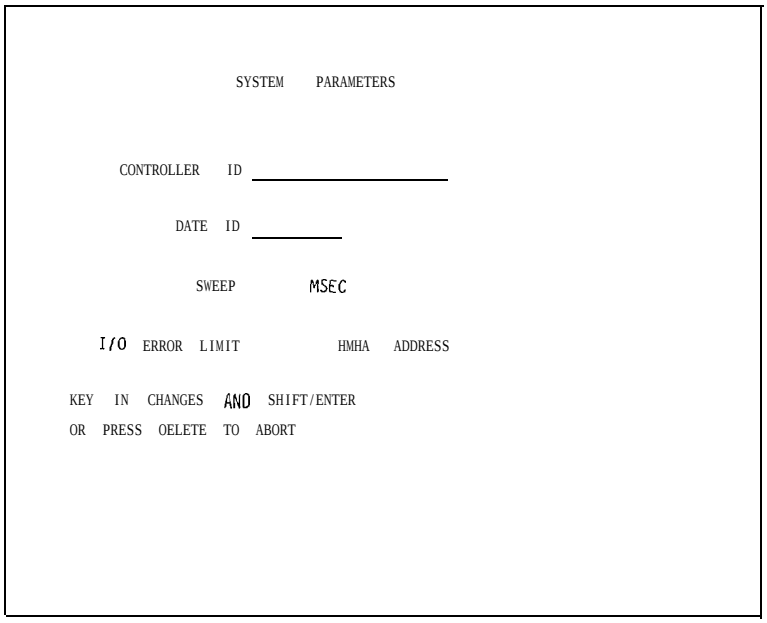
REFERENCED TIMES - shows the number of times that the rack has been referenced by a signal address.

ENTRY - keeps track of the order of rack entries. For instance, the first rack entry created would have an ENTRY number of 1, the next one 2, and so on.

The Rack Definition menu appearing in the upper right part of the screen provides six different functions for viewing and modifying rack entries. **The sequencer must be off-line in order to add, modify, or delete a rack entry.**

- Press 1 and then ENTER to view the next entry in the rack table (if any>.
- Press 2 and then ENTER to view the previous entry in the rack table (if any).
- Press 3 and then ENTER to return to the KIB menu
- Press 4 and then ENTER to add a rack entry to the rack table. The cursor will move to the Channel field and allow you to enter the channel number of the rack you are configuring. The cursor jumps from field to field as you fill each field with data, or you can use the Right and Left Cursor Control keys to position the cursor to the desired field. Pressing DELETE will abort the modifications. After you have entered all values, press SHIFT and then ENTER to to enter the rack entry into the rack table. Note that if a rack entry is currently displayed and you press 4 and ENTER to add another rack entry, the values currently displayed in each field remain there and you can edit them as desired. When you press SHIFT and ENTER, the new rack entry is inserted directly in front of the rack entry that had been displayed.
  All racks within a station must be defined together and all stations within a channel must be defined together as contiguous entries in the rack entry table. The same holds true for channel types (i.e., HMHA I/O, Control Signal Highway I/O, etc. must be similarly defined together in the rack entry table). Failure to do this may result in malfunction of the application software.
- Press 5 and then ENTER to modify the rack entry currently being displayed. ***The Max Groups field cannot be modified.*** The sequence of events for modification is the same as for adding a rack entry.
- Press 6 and then ENTER to delete the rack entry currently being displayed. The rack entry cannot be deleted if any signal addresses reference that rack entry (that is, if a number appears in the Referenced Times field).

3. If you want to display or modify the system parameters, press 2 and then ENTER. The screen will appear as shown below:



The screen contains five fields:

CONTROLLER ID - contains a software identifier, usually reflecting the software revision level.
DATE ID - contains a date, usually reflecting the latest revision date of the software.
SWEEP - indicates the length of time for the sequencer to perform one sweep of all I/O points and to execute the application program. This value must be in multiples of 10 milliseconds.
I/O ERROR LIMIT - indicates how many consecutive I/O errors can occur in a station before the system is notified of an error.
HMHA BASE ADDRESS - the hexadecimal address of the first HMHA channel. This value must be in multiples of 2 Kbytes. This same address value must be configured on the desired HMHA board using the hardware address straps.

If you do not want to modify the system parameters, press DELETE to return to the KIB menu. If you want to modify one or more parameters, key in the changes and press SHIFT and then ENTER. The Right and Left Cursor Control keys can be used to position the cursor in the appropriate lie'ld.

4. If you want to monitor system performance, press 3 and then ENTER. The screen will appear as shown below:

```
                                        KIB MENU:.
MODEL    RELEASE  7.3                   1  DEFINE  RACK'
                                        2  SYSTEM  PARMS
SWEEP     50   10  10   10   1   0      3  SYSTEM PERFM
                                        4  SYSTEM  CDNFIG
USED      800  62  267   0   0  52K     5  RETURN
              INTERNAL  PARAMETER;
DSA       16500  16314  186  1  16314
RNDB      11750  11742   8   1  11742
PRESS  KIB  FUNCTION  KEY  AND  ENTER
```

The various fields are explained below. Note that the values in each field are only samples and will vary between installations.

MODEL - fixed-revision level of the sequencer product software, in this case RELEASE 7.3.
SWEEP - consists of six sweep times in the following order:

- Sweep time that was set in the system parameters (50 MS)
- Shortest sweep time taken by the sequencer (10 MS)
- Average sweep time taken by the sequencer (10 MS)
- Highest sweep time taken by the sequencer (10 MS)
- Number of times the sequencer exceeded the sweep time set in the system parameters (1)
- Number of times the sequencer had to wait for the HMHA board to complete a channel I/O transfer before completing the sweep (0)

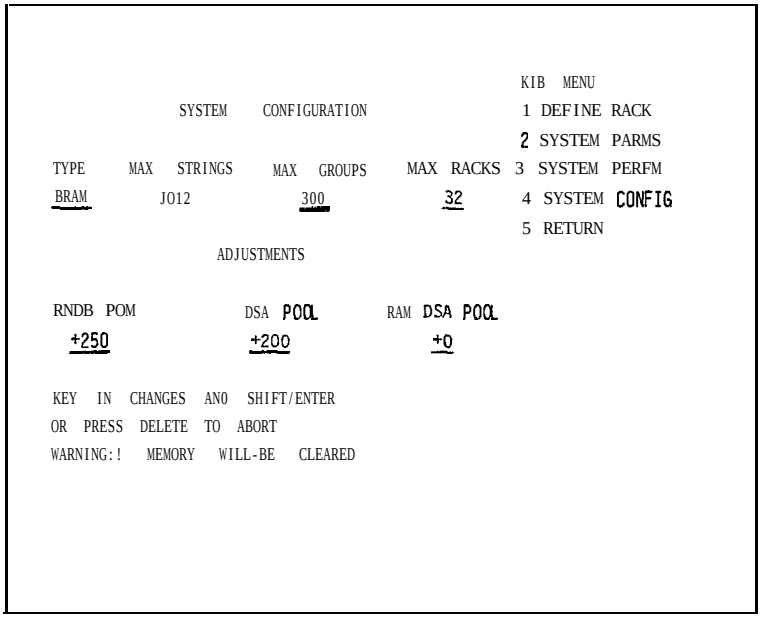
USED - contains values representing the current number of strings (800), I/O racks (62), and I/O groups (267) in the system. The second half of the line specifies the size of RAM DSA memory pool (0) and the current usage of the pool (0). These numbers will be 0 for all BRAM based systems, see the KIB information for BRAM/PROM memory configurations. The last entry on this line is the total usable RAM data area in the system (52K).

INTERNAL PARAMETERS - these are memory usage statistics for the two memory pools in the system — the DSA (Dynamic Storage Allocation) pool and the RNDB (Relay Network Definition Block) pool. These pools are used as buffers during sequencer operation. It is unnecessary for you to understand these statistics, but they are explained below for those who are interested.

There are five statistics reported for each memory pool. The statistics are displayed in the following order:

- The total memory space, in bytes, allocated for each pool (16,500 bytes for DSA and 11,750 bytes for RNDB)
- The memory space remaining for each pool (16,314 bytes in DSA and 11,742 bytes in RNDB)
- The "high water mark" for each pool — that is, the highest byte ever allocated (byte 186 for DSA and byte 8 for RNDB)
- The number of separated memory blocks (holes) currently existing in each pool (1 for both DSA and RNDB)
- The size of the largest free block (hole) in each memory block (16,314 bytes in DSA and 11,742 bytes in RNDB)

5. If you want to view or change the system configuration parameters, press 4 and then ENTER. The screen will appear as shown below. Using this function, you can specify what type of memory the sequencer will contain (BRAM or PROM), change the maximum number of I/O racks and groups, and alter the size of the RNDB, DSA, and RAM DSA memory pools. Be ***aware that sequencer memory will be cleared whenever a configuration change is requested through this function!***



The various fields are explained below. Note that the values in each field are only samples and will vary between installations. All fields except the MAX STRINGS field can be changed by the user.

TYPE - A user-definable field that specifies what type of application code memory the sequencer will contain - BRAM or PROM. Press the ENTER key to change the field from BRAM to PROM or PROM to BRAM. A PROM system provides the capability of storing application code in PROM, but this decreases the maximum number of strings in the system. Currently, application code PROMs can be created only by General Electric Company using customer-generated magnetic tapes; creation of these PROMs by a user is not supported. For more information on creating PROM-based sequencer systems, contact your General Electric representative.

MAX STRINGS - A read-only field that contains the **estimated** maximum number of strings that the sequencer's memory can hold. This number is an estimate determined at configuration time and is based on the amount of memory in the system.

MAX GROUPS and MAX RACKS - User-definable fields that can be used to custom-fit the maximum I/O requirements for a specific application. Since each I/O group contains input (I), output (R), and internal status (S) signals, if you define MAX GROUPS as 100, the system will be able to hold 100 I groups, 100 R groups, and 100 S groups. The system requires 2 racks and 32 groups for its own use; these amounts must be included in the numbers specified for these fields.

RNDB POOL and DSA POOL - User-definable fields that can be used to adjust the size of the RNDB and DSA memory pools. Entering a negative number decreases the pool by that number of bytes, while entering a positive number increases the pool by that number of bytes. The size of these tables are based mainly on the number of strings for which the system is configured. These tables may be re-allocated according to their usage by your particular application. The size and usage of these two memory pools are displayed using the "System Parms" entry of the KIB menu.

RAM DSA POOL - A user-definable field that can be used to adjust the size of the RAM DSA memory pool. This pool is used only in PROM systems for I/O tables and special block extensions and need not be defined in BRAM systems. The RAM DSA pool will not have to be adjusted unless your sequencer contains 28K bytes of memory; on that system it may be to your advantage to decrease the size of the pool to make room for more strings. On all other systems, the size of this pool will not affect the maximum string count. The size and usage of this memory pool are displayed using the "System Parms" entry of the KIB menu.

After changing any of these configuration parameters on the screen, press SHIFT and then ENTER to transmit the changes to the sequencer or press DELETE to abort the process. Memory ***will*** be ***cleared when configuration parameters are changed.*** After *a* configuration change has been made, the display will be updated to show the fmal sequencer configuration. Should the configuration parameters you enter be incompatible with the existing hardware, the sequencer will redefine the parameters to create an acceptable system.

Configuration parameters are dumped to magnetic tape whenever KIB information is dumped. When loading sequencer memory from a tape, you have the option of restoring configuration parameters (if any) along with the KIB, or restoring just the KIB.

6. If you want to return to the Supervisory menu, press DELETE or press 5 and then ENTER.

Print String

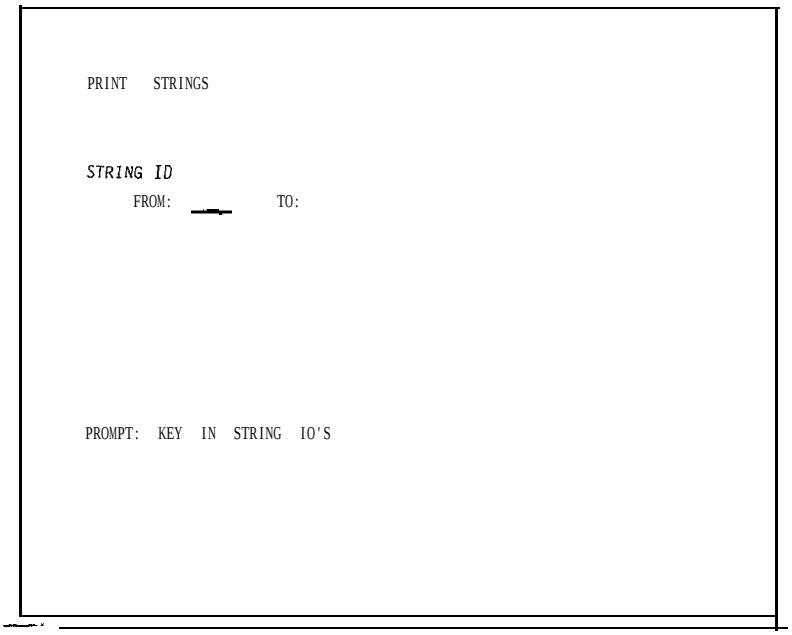
Function:

Prints a relay ladder diagram of one or more application software strings on a compatible printer.

Operation:

1. Make sure the printer is connected to the terminal, powered on, and in a ready condition.

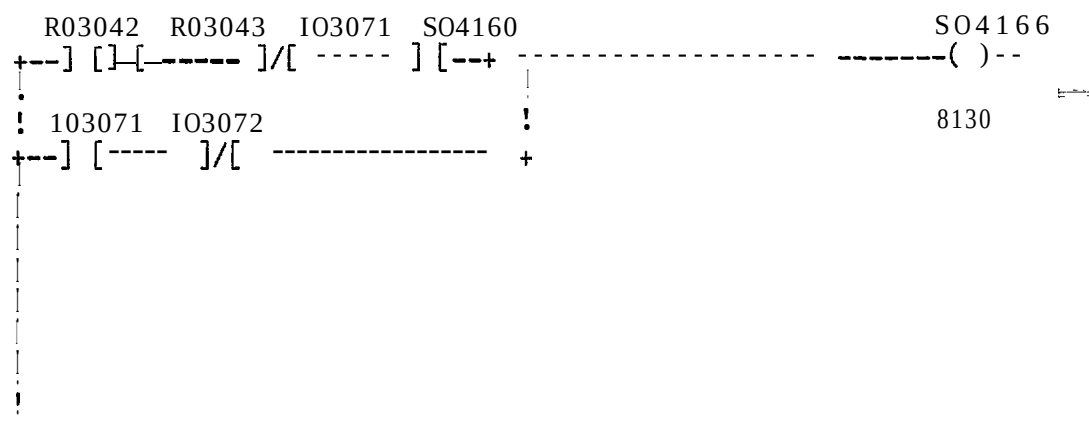
2. Press the SUPVR key to display the Supervisory mode menu, then press the 8 key and the ENTER key. The following information will appear:



3. Enter the FROM and TO string IDs (four decimal digits each) corresponding to the range of strings you wish to print. The cursor "rolls" in the field after you have entered a full field of data. Press the ENTER key after each entry. If you wish to print all strings in the sequencer and are not sure of the string ID of the last string, simply enter FROM 0000 TO 9999.

4. At this point, the screen displays a list of baud rates and asks for a selection as described above under step 4 of "Dump Memory". Enter the number corresponding to the baud rate of the printer you are using (most often 300 baud) and then press the ENTER key to start printing strings. Each string is displayed on the screen as it prints. Press DELETE at any time to abort the operation. When all requested strings have been printed, the screen displays "Operation Complete" and returns to the Supervisory menu.

5. The output of the printer will be similar to the following example:



R03042 8115
R03043 8125
SO4160 8090
SO4166 8195 8180 8170

The printed string is an exact duplicate of the string as it appears on the screen. Below each string is a list of the string's output (R) and internal status (S) signals and the string numbers where related coils or contacts can be found. In the above print-out, the coil for signal R03042 can be found in string 811.5, while related contacts of coil SO4166 can be found in strings 8195, 8180, and 8170.

On/Off Line

Function:

Allows you to place the sequencer on-line or off-line. An off-line sequencer does not communicate with I/O devices and does not execute application software. An on-line sequencer operates normally.

Operation:

1. Press the SUPVR key to display the Supervisory mode menu, then press the 9 key and the ENTER key. The current sequencer condition (Off Line or On Line) will appear on the screen.

2. If you wish to change the on/off line condition press SHIFT and ENTER. The new condition will appear.

3. If you do not wish to change the on/off line condition, press DELETE.

The current state of the sequencer will be reflected on the sequencer's microprocessor card by a red LED if it is an HMPF (HMPC cards do not have this LED). The LED will be off when the sequencer is off line, and it will flash very rapidly if the sequencer *is* on line.

Sequencer Status

Function:

Allows you to enable or disable the sequencer. A disabled sequencer communicates with I/O devices but does not execute application software. An enabled sequencer operates normally.

Operation:

1. Press the SUPVR key to display the Supervisory mode menu, then press the 1 key, the 0 key, and the ENTER key. The current sequencer status (Sequencer Enabled or Sequencer Disabled) will appear on the screen.

2. If you wish to change the sequencer status press SHIFT and ENTER. The new status will appear.

3. If you do not wish to change the sequencer status, press DELETE.

Print I/O Map

Function:

Produces a sequencer I/O map on a compatible printer. An I/O map lists all input, output, and internal status signals used in the sequencer along with a list of string numbers where each is used.

Operation:

1. Make sure the printer is connected to the terminal, powered on, and in a ready condition.

2. Press the SUPVR key to display the Supervisory mode menu, then press the 1 key twice and the ENTER key. At this point, the screen displays a list of baud rates and asks for a selection as described above under step 4 of "Dump Memory". Enter the number corresponding to the baud rate of the printer you are using (most often 300 baud) and then press the ENTER key to start printing the I/O map. Press DELETE at any time to abort the operation. When the complete I/O map has been printed, the screen displays "Operation Complete" and returns to the Supervisory menu.

4. The output of the printer will be similar to the following example:

```
S0807  0  0068  0070  0069
       1  0071  0078  0072
       2  0072  0079  0078
       3  0078  0079  0072
       4  0079  0156  0146  0143  0131
       5  0073  0149  0130  0074
       6  0075  0146  0131  0076
       7  0127  0076  0199  0187  0172  0158

10900  0  0167  0166  0156  2630  0670  0021

R0900  1  1610
       2  0075  0160

10910  B
```

The I/O map contains a listing of all input (I), output (R) and internal status (S) signals in sequencer memory arranged in numeric order by station, rack, and group. Beside each signal address is a list of the points for that signal (usually 0 through 7) arranged vertically. Next to each point is a list of string numbers where that signal is used. In the above example, internal status signal SO8070 is used in strings 0068, 0070, and 0069.

A value-type signal such as a BCD or count signal will be listed on the printout, but where-used information will not be provided. An example of this is signal I091 OB.

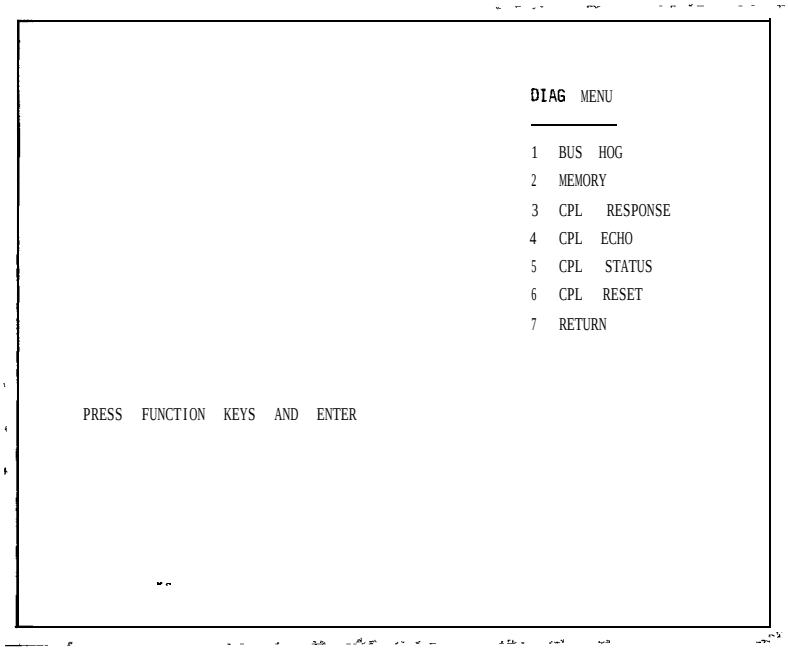
Diagnostics

Function:

The DLC+ Programming Module contains diagnostic tests to help determine whether the sequencer is operating properly and to aid in troubleshooting. There are currently six diagnostics available: a bus hog test, a memory test, and four Control Party Line tests.

Operation:

Press the SUPVR key to display the Supervisory mode menu, then press the 1 key, the 2 key, and the ENTER key. A diagnostic menu will appear on the screen as shown below.

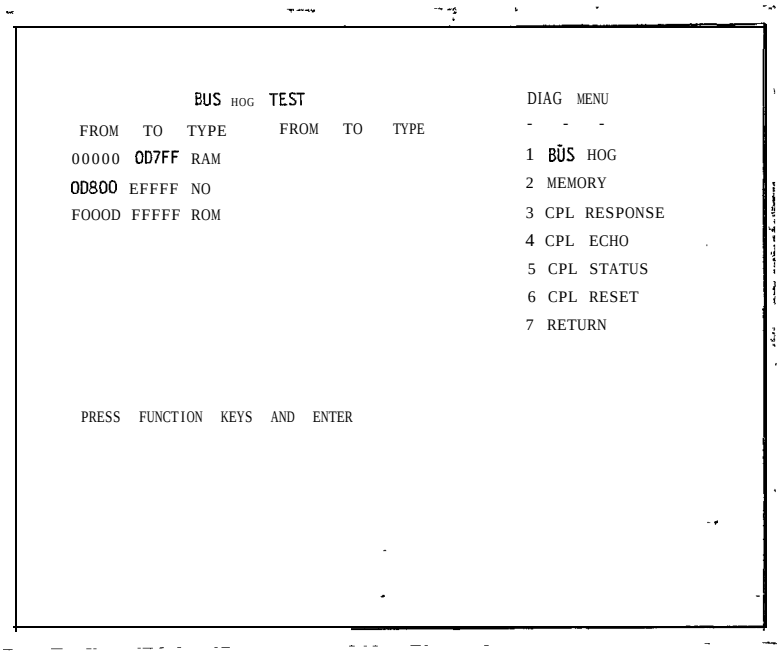


Directions for running each of the diagnostics are included below.

Bus Hog Test

This test scans memory to determine what kind of memory (RAM, ROM, or none> is present in the sequencer. A bus hog will be reflected as a “no memory” condition. Application software in sequencer memory is not damaged by this routine. The test produces a display showing the address range of each type of memory. To operate this diagnostic, follow these instructions:

1. Ensure that the sequencer is off-line. From the diagnostic menu, press the 1 key and then the ENTER key. The Bus Hog test begins running and the screen displays results similar to that shown below.



The screen displays a list of memory address ranges (FROM and TO) in hexadecimal and the type of memory in each address range. In the above example, sequencer memory addresses 00000 through 0D7FF contain RAM memory (this memory may or may not be BRAM memory). Memory addresses OD800 through EFFFF contain no memory, while memory addresses FOOOO through FFFFF contain ROM or PROM memory. This test does not determine whether the memory it finds is defective (i.e., bad RAM will still be flagged as RAM). Since a bus hog condition (caused by a memory board "hogging" the bus) will be signalled as "no memory", it is necessary to know the type of memory in each address range to determine whether a bus hog actually exists.

2. Press DELETE at any time to abort the test and return to the Supervisory menu.

Memory Test

This diagnostic tests a specified memory range to determine whether the RAM memory in that range can be written to and read from. Application software in sequencer memory is not damaged by this routine. The test produces a display showing the status (GOOD or BAD) of adjacent 4 kilobyte sections of memory. To operate this diagnostic, follow these instructions:

1. Ensure that the sequencer is off-line. From the diagnostic menu, press the 2 key and then the ENTER key. At this point the screen asks for a range of memory addresses to be tested. Enter the FROM ADDR and TO ADDR ranges (in hexadecimal), following each entry with the ENTER key. The FROM ADDR value must be smaller than the TO ADDR value. The Memory test begins running and the screen displays results similar to that shown below.

```
                    MEMORY TEST                                  DIAG MENU

 FROM   TO   TYPE      FROM   TO   TYPE
 00000 00FFF GOOD     0BFFF 09FFF GOOD          1  BUS HOG
 00FFF 01FFF GOOD     09FFF 0AFFF GOOD          2  MEMORY
 01FFF 02FFF GOOD     0AFFF 0BFFF GOOD          3  CPL RESPONSE
 0EFFF 03FFF GOOD     0BFFF 0CFFF GOOD          4  CPL ECHO
 03FFF 04FFF GOOD     0CFFF 0DFFF GOOD          5  CPL STATUS
 04FFF 05FFF GOOD     0DFFF 0EFFF GOOD          6  CPL RESET
 05FFF 0CFFF GOOD     0EFFF 0FFFF GOOD          7  RETURN
 0CFFF 07FFF GOOD     0FFFF 10000 GOOD
 07FFF 08FFF GOOD     10000 10FFF BAD
 PRESS FUNCTION KEYS AND ENTER


 LIST FULL, CHANGE ADDRESS RANGE
```

The screen displays a list of memory address ranges (FROM and TO) in hexadecimal and whether the RAM memory in each address range is BAD or GOOD. PROM and ROM memory as well as no memory is flagged as BAD. In the' above example, all of sequencer memory through address 10000 is GOOD, while memory in addresses 10000 through 10FFFF is flagged as BAD. This may indicate that the address range contains PROM or ROM memory, defective RAM memory, or no memory at all. Refer to the results of the Bus Hog test to determine what type of memory is in each address range.

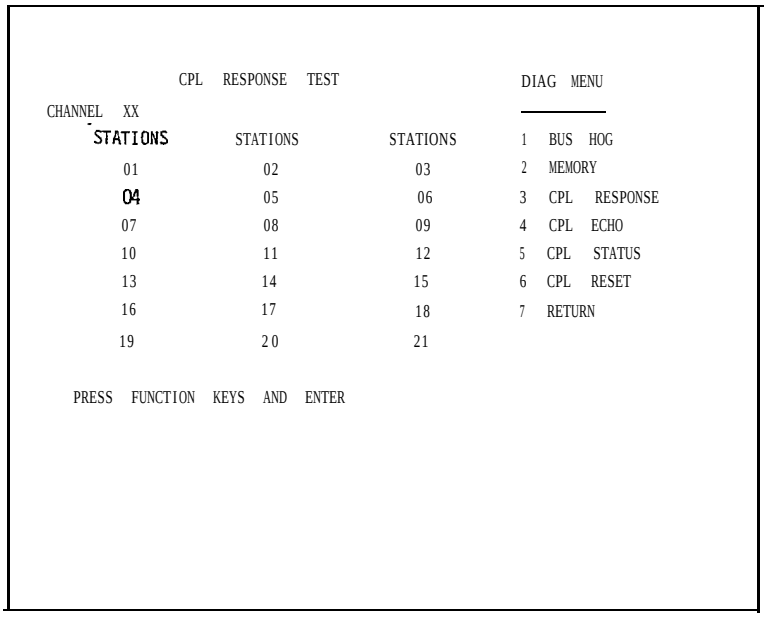
2. If the address range being tested is more than 72 kilobytes (18 screen entries) the message "List Full, Change Address Range" will appear at the bottom of the screen. You must press DELETE and rerun the test using the address range that was not tested the first time.

3. Press DELETE at any time to abort the test and return to the Supervisory menu.

CPL Response Test

This diagnostic helps you identify the stations that exist on a given channel by indicating the stations that respond to a sequencer request. To operate this diagnostic, follow these instructions:

1. Ensure that the sequencer is off-line. From the diagnostic menu, press the 3 key and then the ENTER key. At this point the screen asks for a channel number to be entered. Enter the number of the channel you wish to test, press the SHIFT key, then press the ENTER key. If the screen displays the message "Channel Error Detected," you have entered the number of a channel that does not exist. The diagnostic determines which stations on that channel respond and displays the results as shown below.



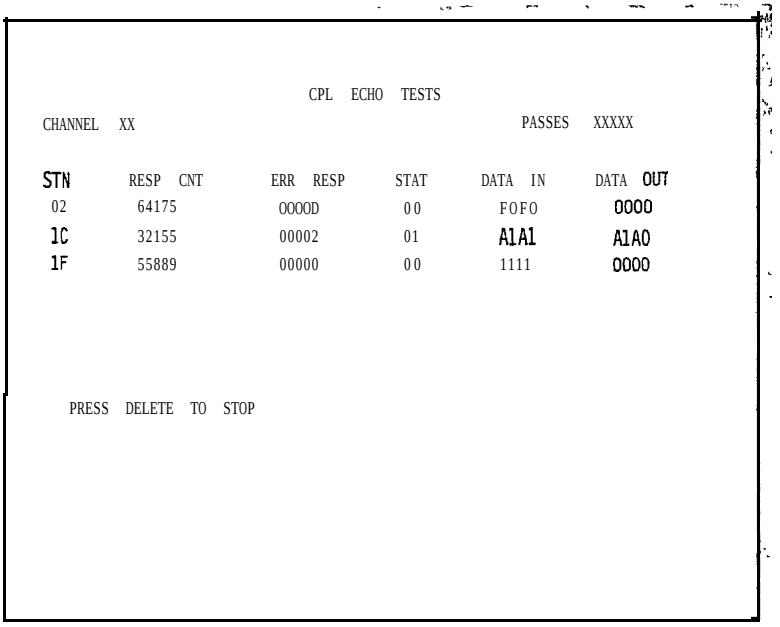
A station number appears for each station that responds to the diagnostic. The diagnostic will look for all stations on a channel whether or not those stations have been configured in the KIB. An existing station which does not respond indicates a problem in the system.

2. At this point you can execute another diagnostic or return to the Supervisory mode menu by pressing the appropriate key (1 through 7) followed by the ENTER key.

CPL Echo Test

This diagnostic helps you verify which of the stations configured in the KIB are responding (echoing data) properly. To operate this diagnostic, follow these instructions:

1. Ensure that the sequencer is off-line. From the diagnostic menu, press the 4 key and then the ENTER key. At this point the screen asks for a channel number to be entered. Enter the number of the channel you wish to test, press the SHIFT key, then press the ENTER key. If the screen displays the message "Channel Error Detected," you have entered the number of a channel that does not exist. The diagnostic determines which stations on that channel are echoing data properly and displays the results as shown below.



A maximum of 10 stations will be displayed on the screen. The diagnostic will only look for stations that have been configured in the KIB. The various fields on the screen are explained below:

PASSES - Shows the number of times (in decimal) the test has been conducted.

STN - Contains the station number (in hexadecimal) of the station being tested.

RESP CNT - Shows the total number (in decimal) of good and bad responses from the station. This field counts to a maximum of 65535 and then resets to 0.

ERR RESP - Shows the number of responses (in decimal) from the station that were in error. This field counts to a maximum of 65535 and then resets to 0.

STAT - Shows the last bad status (in hexadecimal) received from the station.

DATA IN - Shows the last data (in hexadecimal) sent to the station that was echoed in error. The DATA OUT field shows the incorrect data that was echoed from the station.

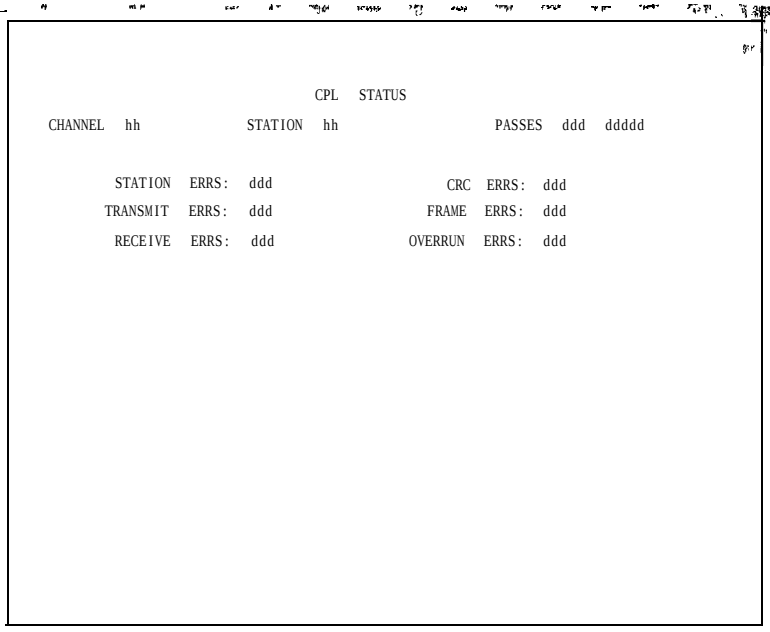
DATA OUT - Shows the last data (in hexadecimal) sent from the station that was echoed in error. The DATA IN field shows the data that was originally sent to the station.

2. This is a continuous test. You can abort the test by pressing the DELETE key, at which time the programming module will display the diagnostic menu and allow you to execute another diagnostic or return to the Supervisory mode menu by pressing the appropriate key (1 through 7) followed by the ENTER key.

CPL Status Test

This diagnostic provides information on the error conditions (status) of a particular station on a channel. To operate this diagnostic, follow these instructions:

1. Ensure that the sequencer is off-line. From the diagnostic menu, press the 5 key and then the ENTER key. At this point the screen asks for a channel number and a station number to be entered. Enter the number of the channel and station you wish to test, press the SHIFT key, then press the ENTER key. If the screen displays the message "Channel Error Detected," you have entered the number of a channel that does not exist. If the screen displays the message "Station Error Detected," you have entered the number of a station that does not exist. The diagnostic determines the status of the station you requested and displays the results as shown below.



The various fields on the screen are explained below:

PASSES - Shows the number of times (in decimal) the test has been conducted. The last five digits count up to 65535, at which time they reset to 0 and increment the counter represented by the first three digits. Since the first three digits can count up to 255, the total counting capacity of the PASSES counter is 255 times 65,535 or 16,711,425 passes.

STATION ERRS: - Shows the number of station errors (in decimal) up to a maximum of 255. Station errors are internal to the slave station.

TRANSMIT ERRS: - Shows the number of transmit errors (in decimal) up to a maximum of 255. Transmit errors occur during transmission of data from the slave station to the master.

RECEIVE ERRS: - Shows the number of receive errors (in decimal) up to a maximum of 255. Receive errors occur during transmission of data from the master to the slave station.

CRC ERRS: - Shows the number of CRC (cyclic redundancy check) errors (in decimal) up to a maximum of 255. A CRC error is a type of receive error that is detected by checking the CRC checksum that is sent along with the data from the master to the slave station.

FRAME ERRS: - Shows the number of frame errors (in decimal) up to a maximum of 255. Frame errors are a type of receive error that is detected when data is garbled during transmission from the master to the slave station.

OVERRUN ERRS: - Shows the number of overrun errors (in decimal) up to a maximum of 255. An overrun error is a type of receive error that is detected when data arrives from the master at a faster rate than the slave can handle.

2. This is a continuous test. You can abort the test by pressing the DELETE key, at which time the programming module will display the diagnostic menu and allow you to execute another diagnostic or return to the Supervisory mode menu by pressing the appropriate key (1 through 7) followed by the ENTER key.

CPL Reset

This is a utility! not a diagnostic, that allows you to reset an I/O channel's status (error counters). To operate this utility, follow these instructions:

1. Ensure that the sequencer is off-line. From the diagnostic menu, press the 6 key and then the ENTER key. At this point the screen asks for a channel number to be entered. Enter the number of the channel whose error counters you wish to reset, press the SHIFT key, then press the ENTER key. If the screen displays the message "Channel Error Detected," you have entered the number of a channel that does not exist. The utility will reset the channel's status and display the message "I/O Status Reset".

2. At this point you can execute another diagnostic or return to the Supervisory mode menu by pressing the appropriate key (1 through 7) followed by the ENTER key.

READ/WRITE Memory

There are 2 diagnostic commands that are purposely not listed on the diagnostic menu. They are:

8) READ/WRITE SEQUENCER MEMORY

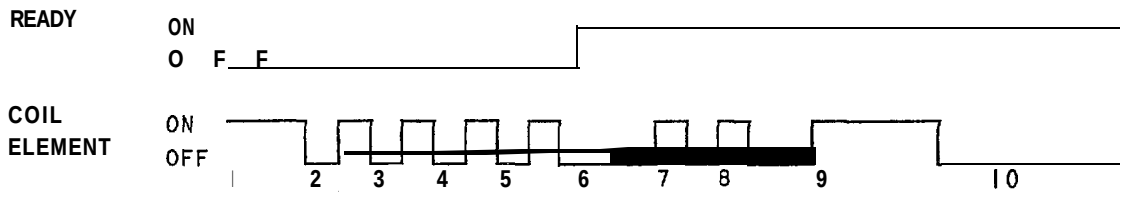
9) READ/WRITE CRT MEMORY

These commands allow inspection and modification of absolute memory locations and should not be used except by trained GE personnel. Modifying absolute memory can cause a system crash and loss of the data base.

CHAPTER 5
VDM AND SEQUENCER POWER UP AND SELFTEST

VDM SELFTEST

The VDM will go through a series of self tests when the power is applied to the module. Each test's success is indicated by a blink of the coil and element light. The ready, element and coil light will blink rapidly if one the tests fail. The following graph indicates the sequence of tests:



1. Power applied to system.
2. 8086 CPU test.
3. HMPC/HMPF RAM memory test.
4. HMPC/HMPF Product code PROM checksum.
5. HMPC/HMPF On board device test.
6. HMPC/HMPF selftest completed.
7. HPMA Product code PROM checksum.
8. HRMB RAM memory test.
9. HVDA screen test.
10. Wait for keyboard interrupt (input) and enter normal processing.

The VDM screen will be blank until the HVDA test has been started. At this time random data patterns will be sent to the screen in both normal and inverse video. After this test completes the following screen is displayed.

```
DLC+ VDM RELEASE 7.3

AUGUST 1 1983

PLEASE PRESS ONE OF THE 'KEYS ON THE KEYBOARD
```

The screen information contains the revision level and release data of the VDM software. Press any key on the keyboard to complete the self tests. The VDM screen will change to the initial screen format described earlier. The VDM is now ready for commands to be entered.

VDM RESET REQUEST FROM THE KEYBOARD

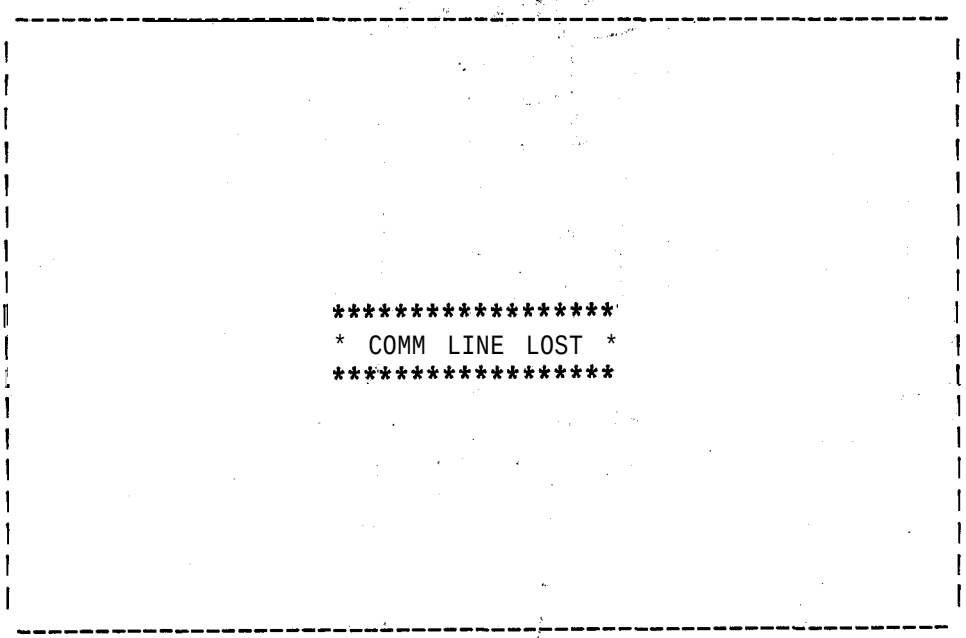
The VDM can be forced to go through the selftest sequence without turning off the power to the VDM module. The VDM will start the selftests at any time if the "F0" function key (bottom left corner) is pressed two times in succession. The following screen is displayed when the command has been accepted.

```
**************
* CRT RESET *
**************
```

The selftest procedure will proceed as described above. The above screen will be displayed until the HVDA test is initiated.

LOSS OF COMMUNICATION WITH SEQUENCER

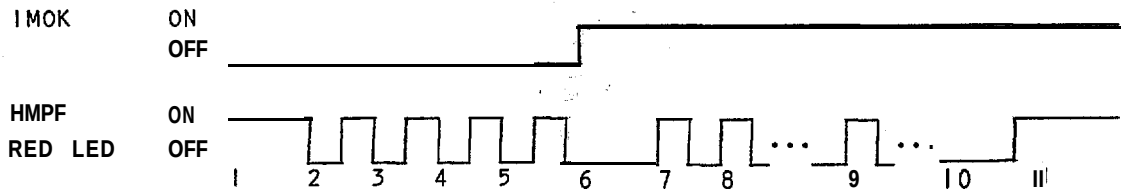
The VDM will reinitialize itself whenever it detects the link between the sequencer and itself is broken. The self tests are not run when this happens, but any operation that was in progress will be aborted and the following screen will be displayed.



The screen will be displayed for several seconds and then the initial screen format will be displayed. The sequencer link should be connected before entering any more commands.

SEQUENCER SELF TEST

The sequencer goes through the same series of selftests as the VDM. The HMPF processor card has a red LED to indicate when a test has completed. The HMPC card does not have the red LED but an error in the self test will be indicated by a blinking amber IMOK light. The blinking pattern for sequencers with the HMPF card is as follows:



1. Power applied to system.
2. 8086 CPU test.
3. HMPC/HMPF RAM memory test.
4. HMPC/HMPF Product code PROM checksum.
5. HMPC/HMPF On board device test.
6. HMPC/HMPF selftest completed.
7. HPMA Product code PROM checksum.
8. HRMB RAM memory test (1 blink for each HRMB board, max 6).
9. HHMA response test (1 blink for each HMHA board, max 4).
10. System initialization and application code validity check
11. Normal processing - On line status of sequencer:

    Off = Off Line

    Fast Blink = On Line

The number of blinks for the self test vary with the memory and IO configuration. Typically the system has 6 HRMB cards and 1 HMHA card, so tests 7 and 8 will be a total of 7 blinks.

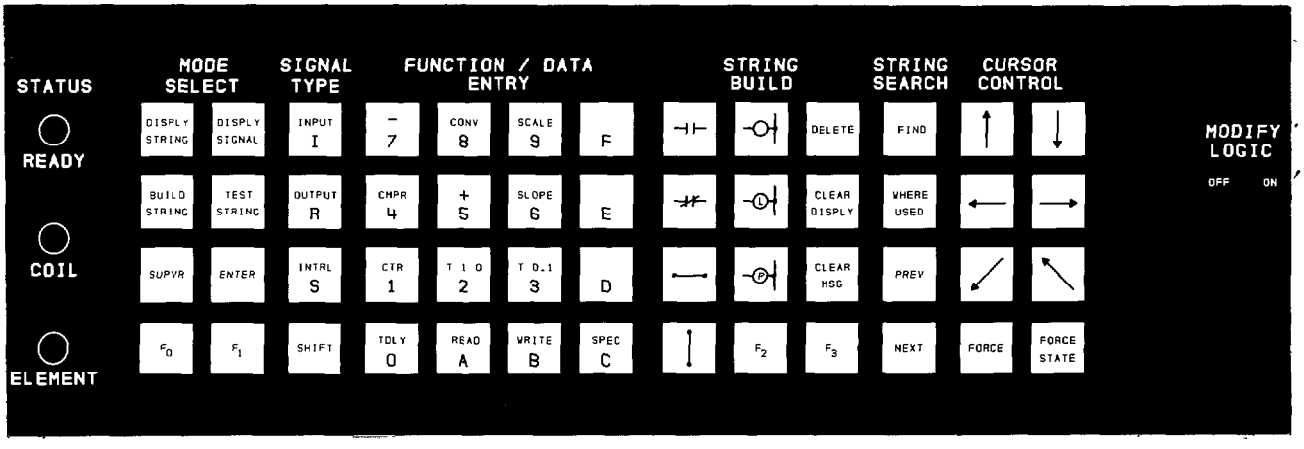
The sequencer will clear the application data base if the system is reset using the test rings on the HMPC/HMPF card during the HRMB RAM memory tests. The sequencer will not clear the data base if power is turned off during this period.

APPENDIX A
PROGRAMMING MODULE KEYBOARD FUNCTIONS

The DLC+ Programming Module keyboard, illustrated below, contains keys, indicators, and a keyswitch used to control all the functions of the module. Some typical module functions are:

- Selecting operating modes
- Entering data
- Building, testing, and placing strings on-line
- Searching for strings
- Controlling the screen's cursor
- Forcing and unforcing signals
- Performing supervisor functions (cassette tape/printer operations, etc.)

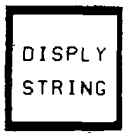
Six types of keys perform all module functions: Mode Select keys, Signal Type keys, Function/Data Entry keys, String Build keys, String Search keys, and Cursor Control keys. These keys are described in this appendix. All of the keys described will automatically repeat their functions if held depressed. This auto-repeat feature, however, is useful only with certain keys such as some of the String Build and Cursor Control keys.



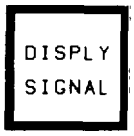
Programming Module Keyboard

MODE SELECT KEYS

These six keys select the operating mode of the Programming Module. A new mode may be selected at any time, except when executing a Supervisory function. If a new mode is selected while a string is under test (in Test String mode), the string remains under test. The ENTER, FO, and F1 keys, although not Mode Select keys, are included in this group.



Puts Programming module into Display String mode, where string logic is displayed in relay ladder diagram form. This mode is for observation rather than programming. The operator chooses the string number to be displayed and can force and unforce contacts and coils. The information displayed includes contact type (normally open, normally closed, etc.), signal addresses for contacts and coils, and contact status (whether open or closed, energized or deenergized, forced or not forced). The String Search keys can be used in this mode to display other strings.



Puts Programming Module into Display Signal mode. The status (energized, deenergized, or forced) of up to eight 16-bit signals is displayed in a tabular form. The operator chooses the signal to be displayed and may force and unforce signals in this mode. Signal status is displayed in both binary and hexadecimal format. Point-type signals (input, output, or internal) are displayed as 8 bit values, while value-type signals (BCD or count) are displayed as 16 bit values.

BUILD
STRING

Puts Programming Module into Build String mode, which allows creation of a new string or modification of an existing string in an off-line memory area. The Function/Data Entry keys and String Build keys are used to create or modify the string. A wide variety of control blocks can be programmed into the string such as time delay strings, latching and pulse relays, timer blocks, slope/offset blocks, etc. The Signal Type keys are used to assign signal types (input, output, internal) to coils and contacts.



This mode can only be entered while in Build String mode. Allows currently displayed string to be tested on-line without any permanent changes being made to the sequencer memory. Errors in the string such as incorrect signal addresses are detected and reported. If the string executes correctly, it can be written to sequencer memory by depressing SHIFT and then ENTER, at which time Test mode is exited. Test mode may be exited without writing the string to memory by pressing TEST STRING again.



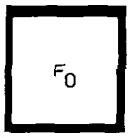
Puts Programming Module into Supervisory mode and displays menu of Supervisory functions. The Supervisory functions, each of which is self-prompting, are:

1 key + ENTER key = OUTPUT STATUS
Enable or disable sequencer outputs
2 key + ENTER key = CLEAR MEMORY
Clear sequencer memory
3 key + ENTER key = RESEQUENCE
Resequence string numbers
4 key + ENTER key = DUMP MEMORY
Dump sequencer memory to cassette tape
5 key + ENTER key = LOAD MEMORY
Load sequencer memory from cassette tape
6 key + ENTER key = COMPARE
Compare data on tape with data in memory
7 key + ENTER key = KIB
Display or define KIB parameters
8 key + ENTER key = PRINT STRING
Print sequencer strings on printer

9 key + ENTER key = ON/OFF LINE
Puts sequencer on-line or off-line
10 keys + ENTER key = SEQ STATUS
Enable or disable sequencer sweep
11 keys + ENTER key = PRINT I/O MAP
Print I/O map on printer
12 keys + ENTER key = DIAGNOSTIC
Display diagnostic menu



Not involved in mode selection. Used primarily in Supervisory and Display Signal modes to execute menu selections or signal addresses. Some critical operations such as clearing memory require pressing SHIFT and then ENTER before the operation will execute. In Test Mode, pressing SHIFT and then ENTER causes the logic string under test to be written to the sequencer's memory, making that string a part of the application program.



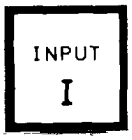
This key will force the VDM into a reset condition. The current operation will be aborted and the VDM self tests will be initiated. This key must be pressed twice in succession before the reset will be activated.



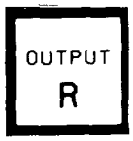
Currently Unused

SIGNAL TYPE KEYS

These four keys specify the type of coil or contact signal address being entered: input, output, or internal. Within a signal address field, one of **these** keys **must** be depressed before entering the numeric signal address. The Shift key, although not a signal type key, is included in this group.



Specifies a signal originating in external hardware such as limit switches, pushbuttons, thumbwheels, and tachometers. When this key is pressed, an I will appear at the cursor location, indicating that the address to follow refers to an Input-type I/O card.



Specifies a signal originating in the sequencer and destined for external hardware such as solenoids, indicators, decimal displays, or speed reference circuits. When this key is pressed, an R will appear at the cursor location, indicating that the address to follow refers to an output-type I/O card.



Specifies a signal used internally by the sequencer. In other words, an S signal is generated within the sequencer and examined by the sequencer. When this key is pressed, an S will appear at the cursor location, indicating that the address to follow refers to an internal coil or contact or to a 16-bit value-type signal.



Not involved in specifying signal type. Used primarily in creating control blocks. Pressing SHIFT before a Function/Data Entry key will cause a control block to be created rather than alphanumeric data to be inserted.

'FUNCTION/DATA ENTRY KEYS

These sixteen keys allow entry of alphanumeric data, selection of Supervisory Mode menu functions, or creation of control blocks. The keys provide alphanumeric data when used alone, or create control blocks if used in conjunction with the SHIFT key. A control block can be created only in Build String mode and only if the message "No Logic Assigned" is visible.



Inserts a numeric 0 at the cursor location when entered by itself. Builds a skeleton time delay block when entered after the SHIFT key.



Inserts a numeric 1 at the cursor location when entered by itself. Builds a skeleton counter block when entered after the SHIFT key.



Inserts a numeric 2 at the cursor location when entered by itself. Builds a skeleton one second timer block when entered after the SHIFT key.



Inserts a numeric 3 at the cursor location when entered by itself. Builds a skeleton one-tenth second timer block when entered after the SHIFT key.



Inserts a numeric 4 at the cursor location when entered by itself. Builds a skeleton compare block when entered after the SHIFT key.



Inserts *a* numeric 5 at the cursor location when entered by itself. Builds a skeleton add block when entered after the SHIFT key.



Inserts a numeric 6 at the cursor location when entered by itself. Builds a skeleton slope/offset block when entered after the SHIFT key.



Inserts a numeric 7 at the cursor location when entered by itself. Builds a skeleton subtract block when entered after the SHIFT key.



Inserts a numeric 8 at the cursor location when entered by itself. Builds a skeleton convert block when entered after the SHIFT key.



Inserts a numeric 9 at the cursor location when entered by itself. Builds a skeleton scale block when entered after the SHIFT key.



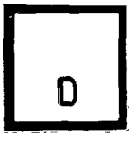
Inserts an alphabetic A at the cursor location when entered by itself. Builds a skeleton read block when entered after the SHIFT key, but this control block is currently not usable.



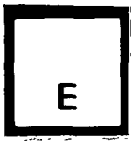
Inserts an alphabetic B at the cursor location when entered by itself. Builds a skeleton write block when entered after the SHIFT key, but this control block is currently not usable.



Inserts an alphabetic C at the cursor location when entered by itself. Builds a skeleton special block when entered after the SHIFT key.



Inserts an alphabetic D at the cursor location when entered by itself. Causes an error when entered after the SHIFT key.



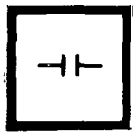
Inserts an alphabetic E at the cursor location when entered by itself. Causes an error when entered after the SHIFT key.



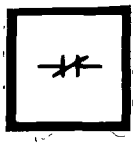
Inserts an alphabetic F at the cursor location when entered by itself. Causes an error when entered after the SHIFT key.

STRING BUILD KEYS

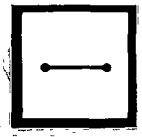
These twelve keys create, modify, and delete skeleton strings and contacts and clear error messages from the screen. Most of the keys are used only in Build String mode. The DELETE and CLEAR MSG keys, however, are used in all modes.



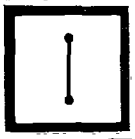
Inserts a normally open contact at the cursor location. If the cursor is on a vertical rather than a horizontal connection, the cursor moves to the next horizontal position without inserting a contact. After inserting the contact, the signal address can be entered by pressing a Signal Type key followed by the rest of the signal address.



Inserts a normally closed contact at the cursor location. If the cursor is on a vertical rather than a horizontal connection, the cursor moves to the next horizontal position without inserting a contact. After inserting the contact, the signal address can be entered by pressing a Signal Type key followed by the rest of the signal address.



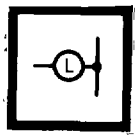
Inserts a horizontal short at the cursor location. If the cursor is on a vertical rather than a horizontal connection, the cursor moves to the next horizontal position without inserting the short.



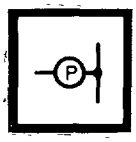
Inserts a vertical short at the cursor location. If the cursor is on a horizontal rather than a vertical connection, the cursor moves to the next vertical position without inserting the short.



Builds a skeleton string (a 5 by 7 matrix of normally open contacts, vertical shorts, and a standard coil as output). This key works only in Build String mode and only if the message "No Logic Assigned" is visible on the screen.



Builds a skeleton string with a latch relay output rather than a standard coil. This keys works only in Build String mode and only if the message "No Logic Assigned" is visible on the screen.



Builds a skeleton string with a pulse relay output rather than a standard coil. This key works only in Build String mode and only if the message "No Logic Assigned" is visible on the screen.



Deletes displayed data, depending on mode. In Build String mode, pressing this key deletes the contact at the current cursor position; SHIFT and then DELETE deletes from sequencer memory the string whose string number appears in the String Number field. In Supervisory mode, pressing this key displays the previous Supervisory function menu. In Display Signal mode, pressing this key deletes the signal address at the current cursor position.



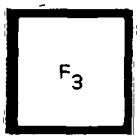
In Build String mode, erases data on screen from cursor position to the end of screen.



Clears error messages in all modes.



Currently unused.



Currently unused.

STRING SEARCH KEYS

These four keys are used to find specific strings in Display String, or Display Signal modes. The strings may be retrieved either by string number or by "Where Used" function. The "Where Used" function finds strings where specific coils or contacts are used.



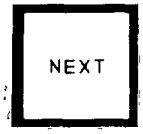
Finds and displays the string whose string number has been entered in the Where Used/Find field. Entries in the Where Used/Find field can be made manually or by using the Where Used key.



This key is used two different ways in both Display Signal and Display String modes:
(1) If the cursor is located on a **contact,** pressing the WHERE USED key causes the Where Used/Find field to display the string number of the string containing the related coil. Pressing the FIND key at this point displays the complete string.
(2) If the cursor is located on a **coil,** pressing the WHERE USED key causes the Where Used/Find field to display the string number of the string containing a related contact. Repeated depressions of the key will display the string numbers of other strings where contacts of that coil are located. Pressing the FIND key at any point will display the complete string whose string number is in the Where Used/Find field.



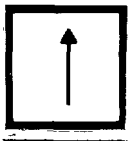
Used only in Display String mode. Finds and displays the string whose string number occurs prior to the currently displayed string.



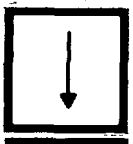
Used only in Display String mode. Finds and displays the string whose string number occurs just after the currently displayed string.

CURSOR CONTROL KEYS

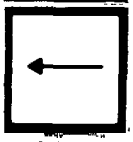
Control movement of the cursor and signal forcing and unforcing.



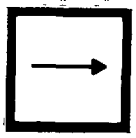
Each depression of this key moves the cursor to the contact or connection directly **above** the current cursor location. If the cursor is on the top line when this key is pressed, the cursor will automatically jump to the bottom line.



Each depression of this key moves the cursor to the contact or connection directly **below** the current cursor location. If the cursor is on the bottom line when this key is pressed, the cursor will automatically jump to the top line.



Each depression of this key moves the cursor to the contact of connection immediately to **the left** of the current cursor location. If the cursor is in the leftmost screen position when this key is pressed, the cursor will automatically jump to the rightmost screen position on that same line.



Each depression of this key moves the cursor to the contact of connection immediately to **the right** of the current cursor location. If the cursor is in the rightmost screen position when this key is pressed, the cursor will automatically jump to the leftmost screen position on that same line.



Each depression of this key moves the cursor to the leftmost position on the next line; in effect, this key acts like a carriage return key. If the cursor is on the bottom line and this key is pressed, nothing happens.



Each depression of this key moves the cursor alternately to the Where Used/Find field or the upper left contact position; in effect, this key acts like a home key.



Used in Display String and Display Signal modes. Places the contact, coil, or signal at the cursor location under force control. The FORCE STATE key is used to energize or deenergize the coil, force a contact closed or open, or change the state of a signal. Pressing SHIFT and then FORCE causes all forced signals to be unforced.



Used in Display String and Display Signal modes. Once a contact, coil, or signal is placed under force control, this key alternately energizes and deenergizes the coil, contact, or signal.

STATUS LIGHTS

Indicate operational status of the Programming Terminal.

READY When on, indicates that the Programming Terminal is powered on and ready for operation.

COIL Monitors the state **of** the displayed string's coil when the programming module is in Display String mode. When the coil is energized, the LED will be on. When the coil is de-energized, the LED will be off. This LED displays coil status in near-real time (after a two-second delay) and will show every state change. The screen image of coils may not show every state change.

ELEMENT Monitors the state of the contact selected by the cursor when the programming module is s in Display String mode. When the contact is closed, the LED will be on. When the contact is open, the LED will be off. If the cursor is not on a contact, the LED will remain off. This LED displays contact status in near-real time (after a two-second delay) and will show every state change. The screen image of contacts may not show every state change.

MODIFY LOGIC KEYSWITCH

Sequencer logic (memory) may be modified when this switch is ON; sequencer memory is protected when this switch is OFF. Memory modifications including entering, testing, deleting, and modifying strings. This switch does not have to be ON to build new strings, only to test or enter them into memory.

UNUSED KEYS

Keys F1 , F2, and F3 are currently undefined.

NOTES

APPENDIX B
SELF-TEST ERROR CODES

Both the DLC+ Sequencer and the DLC+ Programming Module perform automatic self-tests during initialization. Should a DLC+ Sequencer self-test fail, the IMOK light on the affected board will go out. An error message is displayed on the programming module screen if the module is attached to the sequencer and is in Build String mode. Should a DLC + Programming Module self-test fail, the IMOK light on the affected board will go out, but an error message is not displayed unless the module is performing its power-on self-test. In either case the error message reads SELF TEST ERROR NN, where NN in one of the error codes listed below.

If the error is in the programming module, the error message appears in the center of the screen. If the error is in the sequencer, the error message appears at the bottom left of the screen along with an error explanation.

ERROR CODE	MEANING
00	HMPC RAM failure
01	HMPC product software checksum error
03	HMPC USART failure
04	HMPC interrupt controller failure
05	HMPC timer failure
06	HMPC bus hog failure
11	HPMA product software checksum error
21	HRMB/HPMA application software checksum error
22	HRMB/HPMA application software initialization failure - strings cleared from memory
30	HMHA RAM failure (Remote I/O>
31	HMHA self-test failure (Remote I/O)
32	HMHA configuration error (Remote I/O)
40	HVDA RAM failure
50	Application software database consistency check
90	No memory available to complete initialization
91	Out of I/O memory
92	Memory configuration error - RAM data area is not the same as when the sytem was generated.
0C	Special function checksum error
OD	Special function invalid location address

NOTES

APPENDIX C
PROGRAMMING MODULE GENERAL AND ERROR MESSAGES

During programming module operation, descriptive messages may appear in the lower left part of the screen. These messages are of two types: general messages and error messages. General messages indicate that a programming module operation is proceeding normally, while error messages indicate that something is wrong — either an operator mistake or a system problem. Below is an alphabetical list of possible messages, the type of message (general or error), in what mode the message typically occurs, and an explanation of the message.

MESSAGE	TYPE	MODE	EXPLANATION
ADDR OUT OF BOUNDS	Error	All	There is no KIB rack entry for the signal address you have entered.
BAD CONTROLLER MESSAGE	Error	All	The programming module is unable to decipher a message from the sequencer. May indicate system problems.
BAD SIGNAL NNNNNN	Error	Build String	The string you are trying to test has a signal (identified as NNNNNN) that does not reference an existing KIB rack entry.
CHANNEL ERROR DETECTED	Error	Supvr.	A CPL channel error has been detected during a CPL diagnostic test.
COMMAND COMPLETE	General	All	The string under test has been entered into sequencer memory.
CONTROLLER MEMORY EXCEEDED	Error	Build, String	The available application software memory is full. No more strings can be entered into sequencer memory.
CURRENT GROUP IS > 2 FOR HIOB	Error	Supvr.	There can be only 2 groups assigned to a particular HIOB station.
DATA BASE LOST AT NNNN	Error	All	An invalid data base (application code) has been detected while the sequencer was executing at address NNNN. Make a record of the address NNNN and contact General Electric Company for assistance.
DUPLICATE COIL	Error	Build String	A string with this coil signal address already exists in sequencer memory.

MESSAGE	TYPE	MODE	EXPLANATION
DUPLICATE RACK DEFINITION	Error	Supvr.	You have attempted to create a rack entry that duplicates a rack entry already present.
END OF RACK TABLE	General	Supvr.	The KIB rack entry displayed is the last one in the system.
EXEC 00	Error	All	System problem - contact General Electric Company.
EXEC 01	Error	All	System problem - contact General Electric Company.
FORWARD REFERENCE TABLE FULL	Error	Build String	An internal sequencer table is full and no more strings can be entered into sequencer memory.
FROM ADDR IS > TO ADDR	Error	Supvr.	The "From" address you have entered must be smaller than the "To" address.
HMHA ADDRESS INVALID	Error	Supvr.	The HMHA address specified as *a* system parameter must be a multiple of hexadecimal 800.
I/O FAILURE - RETRYING	Error	Display String	Communication between the programming module and sequencer has been interrupted during string status update in Display String mode. The programming module is attempting to retransmit the status request.
I/O FAILU:RE - TRY AGAIN	Error	All	The programming module is unable to communicate with the sequencer. You should attempt the operation again.
ILLEGAL CONTROLLER MESSAGE	Error	All	The programming module is unable to decipher a message from the sequencer. May indicate system problems.
ILLEGAL ENTRY	Error	All	You have pressed a key or attempted a function which cannot be used in the current mode.
ILLEGAL STRING	Error	Build String	You are attempting to test a string whose string number or coil signal address already exists in another string.
INVALID COMMAND	Error	All	The sequencer is unable to decipher a message from the programming module.

MESSAGE	TYPE	MODE	EXPLANATION
IO FAILURE	Error	All	The programming module is unable to communicate with the sequencer or you may be trying to force a signal that does not exist. May indicate an unplugged signal cable, a sequencer without power, or a system problem.
KEY SWITCH NOT ENABLED	Error	All	You are attempting to modify sequencer memory with the MODIFY LOGIC keyswitch in the OFF position. !
LED MONITOR: CRT ERROR	Error	Display String	The programming module has sent bad data to the sequencer. The COIL and ELEMENT LEDs on the keyboard will not reflect proper coil and element status until Display String mode is re-entered.
LED MONITOR: SYCHRONIZATION LOST	Error	Display String	Synchronization has been lost between the programming module and sequencer. This message should disappear once synchronization has been re-established.
LED MONITOR: TRYING AGAIN	Error	Display String	Programming module communication buffers are full. To reset this error, exit Display String mode. If this message appears frequently, contact General Electric Company.
LIST FULL, CHANGE ADDR RANGE	General	Supvr.	The hardware diagnostic has filled the screen with data. To complete the test, change the address range and perform the diagnostic again.
NO ENTRIES IN RACK TABLE	General	Supvr.	No rack entries have been made in the KIB rack table.
NOT IN OFF LINE MODE	Error	Supvr.	You are attempting to perform a KIB function without taking the sequencer off-line.
ON-LINE INITIALIZATION FAILURE	Error	Supvr.	This failure occurs while trying to bring the sequencer on-line. It can be caused by a hardware initialization failure or incompatibilities between product software and hardware. If this error message appears, press BUILD STRING and note the self-test error code that appears on the screen. Appendix B will tell you what the self-test error code means.

MESSAGE	TYPE	MODE	EXPLANATION
OPERATION COMPLETE	General	All	The operation you requested has been completed.
OPERATION STOPPED - INCOMPLETE	Error	All	The current tape operation (read, write, or compare) has been stopped due to a sequencer problem.
PRESS SHIFT AND ENTER	General	All	You must press SHIFT and then ENTER to complete the operation.
PRINTER FAILURE	Error	Supvr.	The printer is not responding to commands from the programming module. May indicate an unplugged signal cable, a printer without power, or a system problem.
RACK DEFINITION TABLE OVERFLOW	Error	Supvr.	There cannot be more than 32 rack entries in the KIB table.
RECEIVE FAILURE	Error	All	The programming module is unable to receive a message that the sequencer is attemping to send.
SEARCHING FOR BLOCK	General	Build or Display String	The sequencer is attempting to retrieve a string from sequencer memory. This message should appear only momentarily.
SEQUENCER NOT OFF LINE	Error	Supvr.	The sequencer is on-line while you are attempting to perform a function that requires the sequencer to be off-line.
SIGNAL INVALID	Error	Display Signal	The signal you have requested to be displayed does not have a corresponding KIB rack en try.
SIGNAL NOT A POINT	Error	Display Signal	You are trying to force a signal that has not been specified as a point in the KIB rack entry table.
SIGNAL NOT DISPLAYED	Error	Display Signal	You are attempting to force a signal that is not currently displayed.
SIGNAL NOT DISPLAYED OR FORCED	Error	Display Signal	You are trying to change the state of a signal that is not currently displayed or one that is not under force control.
STATION ERROR DETECTED	Error	Supvr.	A CPL station error has been detected during a CPL diagnostic test.
STRING ENTERED	General	Build String	A string has been tested and written into sequencer memory, replacing the old version (if any).

MESSAGE	TYPE	MODE	EXPLANATION
STRING NOT IN TEST	Error	Build String	You have attempted to enter a string into sequencer memory or remove a string from test state while that string is not under test.
SWEEP TIME NOT A MULTIPLE OF 10 MSEC	Error	Supvr.	The sweep time parameter must be a multiple of 10 milliseconds.
SYSTEM BUSY - TRY AGAIN	Error	All	The programming module was busy when you made a request. Repeat the request.
TAPE FAILURE	Error	Supvr.	Programming module is unable to properly read from or write to the cassette tape. May indicate no data received from the tape drive, the data received from the tape drive is invalid, the tape is write protected, or no tape has been inserted into the tape drive.
TRANSMIT FAILURE	Error	All	The programming module is unable to send a message to the sequencer.
TRYING TO DELETE RACK WITH REFERENCES	Error	Supvr.	You are attempting to delete a KIB rack entry that is referenced by strings in sequencer memory.
TRYING TO MODIFY MAX GROUP	Error	Supvr.	You are attempting to modify the maximum group value in a KIB rack entry. The only way to do this is to delete all signals that reference the rack entry, delete the rack entry, then recreate the rack entry with a new maximum group value.
UNDEFINED SIGNAL	Error	Display Signal	A rack entry does not exist for the signal you wish displayed.

NOTES

APPENDIX D
PRINTED WIRING BOARD DESCRIPTIONS

This appendix contains brief descriptions of the most common DLC+ Sequencer printed wiring boards. These boards are of two types: sequencer boards and I/O boards. Sequencer boards include microprocessor, memory, communications, and other boards involved with the logical operation of the sequencer. I/O boards are used for signal conditioning of input and output signals. The boards are listed in alphabetical order under each type. Daughter boards are listed directly after their associated mother board. Some boards are considered "modules", which means that they are general purpose boards containing PROM chips that give the board its particular function. Only the highlights of each board are presented. For more detail, consult the Directo-Matic @Plus backsheets for the particular board.

Only those Directo-Matic® Plus boards currently associated with the DLC+ Sequencer module are listed here. New boards may have been added since this manual was written. Check with your General Electric Company sales representative for up-to-date information.

SEQUENCER BOARDS

HACA (Address Customizing Board) Consists of wire wrap pins for customized addressing for up to 18 I/O printed wiring boards in an electronics module (TEM, SEM, or MEM). Addresses within the module can be specified by wire-wrapping appropriate pins together. Berg jumpers can be configured to define sense, freeze, reset, and I/O or slave mode logic for the module.

HACB (Address Customizing Board) A 5 3/4" board which is housed in a card box in the horizontal I/O stick (DS3820HIO). The board contains wire wrap pins for customized addressing for up to 8 I/O printed wiring boards in one field termination module (horizontal stick). Addresses within the module can be specified by wire-wrapping appropriate pins together. Berg jumpers can be configured to define sense, freeze, reset, and I/O or slave mode logic for the module.

DACB (HACB Daughter Board) Enables HACB board to provide customized addressing to 8 I/O printed wiring boards in a VTDD.

HBDA (Buffer Decoder) Interfaces address, data, and control signals between the slave controller and I/O boards in a CPL slave station. Seven logic inputs and outputs can be accessed from the data bus. The HBDA is strapped into one of eight address ranges and can handle up to 16 work addresses in the range.

HCMA (Dual Communications Interface) An 8085 microprocessor-based dual channel serial communication subsystem. It provides an interface to two RS232 full or half duplex serial communication channels. The channels can be programmed for synchronous or asynchronous baud rates up to 19.2 K bits/sec. It accepts either a 2K or 4K byte EPROM to program communication protocol.

HIMA (CPL Isolation Board) Provides optical isolation for CPL master and slave stations. The HIMA board will drive up to 4000 feet of twisted pair wire and is used in the LIMA module.

HIOB (Siltron™ CPL Slave I/O) Interfaces the SILTRON™ drive to the Control Party Line as a CPL slave station.

HKIB (Keyboard Input) A single-chip microcomputer board programmed to monitor a switch-matrix (57 contacts) at 10 ms. intervals. It encodes into 8 bits parallel and generates a 5 microsecond strobe. It interfaces and decodes the keyboard of the DLC + Programming Module.

HLSC (Link Slave Control) A slave device used for information exchange on a serial data link.

HLSD (Link Slave Control) A revised edition of the HLSB that replaces it totally in function. It is used for information exchange on an eight bit parallel data bus, including interrupt generation and handling.

HLXA, HLXB (CSH Modulation-Demodulation) As part of a CSH slave station, these cards provide modulation and demodulation for interface to the Control Signal Highway. They are used with various other card clusters to implement CSH master, slave and test functions.

HMHA (Master Highway Controller) An 8085 microprocessor-based dual channel serial communication subsystem. It has two SDLC/RS422 party line serial channels that can be programmed for baud rates up to 48K bits/sec, synchronous. It interfaces to the L-Bus via a 2K byte dual port RAM. It accepts either a 2K or 4K byte EPROM to program communication protocol.

HMPC (Microprocessor) The heart of the DLC+ Sequencer, consisting of sequencer product software (in PROM) and a complete microcomputer including 4K bytes of RAM memory and provision for 16K bytes of PROM memory, plus an RS-232 communications interface. The board communicates with other boards (memory, I/O, and communication) via the L-bus. The HMPC board, with different product software, is also used in the DLC + Programming Module.

HMPF (Microprocessor) Revision of the HMPC. It includes 8K of RAM and either 16K or 32K bytes of PROM memory. It also has a red LED light on the edge of the card that can be used for startup diagnostics. See HMPC for more details.

HPBA, HPBC (Parallel Line Buffer) Interfaces address, data, and control signals between the CPL slave module and I/O boards. Has two modes of operation: (1) Buffer-Drive mode passes one-for-one all of the F-level signals to the R-level, making a new F-level driving source; (2) Buffer decode mode provides full decoding for up to 32 word addresses. HPBA also checks all of the R-level signals for shorts to P5 or DCOM.

HPMA (32K/64K PROM Board) This board contains up to 64K bytes of Programmable Read Only Memory (PROM), which may contain product or application software.

HRMB (8K Bytes Battery-Backed RAM (BRAM) Memory) Provides 8K bytes of battery-backed CMOS RAM memory for storage of application software. The HRMB board is dual-ported to both the E-bus and L-bus. In case of power failure or removal of the board from an electronics module, the board will retain memory data for up to 8 days (depending on battery charge and temperature). The on-board NiCad battery is automatically charged whenever the DLC+ Sequencer is operating.

HSHA (CPL Slave Highway) Operates as a CPL slave station when connected to a CPL master station via twisted pair wires. Interfaces I/O boards to the Control Party Line.

DSHA (Optional HSHA Daughter Board) Provides LEDs, thumbwheels, and toggle switches for monitoring data flow through the HSHA mother board and performing off-line testing of any I/O board connected to the HSHA.

HVDA (Video Display Interface) Provides an interface between a CRT monitor and the HMPC board and L-bus. This board provides both TTL and composite video signals.

HXPB (HMPC Expander Board) An expander board for the HMPC microprocessor board which provides the following L-bus functions: 4K RAM memory, 4 PROM sockets, digital I/O via front edge connector, 8 interrupts, counters, and a programmable stall timer.

LIMA (CPL Isolation Module) Consists of an HIMA board and a standard telephone jack in a special electronics module for panel mounting and termination of signal cables.

PCLA (CPL Master Station Module) This module consists of an HMHA board plus a PROM chip that enables the board to act as a CPL master station. The PCLA interfaces remote I/O devices via HSHA or HIOB boards to the DLC+ Sequencer over dual twisted pair transmission lines. Communication can be point-to-point or multipoint, and either non-isolated (for distances less than 30 feet) or isolated via the LIMA module (for distances up to 4000 feet).

PRBA (CPL Remote Buffer Module) This module provides a buffer and interface between a host microprocessor and the CPL master station (PCLA). This enables the host microprocessor to communicate with other processors (which look like CPL slaves) over the Control Party Line.

I/O BOARDS

HLIA (Logic (+5VDC) Input) Provides an interface between 16 logic inputs and an S-bit data bus. The 16 inputs are filtered for noise immunity Sixteen LEDs display the status of the inputs. Bus access to the high and low bytes and the high impedance state are controlled by other digital inputs. Accepts 16 logic level (+ 5VDC) signals as non-isolated inputs to the DLC + Sequencer.

HLIC (Logic (+5VDC) Input) Accepts 8 logic level (+5VDC) signals as isolated inputs to the DLC+ Sequencer.

HLOA (Logic (+5VDC) Ouput) Provides buffering, parity check, and LED indication of 16 logic non-isolated inputs from the DLC+ Sequencer. The outputs may be enabled (from the high impedance state) by address decoding logic. The outputs are available both on the backplane and through a front edge connector.

HLOB (Logic (+ 5VDC) Output) Provides 8 logic level (+ 5VDC) signals as non-isolated outputs from the DLC + Sequencer.

HRRB (Reed Relay Output) The card is used to provide 8 normally open reed relay contacts as isolated outputs from the DLC + Sequencer.

HSCA (Signal Conditioning (115VAC, +105VDC) A 115 VAC or 105 VDC isolated signal conditioning card consisting of eight high level optically isolated input circuits, logic output drivers, drive level interface, and a bus isolation circuit. The eight inputs are available after conditioning as logic output signals for direct use or can be read via the drive level interface. Accepts 8 high level signals (11 5VAC or + 105VDC) as isolated inputs to the DLC+ Sequencer.

HSCB (Signal Conditioning (+ 125VDC)) Accepts 8 high level signals (+125VDC) as isolated inputs to the DLC + Sequencer.

HSCC **(Signal Conditioning (+28VDC))** Accepts 16 high level signals (+28VDC) as non-isolated inputs to the DLC + Sequencer.

HSCD (Signal Conditioning (+ 105VDC)) A 105 VDC non-isolated signal conditioning card consisting of sixteen filtered 105V inputs, logic output drivers, drive level interface, and a bus isolation circuit. The input signals are available after conditioning as logic output signals for direct use or can be read via the drive level interface. Used as non-isolated inputs to the DLC+ Sequencer.

HSCE (Signal Conditioning (+ 105VDC)) Accepts 8 high level signals (+ 105VDC) as non-isolated inputs to the DLC+ Sequencer.

HSCF (Signal Conditioning (+28VDC)) Accepts 8 high level signals (+28VDC) as non-isolated inputs to the DLC + Sequencer.

HSDA (Solenoid Driver) Provides 4 fused 115/230VAC solenoid driver signals as isolated outputs from the DLC+ Sequencer.

HSDC (Solenoid Driver) Provides 4 non-fused 115/230VAC solenoid driver signals as isolated outputs from the DLC+ Sequencer.

HSDD (Solenoid Driver) Provides 8 fused 115/230VAC solenoid driver signals as isolated outputs from the DLC+ Sequencer. Includes blown fuse indicators. Fuses may be changed without removing the card from the electronics module.

NADA (Analog to Digital Converter) Provides 12 bit analog to digital conversion for one of eight differential input channels (+ 12.5VDC full scale) selected by the HMPC board.

NDAC (Digital to Analog Converter) A 12-bit binary bipolar analog-to-digital converter (11 bits data plus sign) on a single size board. Includes an 8 channel static analog scanner to accept up to eight ± 10 volt analog signals. Requires externally supplied clock (accepts 2.045 MHz normally from HMPA) . Converted data can be ready 10 microseconds after providing address of point to be converted. Converted output is available externally on buffered data bus.

APPENDIX E
TAPE DRIVE AN-D PRINTER I/O CABLE PIN DEFINITIONS

This appendix provides I/O cable pin definitions for tape drives and printers used in conjunction with the DLC+ Programming Module. This information is provided for customers who may wish to connect their own tape drive or printer to the programming module.

Both the cassette tape drive and printer are connected to the DLC+ Programming Module by an RS-232-compatible I/O cable. The cable terminates in a 25 pin female connector which plugs into a male jack on the programming module HVDA board. The pin definitions of the HVDA connector (configured as DTE - Data Terminal Equipment) are listed below:

PIN	PURPOSE
1	Chassis ground
2	Transmit data (from HVDA)
3	Receive data (to HVDA)
4	Request to send (from HVDA)
5	Clear to send (to HVDA)
6	Data set ready (to HVDA)
7	Signal ground
8-19	No connection
20	Data terminal ready (from HVDA)
21-25	No connection

NOTES

APPENDIX F
GLOSSARY

Add/subtract block
A control block that, when enabled, adds or subtracts two values (A and B) and places the result into a third value (C). A and B can be input, output, or internal status signals, or constants. C can be an output or internal status signal or a constant. If the result of the addition or subtraction is greater than decimal 32,767 (overflow) or less than decimal -32,768 (underflow), the coil is energized.

Application software
Customer-specific software resident within the DLC + Sequencer (either in PROM or BRAM) used to control the customer's process. Relay ladder diagrams displayed on the programming module are a visible representation of application software.

Backplane
Signifies buses, connectors, or other components located to the rear of an electronics module. PWBs plug into the backplane of an electronics module.

BCD signal
Consists of four Binary Coded Decimal digits addressed as two groups (16 bits). BCD signals are used within the DLC+ Sequencer to handle inputs from devices such as thumbwheel switches and outputs to devices such as LED displays.

Bit
Binary digit. A unit of digital information represented as either 0 or 1, corresponding to the opening or closing of a switch. The DLC + Sequencer stores and manipulates data as series of bits.

Block
See Control Block.

Board
See Printed Wiring Board.

BRAM (Battery-backed Random Access Memory)
Semiconductor read-write memory within the DLC+ Sequencer used in cases where frequent application program changes are anticipated. In case of power failure, the memory is kept powered-up by a board-mounted battery. BRAM memory is reprogrammable, while PROM memory is not.

Build String mode
A mode in the DLC+ Programming Module used for creating and modifying strings in an off-line memory area. When building a string, it is necessary to: 1) create a skeleton string or control block, 2) assign signal addresses to contacts and coils, and 3) assign signal addresses or numerical values (constants) to programmable fields within control blocks (if necessary). A created or modified string must be tested before being permanently entered into sequencer memory.

Card rack

A mechanical assembly for holding and interconnecting DLC + Sequencer PWBs.

CBL (Control Block Language)

A General Electric control programming language where traditional control functions, such as integration, are treated as functional blocks. The DLC + Sequencer control blocks are a subset of the full language.

Channel

A port through which a specified number of I/O signals is transferred to and from the DLC + Sequencer. Channels are configured as part of the KIB. The current implementation of the DLC+ Sequencer supports sixteen channels numbered 0 through 15: channel 0 is reserved for module I/O, channel 1 is reserved for Local I/O, channels 2 through 9 are reserved for HMHA (Control Party Line) I/O, channels 10 and 11 are reserved for Control Signal Highway (E-bus) I/O, and channels 12 through 15 are reserved for HMHA slave I/O. Each HMHA board can handle two channels.

Coil

A "software relay" located in a sequencer string. When a current path exists from the left bus to the coil, the coil will energize. The coil can also be energized by a control block (for instance, when a counter reaches a preset count or an add/subtract operation overflows).

Compare block

A control block that, when enabled, compares two values (A and B) according to a specified relationship (equal, not equal, greater than, etc.). The coil is energized when the relationship is true and deenergized when the relationship is false. A and B can be input, output, or internal status signals, or constants.

Constant

A numerical value inserted into the "V*****" field of a control block.

Contact

The DLC+ Sequencer software representation of an relay or switch contact. Contacts may be normally open or normally closed and are activated by the signal whose signal address appears above the contact. Contacts may be activated by inputs (such as external switches) or internal status signals (such as from a coil).

Control Block

Any of the control functions available via the Programming Module's Function/Data Entry keys. These functions can be programmed into the sequencer as strings; they include Add/Subtract, Time Delay, Counter, Compare, Scale, and many other control blocks. Each control block is described separately in this glossary.

Control Party Line (CPL)

A multi-drop communications link connecting a DLC+ Sequencer with local or remote input/output boards.

Control Signal Highway (CSH)

A high speed communications link connecting a Programmable Master Controller (PMC) or other high level computer to local or remote input/output modules. A DLC + Sequencer can be tied into the CSH as such an I/O module.

Convert block

A control block that, when enabled, moves or converts data within the DLC + Sequencer. Depending on the values programmed, this function can move a BCD input to a BCD output, move a count value to another count value, convert a count value to a BCD value, or convert a BCD value to a count value.

Count signal

Consists of an integer value in the range -32,768 to $32,767 addressed as two groups (16 bits). Count signals are used within the **DLC+** Sequencer to handle counting and timing functions.

Counter block

A control block that, when enabled, acts as a presettable up or down counter. When the count value is greater than or equal to the final value, the coil is energized.

Cursor

A blinking, movable, horizontal indicator on the DLC + Programming Module screen that shows where data entered from the keyboard will appear

Display String mode

A mode in the DLC+ Programming Module used for displaying on-line strings, including signal states and where used information. The strings are accessed by string number and cannot be modified in this mode, though signals can be forced and unforced.

Display Signal mode

A mode in the DLC + Programming Module used for displaying input, output, and internal status signal states in a tabular form. Signals may be forced and unforced in this mode.

E-bus (Express bus)

Frontplane data bus used in certain cases for communications between a DLC+ Sequencer processor and other modules. This bus is typically used for communications between the HMPC board and Control Signal Highway.

Electronics module

A module consisting of a card cage, specific printed wiring boards, and, where appropriate, product software. There are three types of electronics modules: the Tiny Electronics Module (TEM), the Small Electronics Module (SEM), and the Medium Electronics Module (MEM).

F-bus (Full bus)

DLC + Sequencer backplane I/O bus containing all address, data, and control lines. Used infrequently.

Field termination module

An assembly containing up to 30 I/O boards in four rows. The design provides wire troughs between I/O boards for ease in running and terminating I/O cables in the field.

Firmware

Computer software that exists in non-volatile, read-only memory, typically in PROM. Firmware, unlike software, is not generally user-programmable. DLC+ Sequencer product software is actually firmware.

Forcing

The ability to force an on-line contact, coil, or signal into a desired state (closed or open, energized or deenergized) regardless of its current state.

Frontplane

Signifies connectors, switches, LEDs, and other components located on the front edges of PWBs in an electronics module. For instance, the IMOK light is located on the frontplane of a DLC+ Sequencer module.

Group

An addressable group of 8 consecutive physical inputs or outputs (called points) that generally exists on one printed wiring board (PWB). Most PWBs contain one group of 8 points, although some can handle 16 points (two groups). When the DLC+ Sequencer addresses a group, it is really addressing a single PWB.

Group address

The single digit identifier within a signal address that specifies the group where the signal originates or terminates. See **Signal address.**

Hexadecimal

A numerical system commonly used in computer systems with a base of 16 rather than a base of 10 (as in our decimal system). One digit can represent a number from 0 to 15. Thus, counting up to 16 in hexadecimal would be as follows: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F, 10.

IMOK light

A yellow LED located on the front edge of microprocessor-based DLC+ Sequencer boards. When lighted, the IMOK light indicates that the board is operating properly. If the IMOK light goes out, a fault has been detected on the board.

Input signal (I prefix)

A signal originating outside of the DLC+ Sequencer in hardware such as limit switches, pushbuttons, thumbwheels, and tachometers.

Internal status signal (S prefix)

A signal within the DLC+ Sequencer that is both generated and used internally. For instance, a signal that links a string coil and a string contact is an internal status signal.

Key Information Block (KIB)

An internal DLC+ Sequencer table used by the programmer to set up the I/O configuration of a particular sequencer. The KIB entry in Supervisory mode allows the programmer to set up system parameters and define the channels, stations, racks, and groups used by the sequencer.

L-bus (Local bus)

Bus contained on the backplane of a DLC + Sequencer module through which the microprocessor board communicates with memory, the CPL master, module I/O, and similar units.

Latch relay string

A skeleton string whose output coil picks up and latches when activated by its string logic. The coil remains energized, even through power outages, until unlatched by its string. The latch relay string is created by the Latch Relay key on the DLC + Programming Module keyboard.

Local I/O

I/O boards located in the same panel lineup as the DLC+ Sequencer module (within 50 feet of the module), but not located within the module. I/O within the DLC+ Sequencer module is considered Module I/O, not Local I/O.

MEM (Medium Electronics Module)

A Directo-Matic ® Plus electronics module with a capacity of 24 single cards (2 rows of 12 cards each), 12 double cards (1 row of 12 cards), or some combination of the two.

Menu

A means of controlling the operation of the programming module by presenting a list of module functions and allowing the operator to choose the desired function. For example, the Supervisory menu allows the operator to choose one of thirteen possible actions, including running diagnostics, sending data to the printer, and so forth.

Mode

A DLC + Programming Module operating state. The module has five key-selected modes, each used for different purposes: Display String, Display Signal, Build String, Test String, and Supervisory modes. Each mode is explained in a separate entry in this glossary.

Mode field

An inverse video field labelled MODE and located in the lower left part of the DLC+ Programming Module display. This field displays the current mode in which the module is operating.

Module I/O

I/O boards located within the DLC + Sequencer module (that is, located within the DLC + Sequencer electronics module).

Off-line

A condition in which the DLC+ Sequencer does not process input or output signals and does not execute the application program.

On-line

The normally-operating condition of the DLC + Sequencer. The sequencer processes input and output signals and executes the application program.

Output signal (R prefix)

A signal originating within the DLC + Sequencer and destined for external hardware such as solenoids, indicator lights, displays, and motors (for speed references).

Physical station

A station whose input or output points exist on a collection of physical I/O boards. When a DLC+ Sequencer addresses a physical station, it is addressing a series of PWBs whose hardware strapping matches the station address.

Point

A single input or output signal wired into an I/O board and individually addressable by the DLC+ Sequencer. An I/O board can typically handle a group of 8 points, though some boards can handle 16 points.

Point address

The single digit identifier within a signal address that specifies the point where the signal originates or terminates. **See Signal address.**

Point-type signals

A single bit DLC + Sequencer signal used for input or output.

PMC (Programmable Master Controller)

A special purpose computer programmed in CBL that coordinates and communicates with sequencers, drives, and operator controls over the Control Signal Highway. As its name implies, it exercises control over a wide range of devices that make up a process/drive control system. It is particularly effective for programming analog and regulating functions.

Product software

Software resident within a General Electric product (usually in PROM) that controls the operation of that product only, not a customer application. Product software is essentially transparent to the customer and cannot be modified by him. The DLC+ Sequencer operating system is an example of product software. Product software is sometimes called firmware.

Programming Module
A portable terminal, consisting of a CRT screen and keyboard, used to program or troubleshoot DLC + Sequencer systems.

PROM (Programmable Read Only Memory)
Semiconductor memory used within the DLC+ Sequencer to store product software. Unlike BRAM memory, PROM memory is not user-programmable.

Pulse relay string
A skeleton string whose output coil is energized for one sequencer sweep when activated by its string logic. The string logic must change state from false to true to generate a pulse.

PWB
Printed Wiring Board.

R-bus (Reduced bus)
A partial decode of the F-bus used to communicate with Local or Remote I/O boards.

Rack
Used in signal addressing to specify a combination of up to 16 groups. Since a group is typically a printed wiring board, a rack is usually a card rack. **See Signal address.**

Rack address
A single character identifier in the signal address that specifies a particular rack.

RAM (Random Access Memory)
Semiconductor read-write memory that is not protected by a battery back-up system. **See BRAM.**

Relay ladder diagram
A method of displaying application software (relay logic) in visual form, using contacts, coils, and horizontal and vertical connections. The DLC+ Programming Module displays strings in relay ladder diagram form.

Remote I/O
I/O boards located outside of the panel line-up containing the DLC+ Sequencer module (more than 50 feet from the module). A Control Party Line is required for communication with remote I/O boards.

Resequence
To change the string numbers (but not the order) of a series of strings so that the string numbers occur in multiples of 10. For instance, the first string of a resequenced string would be 10, the next 20, the next 30, and so on. The Supervisory mode allows resequencing.

Scale block
A control block that, when enabled, performs a scaling calculation on three values (multiplies A times B and divides the result by C) and places the result into a fourth value X. A, B, and C can be input, output, or internal status signals, or constants. X can be an output or internal status signal or a constant. If the result of the scaling operation is greater than decimal +32,767 (overflow) or less than decimal -32,768 (underflow), the coil is energized.

SEM (Small Electronics Module)
A Directo-Matic® Plus electronics module with a capacity of 12 single cards (2 rows of 6 cards each), 6 double cards (1 row of 6 cards), or some combination of the two.

Signal address

A six digit alphanumeric identifier programmed into coils and contacts that specifies: 1) whether the signal is an input, output, or internal status signal and 2) the origin or destination of the signal associated with that coil or contact. The origin or destination is identified by station, rack, and group address, plus the point within that group (or type of signal: BCD, count, etc.). The signal address always appears above the coil or contact in a string.

Signal address field

An inverse video field labelled SIGNAL and located in the lower right part of the DLC+ Programming Module display. The operator enters data into this field in Display Signal mode to specify the signal address of the signal to be displayed.

Skeleton string

A 5 by 7 matrix of programmable relay contacts with a single output coil. Each contact can be assigned to a specific coil by entering a signal address. When a current path exists from the left bus to the coil, the coil is energized.

Slope/offset block

A control block that, when enabled, performs a slope/offset calculation on three values (A x B + C) and places the result into a fourth value (X). A, B and C can be input, output, or internal status signals, or constants. X can be an output or internal status signal or a constant. If the result of the slope/offset calculation is greater than decimal 32,767 (overflow) or less than decimal -32,768 (underflow), the coil is energized.

Station

Used in signal addressing to specify a combination of up to 16 racks. A station may be a physical station or a virtual station. Since a rack is typically a card rack, a **physical** station is usually a combination of card racks, perhaps in a single panel enclosure. A **virtual** station exists only within sequencer software. A single DLC + Sequencer can address up to 254 stations (hexadecimal 01 to FE).

Station address

A two character identifier in the signal address that specifies a particular station (hexadecimal 01 to FE).

String

That portion of a complete ladder diagram visible on the DLC + Programming Module screen. Also may be called “rungs”, “blocks”, or “control blocks.” In the DLC + Sequencer, all strings are numbered for easy recall and manipulation.

String number

A four digit number that identifies a particular string within the DLC + Sequencer. The string number is assigned to the string by the programmer when he builds the string. The number may be resequenced by the resequence function within the Supervisory mode.

String number field

An inverse video field located directly beneath the coil on the DLC + Programming Module screen. This field has two purposes: 1) when building a string, the string number is entered into this field, and 2) the string number of a displayed string is located in this field.

Supervisory mode

A mode in the DLC+ Programming Module used for functions unrelated to building and displaying strings. Some examples of these functions are running diagnostics, clearing memory, modifying the KIB, and controlling printer and tape operations.

Sweep

A complete scan of all input and output points by the DLC+ Sequencer. A scan involves: 1) reading input signals from the I/O boards and updating the affected strings (for example, opening or closing **software** contacts), and 2) reading output signals from the sequencer and updating the affected I/O board (for example, closing a **physical** relay).

TEM (Tiny Electronics Module)

A Directo-Matic @Plus electronics module with a capacity of 6 single cards in one row.

Test String mode

A mode in the DLC + Programming Module used for testing newly-created or modified strings prior to placing them on-line. Such strings must be tested or they cannot be entered into sequencer memory. This mode is actually part of Build String mode.

Test field

An inverse video field labelled TEST and located in the lower middle part of the DLC + Programming Module display. This field displays the string number of a string under test. If there is no string under test, the field is blank.

Time delay string

A skeleton string whose output coil, when enabled, energizes after a programmable time delay. The coil remains energized as long as a current path exists from the left bus to the coil.

Timer block

A control block that, when enabled, acts as a presettable 1.0 second or 0.1 second timer. When the current value is greater than or equal to the final value, the output coil is energized.

Value-type signals

A DLC+ Sequencer internal signal of at least 16 bits (points) requiring two or more groups to address. Some examples of value-type signals are BCD signals, count signals, and analog internal signals. The signal address of a value-type signal will contain an A, B, C, D, E, or F as its point specifier.

Virtual station

A station that exists only within sequencer memory. A virtual station is used for transferring data between sequencers or from a sequencer to a process computer.

Where used /find field

An inverse video field labelled WHERE USED / FIND and located directly beneath the String Address field on the DLC+ Programming Module display. This field has two purposes: 1) displays where used information, and 2) allows operator to specify the string number of a string to find and display.

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